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OCAPI Reference

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Revision History

Table 1: OCAPI Reference Revision History for Software Version 3.0.0

Date	Revision	Comment
July 2025	01	Initial release for software version 3.0.0

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Overview

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1.1 OCAPI Overview

The purpose of the Open Composable Application Programming Interface (OCAPI) is to provide data center provisioning capabilities across all Western Digital and non-Western Digital device technologies using a "normalized" API. The goal of the OCAPI is to configure and manage storage devices via the "control path", while the "data path" delivers data to and from the consuming host systems.

1.2 Industry Standards

The topics in this section describe some of the industry standard modeling and protocols used in the OCAPI.

1.2.1 Common Information Model

The OCAPI follows the industry standard conventions from the Common Information Model (CIM) provided by the Distributed Management Task Force body (dmtf.org). The resource naming conventions and resource attributes follow the CIM model for managed elements. The OCAPI simplifies the CIM model by "flattening" or "compressing" in the following ways.

Collections

Collections are rolled into the Resources as a "plural" of the resource type:

- GET /Volumes returns the list of Volume Resources
- GET /Volumes/{id} returns the specific Volume instance

Services

Services are rolled into the Resources themselves. For example, CIM requires an element called a StorageService to provide creating a new volume. The OCAPI collapses this notion into the resource itself, so that creating a new volume only requires telling the "resource type" (volumes in this case) to create a new volume. Any other manipulation of an existing resource is done directly to the resource. Some examples:

- POST /Volumes (params) creates a new Volume Resource
- PUT /Volumes/{id}?params modifies an existing Volume Resource
- DELETE /Volumes/{id} deletes an existing Volume Resource

Associations

Associations are provided "inherently" with Resource. This means retrieving a particular Resource may also give "navigation links" to other associated resources. This eliminates the need to traverse Association Classes in the CIM Model by providing readily available URIs to Collections and Instances of other resources that are related to this Resource.

Some examples:

- Media contains links to Storage Pools
- Storage Pools contain links to Media that makes up the Pool; contains links to Volumes exported by the Pool

- Volumes contain links to Storage Pools that make up the Volume; contains the links to Storage Endpoints exposing the Volume
- Storage Endpoints contain links to attached Volumes; contains links to Paths to remote/consuming Hosts
- Paths contain links to Storage Endpoints; contains links to Host Endpoints

1.2.2 JSON Data Format Model

The OCAPI uses the standard Javascript Object Notation (JSON) to describe the managed CIM-based elements and their CIM and/or Vendor-Unique-based attributes. The API responses are in a well-formed JSON format as well as request body entries to create or modify a resource. The JSON data format intrinsically determines the data type simply by indicating the “value” within a pair of double-quotations for string types and non-quoted numerical values for number types. No other format descriptions are needed to keep the API as simple as possible.

1.2.3 Protocols

The OCAPI follows strict interpretation of the Representational State Transfer (REST) architectural style put forth by the IETF and the Hypertext Transport Protocol (HTTP) specifications. This requires all API "actions or verbs" be rendered only with HTTP Methods that work directly with the resources.

This API supports the HTTP methods of GET, POST, PUT, DELETE, HEAD, and OPTIONS. This dictates that all Uniform Resource Identifiers (URIs) must be formed as "nouns" or "resources" upon which the methods are applied. No actions or verbs shall be rendered in the URI patterns or within the JSON request body.

1.2.4 HTTP Conditionals and CORS

The OCAPI uses HTTP Conditionals to provide concurrency capabilities among multiple web clients. This “optimistic concurrency” uses the “ETag Header” (entity tag) to provide a resource hash value (32 hexadecimal characters) of the current state of the resource. This ETag value can be used to lower the network traffic when used with HTTP GET requests, and is also used to properly modify or delete a specific resource such that concurrency among web clients is established. The next two subsections describe the API support for GET and PUT/DELETE usage of conditionals.

HTTP GET Conditional

A web client may use the ETag value returned from a GET response to use in the next GET request to the same resource. However, it is not a requirement. And not all resources provide ETags. This is usually the case for resources that change on their own often (e.g., the System Clock, Performance, or other time-based or automatically changing resources). If the web client chooses to use an ETag for a GET request, the “If-None-Match” conditional is used. The web client will understand from the “304 Not Modified” response that the resource or list of resources has not changed since the last time it retrieved the information.

Using Conditional GET significantly reduces network traffic by eliminating unneeded transfer of data if it has not changed over time. This works well with web clients that need to “poll” for status frequently. This means that GET responses are either fully verbose or fully silent if the web client uses the ETag with “If-None-Match” conditional request header.

HTTP PUT & DELETE Conditional

Another form of HTTP conditional allows web clients to take advantage of the HTTP Conditional PUT and DELETE processes. This includes an ETag returned in most GET Request response headers. When used with the HTTP PUT or DELETE method, the web client can determine if the resource has changed since the last GET response before a PUT (modify) or DELETE is requested. This is called "optimistic concurrency". It is the opposite of implementing explicit locking mechanisms to handle multiple web clients ("pessimistic concurrency").

The web client will receive an ETag as part of the GET response header which can be used to pass the "If-Match" conditional to the web service to determine if the resource has changed just before the PUT or DELETE Request.

If the "If-Match" conditional is not sent along with the PUT or DELETE request, the API will return a "428 Precondition Required" to prompt the web client to use the API's HTTP conditional capabilities. This prevents accidental or rogue changes and deletions and makes sure there is an orderly process when two or more web clients work on the same resource at or near the same time.

HTTP Cross-Origin Resource Sharing (CORS)

The OCAPI supports Cross-Origin Resource Sharing (CORS) operations. This is typically needed for web clients that connect to one particular web service which then tells the web client to connect to a different web service, usually in the executing Javascript, for example. The connection focus goes from "same-origin" or "same authority" to a "cross-origin" or different authority. Web clients that support this capability are required to send an OPTIONS method to the remote web service location (cross-origin) with the "Origin" header filled in with the local web service IP address or hostname with the scheme prefix (http or https) to determine of the remote web server will allow a connection from the web client while the focus is still on the local web server. If the remote web service disallows the connection, the OPTIONS response will return a "preflight connection failure" back to the web client indicating no further communication is allowed. If the remote web server allows the connection, it will return headers in the OPTIONS response to indicate what kind of connection, methods, etc., are allowed to the remote web server. The web client can then execute the original HTTP method to the remote web server to complete the transaction. Most modern web browsers support CORS.

1.2.5 HTTPS/SSL/TLS Connections

The OCAPI provides the user the ability to upload their own self-generated SSL/TLS Certificate and Key Pairs.

The OCAPI does provide its own self-signed Certificate and Key Pair, which is just enough to allow a browser to ask the user to proceed with the self-signed but unsecure connection. The user will need to do the following in order to successfully create secure connections between the browser and devices:

1. Create a user-generated Root Authority Certificate and Private Key pair
   - There are several SSL generators available (e.g. OpenSSL)
2. Create a user-generated Leaf Certificate and Key Pair signed by the Root CA Key (from the previous step)
   - The Leaf (or Device) Certificate/Key Pair must have the IP and/or DNS name as part of the Certificate that is within the user's network environment
3. Upload the Leaf Certificate and Key Pair to the Device via the OCAPI – see POST Files/TLSCertandKey in **/Compute/Devices/{id}/Files/{id}/** ***(page 83)*** or **/Compute/Devices/{id}/Files/{id}/** ***(page 261)***.

- The successful upload will automatically restart the API service for the HTTPS listener to use the new Certificate

1.3 Resource Oriented Architecture

The OCAPI implements "Resource-Oriented Architecture" (ROA) which dictates the API to have the following attributes:

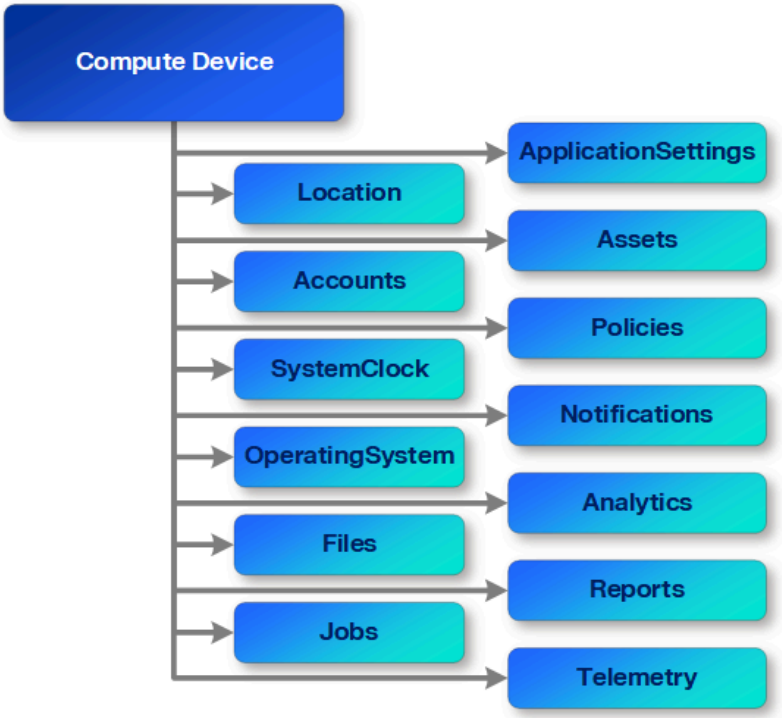
- Addressability - Resources have explicit URIs for direct addressing
- Stateless - Resources do not require previous or next state processing; all operations are atomic
- Linkability - Resources can be reached via "Self" links from other associated Resources; provides navigation among related Resources
- Uniform Interface - Resources respond to supported HTTP Methods

1.4 Modeling Device Resources

The OCAPI architecture requires the control path management scheme to be “resource-oriented”. Any device and its components are all considered resources. For example, a device is a resource as well as its power supply. The management of the device and the power supply have separate resource addressing so that a management client can manage any component resource directly. There are physical resources, logical resources, and virtual resources. The physical resources typically already exist and cannot be created or deleted. However, logical and virtual resources can be created, modified, and deleted. The RMDC Control Node is a located on a Compute Device with modeling includes logical/virtual resources.

The Control Node is located on a Compute Device with modeling that includes logical/virtual resources, as shown in the following diagram.

Figure 1: Control Node Resources Model



1.5 curl

Curl is a free, open-source command line tool and library for transferring data using various network protocols. It can be used to access the OCAPI to request resource information, modify attributes, and create or delete resources using HTTP methods. Example curl commands are provided throughout this API reference; the following are the most commonly used commands and options. For more information on curl usage and syntax, visit **https://curl.haxx.se/**.

Commonly Used Command Options

- `-i` Requests HTTP response headers to be included in the output
- `-u` *`user:password`* Specifies the user name and password for server authentication
- `-X` *`method`* Specifies a custom request method to use when communicating with the HTTP server. Options are GET, POST, PUT, DELETE, HEAD, and OPTIONS
- `-H` *`header`* Specifies an extra header to include in the request, such as Content-Type, If-Match, etc.
- `-d` *`data`* Specifies the body data for a POST or PUT request
- `-F` *`fieldname=@filename`* Specifies a file to upload by emulating the submission of a filled-in form
- `-k` Allows for HTTPS requests

GET

A GET request is the first step in interacting with a resource. The GET curl command provides authentication credentials and the URL of the resource:

```
curl -i -u authentication -X GET resourceURL
```

The response header will include an ETag, which is used in subsequent PUT or DELETE requests to modify or delete the resource.

PUT

A PUT request modifies the resource. The PUT curl command provides authentication credentials, headers (such as Content-Type and If-Match), request body data, and the URL of the resource:

```
curl -i -u authentication -H header -X PUT -d 'data' resourceURL
```

The If-Match conditional in the PUT command requires the ETag from the GET response header in order to modify the resource.

POST

A POST request to a collection creates a resource within that collection. The POST curl command provides authentication credentials, headers (such as Content-Type), request body data, and the URL of the resource:

```
curl -i -u authentication -H header -X POST -d 'data' resourceURL
```

DELETE

A DELETE request deletes a resource. The DELETE curl command provides authentication credentials, headers (such as If-Match), and the URL of the resource:

```
curl -i -u authentication -H header -X DELETE resourceURL
```

The If-Match conditional in the DELETE command requires the ETag from the GET response header in order to delete the resource.

Discovery

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2.1 Discovery Overview

The OCAPI uses a generic doorbell type of URI that queries a particular host and port to determine what kind of device or component is available as a resource. There is also a mechanism to have an OCAPI-based Device do a discovery of all other participating OCAPI-based Devices to generate a dynamic system list of Devices that can be managed through a single OCAPI-based Device as a single-point-of-management.

- **/Query/** - Returns a Summary of the Collection of Devices at this location (IP Address) - see **/Query/** *(page 11)*
- **/System/Query/** - Returns all the "Summary Query" responses from all discovered Collection of Devices across the network - see **/System/Query/** *(page 24)*

2.2 REST-Based API

This API is based on the true REST architectural style, meaning that all actions/verbs will be handled exclusively by the existing HTTP Methods (GET, POST, PUT, DELETE, HEAD, OPTIONS) along with all URI patterns containing only fully qualified collections of resources and resource instantiations (nouns only, no action verbs permitted in the URI).

URI Patterns

Uniform Resource Identifiers (URI) are built with the following pattern:

- *scheme, host, domain, resources, resourceID*

For example:

- `http://ip:port/domain/resources/resourceID[/resource/resourceID]...`

Where:

- *scheme* can be http or https
- *domain* can be “storage”, “compute”, “network”, or “system”
- *resources* can be “devices” “volumes”, “hosts”, “switches”, “servers”, etc.
- *resourceID* can be an index, GUID, UUID, or any type of identifier for the instance of a resource that is unique to at least the local device

The following is a real world example of a particular storage resource:

- `http://localhost:8080/Storage/Devices/12345`

All entries are plural-based so that a URI without an ending GUID or other resource identifier would return the collection or list of a particular element. For example, `http://ip:port/Storage/Devices/12345/Volumes` would return a list of volumes instead of a particular one.

URI representations of physical components and devices must include the fully qualified URI path.

Physical Components:

`http://ip:port/Storage/Devices/id/Adapters/id` is the address of a specific adapter in a specific storage device.

Logical/Virtual Components:

`http://ip:port/Storage/Devices/id/Volumes/id` is the address to a specific volume.

Resource URIs

There are several “management domains” that can be used to describe and manipulate devices. For converged or composed systems, these domains are “compute”, “network”, and “storage”. These typically describe management at the device or component level. The “system” domain entry is used to aggregate the 3 other domains into an all-encompassing management layer to help describe the usage and connectivity of the devices and components as a physical or virtual system. The system-level management typically allows aggregation lists of all participating devices and also any user-defined groupings of the devices and components that make up the system or sub-systems.

The base URI format is as follows:

- /Query – Doorbell URI
- /Storage/Devices – Storage-based device collection URI
- /System/Query – System-based device query aggregation URI
    - Contains all the discovered Query responses for all participating Device within the System

2.3 /Query/

The /Query/ resource (Doorbell) returns the device type and perhaps its higher level system membership and/or any other device(s) managed by this device via proxy, if any.

HTTP Methods: GET HEAD OPTIONS

Table 2: Resource Attributes

Attribute	Data Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
SystemQuery	String	Mandatory	No	Free-Form URI Format with IP plus "/System/Query"	Link to ask this Device to discover other devices
InformationStructure.Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Link to use to display or modify the InformationStructure
InformationStructure.Name	String	Mandatory	No	Free-Form 64 Chars	The API name
InformationStructure.HttpPort	Number	Mandatory	Yes	80, or 1024 - 65535 range	Port number for HTTP listener; default = 80
InformationStructure.HttpsPort	Number	Mandatory	Yes	443, or 1024 - 65535 range	Port number for HTTPS listener; default = 443
InformationStructure.ID	String	Mandatory	No	Free-Form 256 Chars	The query identifier represents the main device ID
InformationStrucuture.AuthenticationType.ID	Number	Mandatory	Yes	0 = Basic; 1 = Digest	The current authentication type ID
InformationStrucuture.AuthenticationType.Name	String	Mandatory	No	Basic or Digest	The current authentication type Name
InformationStructure.StructureDescription	String	Mandatory	No	Free-Form 256 Chars	The API Schema Description
InformationStructure.OwningOrganization	String	Mandatory	No	Free-Form 64 Chars	The API Schema Owner
InformationStructure.Status	String	Mandatory	No	Free-Form 64 Chars	The API Schema Version Release Status
InformationStructure.URI	String	Mandatory	No	Free-Form URI Format	The API Schema URI starting point

Attribute	Data Type	Mandatory or Optional	User Config-urable	Possible Values	Description
InformationStructure. Version	String	Mandatory	No	Major.Minor.Release Format	The API Schema Version Number
InformationStructure. ClientFilter	String	Optional	Yes	Default: All Clients if empty; REGEX of IP Address ranges with "|" delimiter	Creates a "white list filter" of allowed Clients based on REGEX pattern of IP Addresses. Multiple entries with "|" delimiter.
InformationStructure. MaximumThreads	Number	Optional	Yes	Default: 5	Throttles the total number of web client connections. Returns a 429 Too Many Requests if maximum number reached. Try again.
Devices. Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Devices. Members	String Array	Optional	No	One or more /Devices bodies	Collection of Device Summaries

GET /Query/

GET /Query/ returns the query information along with the “links” for further management navigation into this Device.

Example GET curl Command

```
curl http://10.20.30.40/Query/
```

Note: User authentication is not required, since the GET /Query/ is the "doorbell" to initiate first contact with the Device. The JSON response includes URI links to go further into the Device, which will require user authentication.

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Query/",
    "SystemQuery": "http://10.20.30.40/System/Query/",
    "SystemGroups": "http://10.20.30.40/System/Groups/",
    "SystemJobs": "http://10.20.30.40/System/Jobs/",
    "InformationStructure": {
        "Self": "http://10.20.30.40/Query/InformationStructure/",
        "AuthenticationType": {
            "ID": 0,
            "Name": "Basic"
        },
        "HTTPPort": 80,
        "HTTPSPort": 443,
        "LogLevel": "info",
```

```
        "MaximumThreads": 50,
        "Name": "Open Composable API",
        "OwningOrganization": "Copyright 2020-2022 Western Digital Corporation or its
 affiliates. All rights reserved.",
        "Status": "Preliminary Proof-of-Concept",
        "StructureDescription": "REST-based API for Device Management. Use HTTP OPTIONS
 with header {\"Documentation\": \"Schema\"} to get resource schema information based
 on URI. Use HTTP OPTIONS with header {\"Documentation\": \"Info\"} to get general
 information based on URI. ",
        "URI": "/Query/",
        "TimeoutMultiplier": 1,
        "Discovery": {
            "DiscoveryCycleTime": 60,
            "DiscoveryTimeout": 5,
            "QueryTimeout": 20,
            "Threads": 128,
            "Netmasks": []
        },
        "Orchestration": {
            "Enabled": false
        },
        "Version": "RMDC-1.4.4-224",
        "LDAPSettings": {}
    },
    "Devices": {
        "Self": "http://10.20.30.40/Devices/",
        "Members": [
            {
                "Self": "http://10.20.30.40/Compute/Devices/4C4C4544-004E-5410-8035-
B5C04F374733/",
                "SystemType": {
                    "ID": 1,
                    "Name": "Compute"
                },
                "Name": "5NT57G3",
                "ID": "4C4C4544-004E-5410-8035-B5C04F374733",
                "OperatingSystem": {
                    "Self": "http://10.20.30.40/Compute/Devices/4C4C4544-004E-5410-8035-
B5C04F374733/OperatingSystem/",
                    "Name": "Resource Manager Data Center",
                    "OSType": {
                        "ID": 18,
                        "Name": "Winnt"
                    },
                    "Version": "0.4.224"
                },
                "SerialNumber": "0000",
                "Model": "Resource Manager Data Center",
                "Manufacturer": "WDC",
                "Capabilities": {
                    "Members": [
                        {
                            "ID": 9,
                            "Name": "Access Server",
                            "CapabilityDescription": "This device is a server providing
 access to resources directly attached or in a network."
                        },
                        {
                            "ID": 14,
```

```
                            "Name": "Management",
                            "CapabilityDescription": "This device provides software
 system management capabilities."
                        }
                    ]
                },
                "Status": {
                    "State": {
                        "ID": 16,
                        "Name": "In service"
                    },
                    "Health": [
                        {
                            "ID": 5,
                            "Name": "OK"
                        }
                    ],
                    "Details": [
                        "None"
                    ]
                },
                "IPAddresses": {
                    "Members": [
                        {
                            "IPAddress": "10.20.30.40"
                        }
                    ]
                }
            }
        ]
    }
}
```


Table 3: GET Response Codes

Response Code	Name	Description
200	OK	OK & Support resource information in the response body
304	Not Modified	If-None-Match Etag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

HEAD /Query/

The HEAD /Query/ pings the Query resource.

Table 4: HEAD Response Codes

Response Code	Name	Description
200	OK	OK with no response body + Header "Server: OpenFlex API" to help determine that the IP address response to the doorbell request

Response Code	Name	Description
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

OPTIONS /Query/

The OPTIONS /Query/ returns the allowable HTTP Methods list for this resource.

Table 5: OPTIONS Request Headers

Request Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	Etag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 6: OPTIONS Response Codes

Response Code	Name	Description
200	OK	OK with no response body + Header "Server: OpenFlex API" to help determine that the IP address response to the doorbell request
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

2.4 /Query/InformationStructure/

The /Query/InformationStructure/ resource provides detailed information about the API.

HTTP Methods: GET PUT HEAD OPTIONS

Table 7: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
SystemQuery	String	Mandatory	No	Free-Form URI Format with IP plus "/System/Query"	Link to ask this Device to discover other devices
InformationStructure. Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Link to use to display or modify the InformationStructure
InformationStructure. Name	String	Mandatory	No	Free-Form 64 Chars	The API name
InformationStructure. HTTPPort	Number	Mandatory	Yes	80, or 1024 - 65535 range	Port number for HTTP listener; default = 80
InformationStructure. HTTPSPort	Number	Mandatory	Yes	443, or 1024 - 65535 range	Port number for HTTPS listener; default = 443
InformationStructure. ID	String	Mandatory	No	Free-Form 256 Chars	The query identifier represents the main device ID
InformationStrucuture. AuthenticationType.ID	Number	Mandatory	Yes	0 = Basic; 1 = Digest	The current authentication type ID
InformationStrucuture. AuthenticationType. Name	String	Mandatory	No	Basic or Digest	The current authentication type Name
InformationStructure. StructureDescription	String	Mandatory	No	Free-Form 256 Chars	The API Schema Description
InformationStructure. OwningOrganization	String	Mandatory	No	Free-Form 64 Chars	The API Schema Owner
InformationStructure. Status	String	Mandatory	No	Free-Form 64 Chars	The API Schema Version Release Status
InformationStructure. URI	String	Mandatory	No	Free-Form URI Format	The API Schema URI starting point
InformationStructure. Version	String	Mandatory	No	Major.Minor.Release Format	The API Schema Version Number

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
InformationStructure. ClientFilter	String	Optional	Yes	Default: All Clients if empty; REGEX of IP Address ranges with "|" delimiter	Creates a "white list filter" of allowed Clients based on REGEX pattern of IP Addresses. Multiple entries with "|" delimiter.
InformationStructure. MaximumThreads	Number	Optional	Yes	Default: 5	Throttles the total number of web client connections. Returns a 429 Too Many Requests if maximum number reached. Try again.
RestartAPI	Boolean	WriteOnly	Yes	true	Directs the device to restart the API service
LogLevel	String	Optional	Yes	"debug" | "info" | "warn" | "error"	The log level setting for writing into /var/log/vstore.log
TimeoutMultiplier	Number	Optional	Yes	1 - n	The timeout multiplier value to increase the API timeout values
Discovery. DiscoveryCycleTime	Number	Optional	Yes		
Discovery. DiscoveryTimeout	Number	Optional	Yes	1 to n; default = 5	The time in seconds to wait for HEAD / Query responses from remote devices during background GET / System/Query scans
Discovery. QueryTimeout	Number	Optional	Yes	1 to n; default = 20	The time in seconds to wait for GET / Query responses from remote devices during background GET / System/Query scans
Discovery. Threads	Number	Optional	Yes	1 to n; default = 128	The number of concurrent HEAD or GET /Query requests at once

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Discovery. Netmasks[]. Netmask	String	Optional	Yes	IPv4 IP Address format with CIDR Netmask value. nnn.nnn.nnn.nnn/mm where: nnn = 0 to 255 (IPv4 Address octet value) mm = 0 to 32 (CIDR subnet mask value)	The IPv4 IP Address/ CIDR netmask range for a given scan entry in the Netmasks[] Object Array. Default entry if the Netmasks[] object array is empty = "this RMDC server's IP Address and subnet"
Discovery. Netmasks[]. Enabled	Boolean	Optional	Yes	true | false	Indicates if the Netmask value is enabled for the scan range list of Netmasks
Discovery. DiscoveryInvFrequency	Number	Optional	Yes	2 to 10 minutes	Background Discovery interval in minutes
Discovery. DiscoveryFullInvMultiple	Number	Optional	Yes	1 to n multiplier to DiscoveryInvFrequency. Minimum allowed is 10 minutes	Background Discovery interval multiplier value
Discovery. PolicyESReadInv Frequency	Number	Optional	Yes	1 to n minutes	Background Policy DB interval update check
Discovery. PolicyESReadFullInv Frequency	Number	Optional	Yes	1 to n minutes	Background Policy DB interval for Full data update check
Orchestration. Enabled	Boolean	Optional	No	true | false	Indicates if the overall Orchestration capability is "Enabled" if all the subsequent "applications" are successfully installed and available
Orchestration. ApplicationsInstalled. Ansible	Boolean	Optional	No	true | false	Indicates if the "Ansible" application is enabled by being successfully installed and available
Orchestration. ApplicationsInstalled. Python3	Boolean	Optional	No	true | false	Indicates if the "Python3" application is enabled by being successfully installed and available

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Orchestration. ApplicationsInstalled. WDCModuleInfo	Boolean	Optional	No	true | false	Indicates if the "WDCModuleInfo" for the Ansible application is enabled by being successfully installed and available
Orchestration. ApplicationsInstalled. WDCModuleCommand	Boolean	Optional	No	true | false	Indicates if the "WDCModuleCommand" for the Ansible application is enabled by being successfully installed and available
Orchestration. ApplicationsInstalled. Nslookup	Boolean	Optional	No	true | false	Indicates if the "nslookup" application is enabled by being successfully installed and available
Version	String	Optional	No	maj.min.patch-build format	The current OCAPI version running on this system
LDAPSettings	Object	Optional	Yes		Remote Account LDAP/AD settings

GET /Query/InformationStructure/

GET /Query/InformationStructure/ retrieves the Information Structure information.

Table 8: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization.
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header.

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Query/InformationStructure/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Query/InformationStructure/",
    "AuthenticationType": {
        "ID": 0,
```

```
        "Name": "Basic"
    },
    "HTTPPort": 80,
    "HTTPSPort": 443,
    "LogLevel": "debug",
    "MaximumThreads": 50,
    "Name": "Open Composable API",
    "OwningOrganization": "Copyright 2020-2023 Western Digital Corporation or its
affiliates. All rights reserved.",
    "Status": "Released",
    "StructureDescription": "REST-based API for Device Management. Use HTTP OPTIONS with
header {\"Documentation\": \"Schema\"} to get resource schema information based on URI.
Use HTTP OPTIONS with header {\"Documentation\": \"Info\"} to get general information
based on URI. ",
    "URI": "/Query/",
    "TimeoutMultiplier": 1,
    "Discovery": {
        "DiscoveryCycleTime": 60,
        "DiscoveryTimeout": 5,
        "QueryTimeout": 20,
        "Threads": 128,
        "Netmasks": [
            {
                "Netmask": "10.202.239.0/24",
                "Enabled": true
            },
            {
                "Netmask": "10.203.111.0/22",
                "Enabled": true
            },
            {
                "Netmask": "10.206.137.0/24",
                "Enabled": true
            },
            {
                "Netmask": "10.206.136.0/24",
                "Enabled": true
            },
            {
                "Netmask": "10.206.111.0/22",
                "Enabled": true
            },
            {
                "Netmask": "10.202.237.0/24",
                "Enabled": true
            }
        ],
        "DiscoveryInvFrequency": 5,
        "DiscoveryFullInvMultiple": 2,
        "PolicyESReadInvFrequency": 10,
        "PolicyESReadFullInvFrequency": 60
    },
    "Orchestration": {
        "Enabled": true,
        "ApplicationsInstalled": {
            "Ansible": true,
            "Python3": true,
            "WDCModuleInfo": true,
            "WDCModuleCommand": true,
```

```
            "Nslookup": true
        }
    },
    "Version": "1.5.8-103",
    "LDAPSettings": {}
}
```

Table 9: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Support resource information in the response body
304	Not Modified	If-None-Match Etag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
403	Forbidden	ClientFilter is preventing access and resource is not allowed to retrieved
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy, client should try again later

PUT /Query/InformationStructure/

PUT /Query/InformationStructure/ modifies an attribute of the Information Structure.

Table 10: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization.
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header.

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Query/InformationStructure/
    --data '{
        "LogLevel": "info"
    }'
```

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

Example PUT Request Body

```
{
    "HTTPPort": number,
    "HTTPSPort": number,
    "AuthenticationType": {
        "ID" : 0 | 1 | 2
    },
    "ClientFilter" : REGEX,
    "MaximumThreads": number,
    "RestartAPI": true,
    "LogLevel": "debug"|"info"|"warn"|"error",
    "TimeoutMultiplier": 1 - n (default: 1),
    "Discovery": {
        "DiscoveryCycleTime": 60,
        "DiscoveryTimeout": 5,
        "QueryTimeout": 20,
        "Threads": 128,
        "Netmasks": [
            {
                "Netmask": "10.202.239.0/24",
                "Enabled": true
            }, {
                "Netmask": "10.203.111.0/22",
                "Enabled": true
            }
        ],
        "DiscoveryInvFrequency": 5,
        "DiscoveryFullInvMultiple": 2,
        "PolicyESReadInvFrequency": 10,
        "PolicyESReadFullInvFrequency": 60
    }
}
```

Table 11: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Volume information in the response body + ETag Header
301	Moved Permanently	URI Address has moved to a new location if the Port Number(s) are changed; provides new URI in "Location" header in the response
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
412	Precondition Failed	Indicates the “If-Match” Etag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an Etag value in the Request Header
429	Too Many Requests	Indicates API service is Busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Query/InformationStructure/

HEAD /Query/InformationStructure/ pings the Information Structure resource.

Table 12: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body + Header "Server: OpenFlex API" to help determine that the IP address response to the doorbell request
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Query/InformationStructure/

OPTIONS /Query/InformationStructure/ returns the allowable HTTP Methods list for this resource.

Table 13: OPTIONS Request Headers

Header	Mandatory or Optional	Description
None	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain tex
Documentation	Optional	Schema - Returns the schema information in formatted JSON text as plain text
If-None-Match	Optional	Etag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 14: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match Etag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

2.5 /System/Query/

The /System/Query/ resource returns the dynamically-discovered list of device Query Doorbell responses in a single response body, to provide a single-point-of-management entry point.

HTTP Methods: GET HEAD OPTIONS

Table 15: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Members	String Array	Optional	No	One or more discovered / Query bodies	Collection of discovered / Query doorbell resources

GET /System/Query/

GET /System/Query/ returns the information of the System Query resource.

Table 16: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 17: GET Query Parameters

Parameter	Description
Netmask = {ip_address/cidr}	The IP address and CIDR code range for the sweep/scan process. Default sweep is system IP Address subnet (current IP subnet 10.20.30.xx, CIDR = 24). Note: CIDR determines "ip mask" range to scan; 0 - 32, smaller the number, the wider the scan range.
DiscoveryTimeout = {1-n}	The timeout value in seconds for the discovery process. Default = 5 seconds. 3 or more recommended.
QueryTimeout = {1-n}	The timeout value in seconds for the query process. Default = 20 seconds. 8 or more recommended.
Threads = {1-n}	The number of parallel /Query doorbell requests based on the number of IP Address/CIDR to send to the network (multi-threaded unicast). Default = 128. 1024 or less recommended.

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/System/Query/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/System/Query/",
    "Members": [
        {
            "Self": "http://10.20.30.40/Query/",
            "SystemQuery": "http://10.20.30.40/System/Query/",
            "InformationStructure": {
                "Self": "http://10.20.30.40:80/Query/InformationStructure/",
                "AuthenticationType": {
                    "ID": 0,
                    "Name": "Basic"
                },
                "Name": "OpenFlex API",
                "HTTPPort": 80,
                "HTTPSPort": 443,
                "ID": "0123456789",
                "StructureDescription": "REST-based API for Device or System Management.
 Use HTTP OPTIONS to get resource schema information based on URI.",
                "OwningOrganization": "WDC",
                "Status": "Preliminary",
                "URI": "/Query/",
                "Version": "0.9.11"
            },
            "Devices": {
                "Self": "http://10.20.30.40/Devices/",
                "Members": [
                    {
                        "Self": "http://10.20.30.40/Storage/Devices/0123456789/",
                        "SystemType": {
                            "ID": 2,
                            Name": "Storage"
                        },
                        "Name": "opnflx-6789",
                        "ID": "blade_uuid_0123456789",
                        "OperatingSystem": {
                            "Self": "http://10.20.30.40/Storage/Devicess/0123456789/
OperatingSystem/",
                            "Name": "Vendor Firmware",
                            "OSType": {
                                "ID": 0,
                                "Name": "Unknown"
                            },
                            "Version": "blade-FW-Rev0"
                        },
                        "SerialNumber": "blade-SN-0123456789",
                        "Model": "blade-MN-0123456789",
                        "Manufacturer": "WDC",
                        "Capabilities": {
                            "Members":[
                                {
                                    "ID": 3,
                                    "Name": "Storage",
                                    "CapabilityDescription": "This device is a storage
 provider."
                                },
                                {
                                    "ID": 15,
```

```
                                    "Name": "Block Server",
                                    "CapabilityDescription": "This device provides block
 storage."
                                },
                                {
                                    "ID": 28,
                                    "Name": "Management Controller",
                                    "CapabilityDescription": "This device provides
 specialized hardware dedicated to systems management."
                                },
                                {
                                    "ID": 40,
                                    "Name": "Blade",
                                    "CapabilityDescription": "This device is a blade
 device."
                                }
                            ]
                        },
                        "Status": {
                            "State": {
                                "ID": 16,
                                "Name": "In service"
                            },
                            "Health": [
                                {
                                    "ID": 5,
                                    "Name": "OK"
                                }
                            ],
                            "Details": [
                                "None"
                            ]
                        },
                        "IPAddresses": {
                            "Members": [
                                {
                                    "IPAddress": "10.20.30.40"
                                },
                                {
                                    "IPAddress": "192.168.0.9"
                                }
                            ]
                        }
                    }
                ]
            }
        }
    ]
}
```

Example GET curl Command Across Disparate Subnets

Note: This example scans 10.20.30.0 thru 10.20.30.255 (1 subnet) and 10.202.236.0 thru 10.202.239.255 (4 subnets). The user can also enter a Netmask that does not include the Device doing the GET /System/Query. In that case, the response will not include that Device, but all other discovered Devices in the scan.

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/System/Query/?
Netmask=10.20.30.40/24&Netmask=10.202.236.0/22
```

Table 18: GET Response Codes

Response Code	Name	Definition
200	OK	OK & System Query resource information in the response body
304	Not Modified	If-None-Match Etag value matched, therefore no Response Body will be returned. **Note**: for /System/Query, the order of devices coming back may be different, so it is rare to get a 304.
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /System/Query/

HEAD /System/Query/ pings the System Query resource.

Table 19: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	

Table 20: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /System/Query/

OPTIONS /System/Query/ returns the allowable HTTP Methods list for this resource.

Table 21: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	Etag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 22: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match Etag value matched, therefore no Response Body will be returned. **Note**: for /System/Query, the order of devices coming back may be different, so it is rare to get a 304.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

Central Service

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3.1 System Level Resources

The OCAPI incorporates "system-level" resources to provide multiple "device resource" activities and operations such as FW Update, Reboot, etc. This requires the concept of "grouping device resource" into a new virtual resource called "System Groups" to allow the system to apply certain operations across multiple device resources at the same time. The System Jobs resource describes the current and completed "Jobs" that were enacted on a System Group.

- **/System/Groups/** - Returns the collection of System Groups - see **/System/Groups/** *(page 31)*
- **/System/Jobs/** - Returns the collection of System Jobs - see **/System/Jobs/** *(page 42)*

3.1.1 /System/Groups/

The Collection-level methods and attributes of /System/Groups/ return information about the System Groups collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 23: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/System/Groups/{id}/** *(page 35)* for Instance Attribute details

GET /System/Groups/

GET /System/Groups/ returns the Collection of System Groups.

Table 24: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/System/Groups/
```

Table 25: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of System Groups + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /System/Groups/

POST /System/Groups/ creates a new System Group.

Table 26: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

POST /System/Groups/ creates a new system group.

Example POST Request Body

```
{
    "Name": "name",
    "Description": "description",
    "Resources": [
        {
            "Self": "uri1"
        },
        {
            "Self": "uriN"
        }
    ]
}
```

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/System/Groups/
    --data '{
        "Name":"Data102 Group",
        "Description":"Data102 Group for Bulk FW Update"
        "Resources": [
            {
                "Self": "https://10.20.30.40/Storage/Devices/0123456789/?
IPAddress=10.20.30.50"
            },
            {
                "Self": "https://10.20.30.40/Storage/Devices/012345678912/?
IPAddress=10.20.30.51"
            },
            {
                "Self": "https://10.20.30.40/Storage/Devices/012345678933/?
IPAddress=10.20.30.52"
            }
        ],
    }'
```

Table 27: POST Response Codes

Response Code	Name	Definition
201	Created	Created & System Group information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /System/Groups/

HEAD /System/Groups/ pings the System Groups Collection.

Table 28: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 29: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body

Response Code	Name	Definition
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /System/Groups/

OPTIONS /System/Groups/ returns the allowable HTTP Methods list for this resource.

Table 30: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 31: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.1.2 /System/Groups/{id}/

The Device-level methods and attributes of /System/Groups/{id}/ return information about a specific System Group and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 32: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier this particular System Group
Name	String	Mandatory	Yes	Free-Form 64 Characters	Name of the Group
Description	String	Optional	Yes	Free-Form 256 Characters	Description of the Group
CreateDate	Datetime	Mandatory	No	"23-Feb-2022 09:35:47"	Creation datetime auto-generated upon POST /System/Groups
LastModified	Datetime	Mandatory	No	"23-Feb-2022 09:35:47"	LastModified datetime auto-generated upon POST /System/Groups or PUT /System/Groups/{id}
MixedResourceTypes	Boolean	Mandatory	No	true | false	Indicates if there are members of the group that have differing Resource Type and/or "OOB Interface" types (e.g. OCAPI versus Redfish)
Resources[x]. Self	URI String	Mandatory	Yes	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Resources[x]. IPAddress	String	Mandatory	No	Free-Form 128 Characters	IP Address that is provided from the Self URI repeated here (auto-generated)

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Resources[x]. Hostname	String	Optional	Yes	Free-Form 128 Characters	Hostname based on the DNS lookup of the IP Address provided in the Self URI (auto-generated) or sent by web client as an option
Resources[x]. ConnectionStatus	String	Mandatory	No	"OK" | "Unauthorized" | "Unknown"	Current connection status to determine if the Resource is available to connect to (e.g. OK or Unauthorized requiring credential update)

GET /System/Groups/{id}/

GET /System/Groups/{id}/ returns information regarding a specific System Group based on the {id} in the URI address string.

Table 33: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/System/Groups/087d9191-9cd0-7107-b24a-0afd619ef10f
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/System/Groups/087d9191-9cd0-7107-b24a-0afd619ef10f/",
    "ID": "087d9191-9cd0-7107-b24a-0afd619ef10f",
    "Name": "Data102 Group",
    "Description": "",
    "CreateDate": "20-Jul-2022 22:55:42",
    "LastModified": "20-Jul-2022 22:55:42",
    "MixedResourceTypes": false,
    "Resources": [
        {
            "Self": "http://10.20.30.40/Storage/Devices/THCLS00220EA0045-A/?
IPAddress=10.202.239.228",
            "IPAddress": "10.202.239.228",
```

```
            "Hostname": "oobm-00-0c-ca-07-0a-e3.hgst.com.",
            "ConnectionStatus": "OK"
        },
        {
            "Self": "http://10.20.30.40/Storage/Devices/USCSJ04217EA0004/?
IPAddress=10.202.239.176",
            "IPAddress": "10.202.239.176",
            "Hostname": "oobm-00-0c-ca-04-00-28.hgst.com.",
            "ConnectionStatus": "OK"
        },
        {
            "Self": "http://10.20.30.40/Storage/Devices/USWSJ01119EZ002C/?
IPAddress=10.202.239.246",
            "IPAddress": "10.202.239.246",
            "Hostname": "oobm-00-0c-ca-08-1d-e8.hgst.com.",
            "ConnectionStatus": "Unauthorized"
        }
    ]
}
```


Table 34: GET Response Codes

Response Code	Name	Definition
200	OK	OK & System Group information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /System/Groups/{id}/

PUT /System/Groups/{id}/ modifies an attribute or executes an operation for this System Group.

Table 35: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Optional	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H 'If-Match: 3677651d2011428f32120a029b015e81'
    -H 'Content-Type: application/json'
    -X PUT http://10.20.30.40/System/Groups/
    --data '{
        "Resources": [
            {
                "Self": "https://10.20.30.40/Storage/Devices/0123456789/?
IPAddress=10.20.30.50"
            },
            {
                "Self": "https://10.20.30.40/Storage/Devices/012345678912/?
IPAddress=10.20.30.51"
            },
            {
                "Self": "https://10.20.30.40/Storage/Devices/012345678933/?
IPAddress=10.20.30.52"
            },
            {
                "Self": "https://10.20.30.40/Storage/Devices/012345678933/?
IPAddress=10.20.30.53"
            }
        ],
       "Description": "Updated Data102 Group for Bulk FW Update"
    }'
```

Example PUT Request Body

```
{
    "Name": "name",
    "Description": "description",
    "Resources": [
        {
            "Self": "uri1",
        },
        {
            "Self": "uriN",
        }
    ],
}
```

Table 36: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /System/Groups/{id}/

DELETE /System/Groups/{id}/ deletes this System Group.

Table 37: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/System/Groups/12345
```

Table 38: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /System/Groups/{id}/

HEAD /System/Groups/{id}/ pings this System Group.

Table 39: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 40: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /System/Groups/{id}/

OPTIONS /System/Groups/{id}/ returns the allowable HTTP Methods list for this resource.

Table 41: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 42: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.1.2.1 Bulk Operations for a Resource Group

The following are example curl commands to perform bulk operations on a resource group.

Bulk Firmware Update

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/System/Groups/0123456/OperatingSystem/
    --data '{
        "FirmwareFile": "http://location/of/file"
    }'
```

Bulk Reboot

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/System/Groups/0123456/Reboot/
    --data '{
        "Reboot": true
    }'
```

Bulk Power State Change

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/System/Groups/0123456/PowerState/
    --data '{
        "ID": 4
    }'
```

Note: ID=2 is "PowerOn", ID=4 is "Sleep".

Bulk Locate LED Toggle

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/System/Groups/0123456/IndicatorLED/
    --data '{
        "ID": 2
    }'
```

Note: ID=2 is "Lit", ID=4 is "Off".

3.1.3 /System/Jobs/

The Collection-level methods and attributes of /System/Jobs/ return information about the System Jobs collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 43: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/System/Jobs/{id}/** *(page 45)* for Instance Attribute details

GET /System/Jobs/

GET /System/Jobs/ returns the Collection of System Jobs.

Table 44: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/System/Jobs/
```

Table 45: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of System Jobs + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /System/Jobs/

HEAD /System/Jobs/ pings the System Jobs collection.

Table 46: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 47: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /System/Jobs/

OPTIONS /System/Jobs/ returns the allowable HTTP Methods list for this resource.

Table 48: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 49: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.1.4 /System/Jobs/{id}/

The Device-level methods and attributes of /System/Jobs/{id}/ return information about a specific System Job and enable management operations.

HTTP Methods: GET DELETE HEAD OPTIONS

Table 50: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 64 Chars UUID Format	The Identifier this particular System Job. This ID is based on the prefix "GroupOperation" with a suffix of UNIX Epoch Time for uniqueness
StartTime	String	Mandatory	No	23-Feb-2022 09:35:47	The job start timestamp
ElapsedTime	String	Mandatory	No	23-Feb-2022 09:46:04	The job elapsed timestamp; updates during process; displays "end time" when job completed
Owner	URI String	Mandatory	No	Free-Form URI Format 256 Chars	The System Group URI used for this System Job
PercentComplete	Number	Mandatory	No	0 - 100	The percent complete status of the System Job

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service" | "Inoperative" | "Write Protected" | "Reboot Needed" | "Activate Needed" | "Activate with reboot needed" | "Activate invalidated by reboot"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16 | 65537 - 65542	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries.
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65537 - 65541	The current health value map of this Resource. There can be 1 or more ID entries.

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Mandatory	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Resource. For System Jobs, the following Details may be rendered: • 1st entry describes Operation Type • 2nd to Nth entry describes each of the System Group entries status/ progress

GET /System/Jobs/{id}/

GET /System/Jobs/{id}/ returns information regarding a specific System Job based on the {id} in the URI address string.

Table 51: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/System/Jobs/GroupOperation1812345
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/System/Jobs/GroupOperation1812345",
    "ID": "GroupOperation1812345",
    "StartTime: "23-Feb-2022 09:35:47",
    "ElapsedTime": "23-Feb-2022 09:46:04",
    "Owner": "http://10.20.30.40/System/Groups/12345",
    "PercentComplete": 100,
    "Status": {
        "State": {
            "ID": 8,
            "Name": "Completed"
        },
        "Health": [
```

```
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "OPERATION: System Group OS Update: http://fileserver/users/bob/
bundle4_0_13.tar.gz",
            "SUCCESS: Resource:oobm-00-0c-ca-08-1d-e8, Message:Action was successful,
 StatusCode:0",
            "SUCCESS: Resource:oobm-00-0c-ca-07-09-20, Message:Action was successful,
 StatusCode:0",
            "SUCCESS: Resource:oobm-00-0c-ca-07-09-02, Message:Action was successful,
 StatusCode:0"
        ]
    }
}
```


Table 52: GET Response Codes

Response Code	Name	Definition
200	OK	OK & System Job information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /System/Jobs/{id}/

DELETE /System/Jobs/{id}/ deletes this System Job.

Table 53: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/System/Jobs/GroupOperation1812345
```

Note: System Jobs cannot be deleted when in-progress (i.e. Percent Complete is less than 100%).

Note: System Jobs are automatically created with unique IDs when a System Group is used in a Group Operation (e.g. FW Update, Reboot, Power State, or Locate LED).

Table 54: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /System/Jobs/{id}/

HEAD /System/Jobs/{id}/ pings this System Job.

Table 55: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 56: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /System/Jobs/{id}/

OPTIONS /System/Jobs/{id}/ returns the allowable HTTP Methods list for this resource.

Table 57: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 58: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.2 Physical Devices

The OCAPI's physical resource model includes the overall Compute Proxy Device. The topics in this section provide information for accessing and managing the features of these physical resources.

3.2.1 /Compute/Devices/

The Collection-level methods and attributes of /Compute/Devices/ return information about the Devices collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 59: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/** *(page 54)* for Instance Attribute details

GET /Compute/Devices/

GET /Compute/Devices/ returns the Collection of Devices.

Table 60: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/
```

Table 61: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Devices + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/

HEAD /Compute/Devices/ pings the Devices collection.

Table 62: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 63: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/

OPTIONS /Compute/Devices/ returns the allowable HTTP Methods list for this resource.

Table 64: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 65: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.2.2 /Compute/Devices/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/ return information about a specific Device and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 66: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
SystemType. Name	String	Mandatory	No	"Compute"	The domain type string indicating the type of Device Resource
SystemType. ID	Number	Mandatory	No	1	The domain type number indicating the type of Device Resource
Name	String	Mandatory	Yes	Free-Form 256 Chars	The user-defined Friendly Name for this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
OperatingSystem. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Operating System Resource
OperatingSystem. Name	String	Mandatory	No	Free-Form URI	Resource Manager Data Center Edition Application Name
OperatingSystem. OSType. Name	String	Mandatory	No	"Winnt" | "Linux"	Resource Manager Data Center Edition running on this OS Type string value
OperatingSystem. OSType. ID	Number	Mandatory	No	18 | 36	Resource Manager Data Center Edition running on this OS Type number value
OperatingSystem. Version	String	Mandatory	No	Free-Form URI	Resource Manager Data Center Edition Application Version

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Model	String	Mandatory	No	"Resource Manager Data Center"	Resource Manager Data Center Edition application model
Manufacturer	String	Mandatory	No	"WDC"	Resource Manager Data Center Edition manufacturer
Capabilities. Members[]. Name	String	Mandatory	No	"Access Server" | "Management"	The capabilities information value(s) regarding the Resource
Capabilities. Members[]. ID	Number	Mandatory	No	9 | 14	The capabilities information value map(s) regarding the Resource
Capabilities. Members[]. CapabilitiesDescription	String	Mandatory	No	Free-form characters	Entry describing the capability
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries.
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536 - 65538	The current health value map of this Resource. There can be 1 or more ID entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
IPAddresses. Members[]. IPAddress	String Array	Mandatory	Yes	IPv4 or IPv6 pattern	The IP Address list for this Resource
InformationStructure. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Information Structure Resource
ApplicationSettings. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Application Settings Resource
Location. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Location Resource
Accounts. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Accounts Resource
Files. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Files Resource
Jobs. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Jobs Resource
SystemClock. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the System Clock Resource
Assets. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Assets Resource Collection
Notifications. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Notifications Resource
Policies. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Policies Resource
Analytics. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Analytics Resource
Reports. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Reports Resource
Telemetry. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Telemetry Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
FactoryReset	Boolean	WriteOnly	Yes	true	Directs the Application to return to factory fresh settings

GET /Compute/Devices/{id}/

GET /Compute/Devices/{id}/ returns information regarding a specific Device based on the {id} in the URI address string.

Table 67: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/",
    "SystemType": {
        "ID": 1,
        "Name": "Compute"
    },
    "Name": "rmdc-server",
    "ID": "0123456789",
    "OperatingSystem": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/OperatingSystem/",
        "Name": "Resource Manager Data Center",
        "OSType": {
            "ID": 36,
            "Name": "Linux"
        },
        "Version": "1.0.0"
    },
    "Model": "Resource Manager Data Center",
    "Manufacturer": "WDC",
    "Capabilities": {
        "Members": [
            {
                "ID": 9,
                "Name": "Access Server",
```

```
                "CapabilityDescription": "This device is a server providing access to
resources directly attached or in a network."
            },
            {
                "ID": 14,
                "Name": "Management",
                "CapabilityDescription": "This device provides software system
management capabilities."
            }
        ]
    },
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "None"
        ]
    },
    "IPAddresses": {
        "Members": [
            {
                "IPAddress": "10.20.30.40"
            },
            {
                "IPAddress": "192.168.0.6"
            }
        ]
    },
    "InformationStructure": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/InformationStructure/"
    },
    "ApplicationSettings": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/ApplicationSettings/"
    },
    "Location": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Location/"
    },
    "Accounts": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Accounts/"
    },
    "Files": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/"
    },
    "Jobs": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Jobs/"
    },
    "SystemClock": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/SystemClock/"
    },
    "Assets": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Assets/"
```

```
    },
    "Notifications": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Notifications/"
    },
    "Policies": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Policies/"
    },
     "Analytics": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Analytics/"
    },
     "Reports": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Reports/"
    },
    "Telemetry": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Telemetry/"
    }
}
```

Table 68: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Device information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/

PUT /Compute/Devices/{id}/ modifies an attribute or executes an operation for this Compute Device.

Table 69: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789
    --data '{
        "FactoryReset" : true
    }'
```

Table 70: PUT Response Codes

Response Code	Name	Definition
200	OK	OK
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	Resource could not execute the request

HEAD /Compute/Devices/{id}/

HEAD /Compute/Devices/{id}/ pings this Storage Device.

Table 71: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 72: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/

OPTIONS /Compute/Devices/{id}/ returns the allowable HTTP Methods list for this resource.

Table 73: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 74: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

Note: There may be some attributes returned in the OPTIONS response schema that may not be directly supported by this resource.

3.3 Logical Devices

The OCAPI's logical and virtual resources model include those resources that are logical or virtual in nature. The topics in this section provide information for accessing and managing the features of these logical and virtual resources.

3.3.1 /Compute/Devices/{id}/Accounts/

The Collection-level methods and attributes of /Compute/Devices/{id}/Accounts/ return information about the Accounts collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 75: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Accounts/{id}/** *(page 66)* for Instance Attribute details

GET /Compute/Devices/{id}/Accounts/

GET /Compute/Devices/{id}/Accounts/ returns the Collection of Accounts.

Table 76: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/ComputeDevices/0123456789/Accounts/
```

Table 77: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Accounts + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Accounts/

POST /Compute/Devices/{id}/Accounts/ creates a new Account.

Table 78: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/Compute/Devices/01234567890/Accounts/
    --data '{
        "UserID":"new_username",
        "BasicAuthentication": {
            "UserPassword":"password"
        },
        "Role": {
            "ID": 0
        }
    }'
```

Example POST Request Body

```
{
    "UserID": "user_id",
    "BasicAuthentication": {
       "UserPassword": "password"
    },
    "DigestAuthentication": {
        "UserPassword": "password"
    },
    "JWTAuthentication": {
        "UserPassword": "password"
    },
    "Role": {
        "ID": 0 | 1
    }
}
```

Table 79: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Account information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Accounts/

HEAD /Compute/Devices/{id}/Accounts/ pings the Accounts collection.

Table 80: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 81: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Accounts/

OPTIONS /Compute/Devices/{id}/Accounts/ returns the allowable HTTP Methods list for this resource.

Table 82: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 83: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.2 /Compute/Devices/{id}/Accounts/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Accounts/{id}/ return information about a specific Account and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 84: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 64 Chars Valid characters include: "a-z A-Z 0-9 _ - "	The Identifier of this particular Account
BasicAuthentication. UserPassword SHA1Checksum	String	Mandatory	No	Free-Form 128 Characters	The basic authentication checksum
BasicAuthentication. UserPassword	String	Optional	Yes	Free-Form 128 Characters	The user password used to create the checksum (not shown in the GET response)
DigestAuthentication. UserPassword MD5Checksum	String	Mandatory	No	Free-Form 128 Characters	The digest authentication checksum
DigestAuthentication. UserPassword	String	Optional	Yes	Free-Form 128 Characters	The user password used to create the checksum (not shown in the GET response)
DigestAuthentication. Realm	String	Optional	No	Free-Form 128 Characters	The digest authentication realm
Role. Name	String	Mandatory	No	"Admin"	The Role type string for this Account
Role. ID	Number	Mandatory	No	0	The Role type value for this Account
UserID	String	Mandatory	Yes	Free-Form 64 Chars	The user identifier for this Account

GET /Compute/Devices/{id}/Accounts/{id}/

GET /Compute/Devices/{id}/Accounts/{id}/ returns information regarding a specific Account based on the {id} in the URI address string.

Table 85: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Accounts/admin
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Accounts/admin/",
    "ID": "admin",
    "BasicAuthentication": {
        "UserPasswordSHA1Checksum": "{SHA}0DPiKuNIrrVmD8IUCuw1hQxNqZc="
    },
    "DigestAuthentication": {
        "UserPasswordMD5Checksum": "97ebad852d0dabfd6b71ae26fff61fa3",
        "Realm": "Western Digital Corporation"
    },
    "JWTAuthentication": {
        "UserPasswordMD5Checksum": "b80648545a30e0999892b517127c0e17"
    },
    "Role": {
        "ID": 0,
        "Name": "Admin"
    },
    "UserID": "admin"
}
```


Table 86: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Accounts/{id}/

PUT /Compute/Devices/{id}/Accounts/{id}/ modifies an attribute or executes an operation for this Account.

Table 87: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Accounts/test
    --data '{
        "UserID": "test",
        "BasicAuthentication": {
            "UserPassword": "new_password"
        },
        "Role": {
            "ID": 1
        }
    }'
```

Example PUT Request Body

```
{
    "UserId": "userId",
    "BasicAuthentication": {
        "UserPassword":"new_password"
    },
    "DigestAuthentication": {
        "UserPassword":"new_password"
    },
    "JWTAuthentication": {
        "UserPassword":"new_password"
    },
    "Role": {
        "ID": 0 | 1
    }
}
```

Table 88: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Accounts/{id}/

DELETE /Compute/Devices/{id}/Accounts/{id}/ deletes this account.

Table 89: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Accounts/test
```

Note: The last remaining "Admin" role account cannot be deleted to prevent loss of communication with the Device.

Table 90: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist

Response Code	Name	Definition
409	Conflict	In the case of only one account, that last one cannot be deleted
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is Busy, client should try again later

HEAD /Compute/Devices/{id}/Accounts/{id}/

HEAD /Compute/Devices/{id}/Accounts/{id}/ pings this Account.

Table 91: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 92: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Accounts/{id}/

OPTIONS /Compute/Devices/{id}/Accounts/{id}/ returns the allowable HTTP Methods list for this resource.

Table 93: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 94: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.3 /Compute/Devices/{id}/OperatingSystem/

The methods and attributes of /Compute/Devices/{id}/OperatingSystem/ return information about the Operating System (Application) and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 95: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware name
OSType. Name	String	Mandatory	No	"Winnt" | "Linux"	The Operating System Type value
OSType. ID	Number	Mandatory	No	18 | 36	The Operating System Type value map
Version	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Application Version

GET /Compute/Devices/{id}/OperatingSystem/

GET /Compute/Devices/{id}/OperatingSystem/ returns the information of the Operating System resource.

Table 96: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/OperatingSystem/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/OperatingSystem/",
    "Name": "Resource Manager Data Center",
    "OSType": {
        "ID": 36,
        "Name": "Linux"
    },
    "Version": "3.0.0"
}
```

Table 97: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Operating System resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/OperatingSystem/

HEAD /Compute/Devices/{id}/OperatingSystem/ pings the Operating System resource.

Table 98: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 99: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/OperatingSystem/

OPTIONS /Compute/Devices/{id}/OperatingSystem/ returns the allowable HTTP Methods list for this resource.

Table 100: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 101: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.4 /Compute/Devices/{id}/ApplicationSettings/

The methods and attributes of /Compute/Devices/{id}/ApplicationSettings/ return information about the Application Settings and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 102: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 64 Chars	The Name of this application
Version	String	Mandatory	No	Free-Form 64 Chars	The current Version of this application in the major.minor.patch format
ClusterSettings. Nodes	Object	Mandatory	No	ClusterSettings.Nodes	The Cluster Node settings for the ElasticSearch Database. Up to 3 Nodes are supported.
ClusterSettings. Nodes.IPAddress	String	Mandatory	Yes	Free-Form 64 Chars	The IP Address for this ElasticSearch Cluster Node (typically "localhost")
ClusterSettings. Nodes.Username	String	Mandatory	Yes	Free-Form 64 Chars	The Username credential for this ElasticSearch Cluster Node
ClusterSettings. Nodes.Password	String	Mandatory	Yes	Free-Form 64 Chars	The Password credential for this ElasticSearch Cluster Node

GET /Compute/Devices/{id}/ApplicationSettings/

GET /Compute/Devices/{id}/ApplicationSettings/ returns the Application Settings information.

Table 103: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/ApplicationSettings/
```

Example GET Resonse Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/ApplicationSettings/",
    "Name": "Resource Manager Data Center",
    "Version": "3.0.0",
    "ClusterSettings": {
        "Nodes": [
            {
                "IPAddress": "localhost",
                "Username": "elastic",
                "Password": "password"
            }
        ]
    }
}
```

Table 104: GET Response Codes

Response Code	Name	Definition
200	OK	OK & ApplicationSettings information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/ApplicationSettings/

PUT /Compute/Devices/{id}/ApplicationSettings/ modifies an attribute of the Application Settings.

Table 105: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/ApplicationSettings/
    --data '{
        "ClusterSettings": {
            "Nodes": [
                {
                    "IPAddress": "computer1",
                    "Username": "elastic",
                    "Password": "password"
                },
                {
                    "IPAddress": "computer2",
                    "Username": "elastic",
                    "Password": "password"
                },
                {
                    "IPAddress": "computer3",
                    "Username": "elastic",
                    "Password": "password"
                }
            ]
        }
    }'
```

Example PUT Response Body

```
{
    "Self": "http://10.20.30.40:80/Compute/Devices/0123456789/ApplicationSettings/",
    "Name": "Resource Manager Data Center",
    "Version": "3.0.0",
    "ClusterSettings": {
        "Nodes": [
            {
                "IPAddress": "computer1",
                "Username": "elastic",
                "Password": "password"
            },
            {
                "IPAddress": "computer2",
                "Username": "elastic",
                "Password": "password"
            },
            {
```

```
                "IPAddress": "computer3",
                "Username": "elastic",
                "Password": "password"
            }
        ]
    }
}
```

Table 106: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & ApplicationSettings information in the response body
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	Resource could not execute the request

HEAD /Compute/Devices/{id}/ApplicationSettings/

HEAD /Compute/Devices/{id}/ApplicationSettings/ pings the Application Settings.

Table 107: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 108: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/ApplicationSettings/

OPTIONS /Compute/Devices/{id}/ApplicationSettings/ returns the allowable HTTP Methods list for this resource.

Table 109: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 110: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.5 /Compute/Devices/{id}/Files/

The Collection-level methods and attributes of /Compute/Devices/{id}/Files/ return information about the Files collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 111: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Files/{id}/** ***(page 83)*** for Instance Attribute details

GET /Compute/Devices/{id}/Files/

GET /Compute/Devices/{id}/Files/ returns the Collection of Files.

Table 112: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Files/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/",
    "Members": [
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/auditlog/",
            "ID": "auditlog",
            "Description": "Audit log"
        },
        {
```

```
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/notices/",
            "ID": "notices",
            "Description": "Notices"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/eula/",
            "ID": "eula",
            "Description": "EULA"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/telemetry/",
            "ID": "telemetry",
            "Description": "Telemetry"
        }
    ],
    "TLSCertAndKey": {},
    "LDAPCert": {}
}
```


Table 113: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Files + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Files/

HEAD /Compute/Devices/{id}/Files/ pings this Files collection.

Table 114: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 115: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Files/

OPTIONS /Compute/Devices/{id}/Files/ returns the allowable HTTP Methods list for this resource.

Table 116: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 117: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.6 /Compute/Devices/{id}/Files/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Files/{id}/ return information about a specific File and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 118: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier
Name	String	Mandatory	No	Free-Form 256 Chars	The name of the file

GET /Compute/Devices/{id}/Files/{id}/

GET /Compute/Devices/{id}/Files/{id}/ returns information regarding a specific File based on the {id} in the URI address string.

Table 119: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command – Audit Log

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Files/auditlog
```

Example GET Response Body – Audit Log

```
index:000001|
addr:[10.88.94.81:52974]|
type:Response|
statuscode:204|
user:admin|
path:/Compute/Devices/000af7957e01/Jobs/GroupOperation15798/
```

Table 120: Available Files

File Type	Description
auditlog	Returns the current audit log entries as text to the requestor (content-type: text/plain)
notices	Returns the 3rd Party open-source software entries and their licensing used in this application
eula	Returns the End-User License Agreement documentation
telemetry	Returns a downloaded compressed bundle file called "telemetry_Compute_<id>_<timestamp>.tgz" to the requestor containing instantaneous GET of all the device info
cert_pem (204 No Content or 200 OK)	If the User uploaded a custom Cert/Key pair, returns the current SSL certificate entry as text to the requestor (content-type: text/plain)
key_pem (405 Method Not Allowed)	If the User uploaded a custom Cert/Key pair, SSL key GET not allowed for security of the key (403 Forbidden response)
ldapcert_pem	If the User uploaded an LDAP Client Certificate, returns the current LDAP certificate entry as text to the requestor (content-type: text/plain)

Table 121: GET Response Codes

Response Code	Name	Definition
200	OK	OK & File information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Files/{id}/

POST /Compute/Devices/{id}/Files/{id}/ updates the specific file.

Table 122: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
Content-Type: multipart/form-data	Mandatory	The FW image file is of content type "multipart/form-data"

Example POST curl Command – Adding an SSL Certificate & Key

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Files/TLSCertAndKey/
    -H "Content-Type: multipart/form-data"
    -F "cert.pem=@cert.pem"
    -F "key.pem=@key.pem"
```

Example POST Request Body – Adding an SSL Certificate & Key

```
Content-Disposition: form-data; cert.pem="{file_name}"
{file body}
Content-Disposition: form-data; key.pem="{file_name}"
{file body}
```

Note: Both Certificate and Key files MUST be sent together in the same POST request.

Note: Once the Certificate and Key files are uploaded, the API will restart for the certificate/key to take effect. See PUT **/Query/InformationStructure/** *(page 16)* {"RestartAPI" : true}.

Note: The Certificate and Key files cannot be validated, therefore the user must make sure the files are valid before uploading. If they are invalid, the result is the HTTPS service will not operate.

Example POST curl Command – Adding an LDAP Client SSL Certificate

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Files/LDAPCert/
    -H "Content-Type: multipart/form-data"
    -F "cert.pem=@ldapcert.pem"
```

Example POST Request Body – Adding an LDAP Client SSL Certificate

```
Content-Disposition: form-data; ldapcert.pem="{file_name}"
{file body}
```

Note: The Certificate file cannot be validated, therefore the user must make sure the files are valid before uploading. If they are invalid, the result is the LDAP service will not operate properly.

Table 123: POST Response Codes

Response Code	Name	Definition
201	Created	Created & File information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Files/{id}/

HEAD /Compute/Devices/{id}/Files/{id}/ pings a specific Compute File.

Table 124: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 125: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Files/{id}/

OPTIONS /Compute/Devices/{id}/Files/{id}/ returns the allowable HTTP Methods list for this resource.

Table 126: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 127: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
204	No Content	No Content & "Allow" list of methods in the response header. Some files will return 204 No Content indicating that there is no entity body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.7 /Compute/Devices/{id}/Location/

The methods and attributes of /Compute/Devices/{id}/Location/ return information about the resource Location and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 128: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Country	String	Optional	Yes	Free-Form 64 Chars	The Country name where this Device is located
Territory	String	Optional	Yes	Free-Form 64 Chars	The Territory name where this Device is located
State	String	Optional	Yes	Free-Form 64 Chars	The State name where this Device is located
City	String	Optional	Yes	Free-Form 64 Chars	The City name where this Device is located
Address1	String	Optional	Yes	Free-Form 64 Chars	The first Address information where this Device is located
Address2	String	Optional	Yes	Free-Form 64 Chars	The second Address information where this Device is located as needed
Address3	String	Optional	Yes	Free-Form 64 Chars	The third Address information where this Device is located as needed
PostalCode	String	Optional	Yes	Free-Form 32 Chars	The Postal or Zip Code name where this Device is located
SiteName	String	Optional	Yes	Free-Form 32 Chars	The friendly Site Name
Building	String	Optional	Yes	Free-Form 32 Chars	The Building name or number where this Device is located
Room	String	Optional	Yes	Free-Form 32 Chars	The Room name or number in the Building where this Device is located

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Pod	String	Optional	Yes	Free-Form 32 Chars	The Pod name or number in the Room this Device is located as needed
Row	String	Optional	Yes	Free-Form 32 Chars	The Row name or number in the Room or Pod this Device is located
Rack	String	Optional	Yes	Free-Form 32 Chars	The Rack name or number in the Row this Device is located
Shelf	String	Optional	Yes	Free-Form 32 Chars	The Shelf name or number of the Rack this Device is located
Device	String	Optional	Yes	Free-Form 32 Chars	The Device name or number of the Rack this Device is located
Item	String	Optional	Yes	Free-Form 32 Chars	The Item name indicating the Shelf units or dimension for this Device
GPSCoords	String	Optional	Yes	Decimal Latitude, Decimal Longitude Format	The GPS coordinates of the location of this Device
OtherLocationInfo	String	Optional	Yes	Free-Form 64 Chars	A place-holder for other location information

GET /Compute/Devices/{id}/Location/

GET /Compute/Devices/{id}/Location/ returns the Location information.

Table 129: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Location/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Location/",
    "Country": "USA",
    "Territory": "",
    "State": "Colorado",
    "City": "Colorado Springs",
    "Address1": "9950 Federal Drive",
    "Address2": "Suite 100",
    "Address3":"",
    "PostalCode": "80921",
    "SiteName": "WD COS DataCenter 1",
    "Building": "9950",
    "Room": "159",
    "Pod": "p2",
    "Rack": "r02",
    "Shelf" : "30",
    "Device":"5",
    "Item": "Rack Units",
    "GPSCoords": "38.9838684,-104.8040493",
    "OtherLocationInfo": "This is other info"
}
```

Table 130: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Location information + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Location/

PUT /Compute/Devices/{id}/Location/ modifies an attribute of the Location.

Table 131: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Location/
    --data '{
        "City" : "COS"
    }'
```

Example PUT Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Location/",
    "City": "COS",
}
```

Table 132: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Location information with no response body
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	Resource could not execute the request

DELETE /Compute/Devices/{id}/Location/

DELETE /Compute/Devices/{id}/Location/ deletes Location information.

Table 133: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Location/
```

Table 134: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Compute/Devices/{id}/Location/

HEAD /Compute/Devices/{id}/Location/ pings the Location resource.

Table 135: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 136: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Location/

OPTIONS /Compute/Devices/{id}/Location/ returns the allowable HTTP Methods list for this resource.

Table 137: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 138: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.8 /Compute/Devices/{id}/SystemClock/

The methods and attributes of /Compute/Devices/{id}/SystemClock/ return information about the System Clock and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 139: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Uptime. Duration	String	Mandatory	No		The current uptime value in ISO 8601 format
Uptime. Days	Number	Mandatory	No	0 - n	The current uptime Days since boot
Uptime. Hours	Number	Mandatory	No	0 - 23	The current uptime Hours since boot
Uptime. Minutes	Number	Mandatory	No	0 - 59	The current uptime Minutes since boot
Uptime. Seconds	Number	Mandatory	No	0 - 59	The current uptime Seconds since boot

GET /Compute/Devices/{id}/SystemClock/

GET /Compute/Devices/{id}/SystemClock/ returns the System Clock information.

Table 140: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/SystemClock/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/SystemClock/",
    "Date": "17-Aug-2022",
    "Time": "12:34:13",
    "TimeZone": "(UTC-6.00) MDT",
    "TimeZoneSetting": "MDT",
    "DST": "Enabled",
    "Uptime": {
        "Duration": "P82DT1H26M51S",
        "Days": 82,
        "Hours": 1,
        "Minutes": 26,
        "Seconds": 51
    }
}
```

Table 141: GET Response Codes

Response Code	Name	Definition
200	OK	OK & System Clock information + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned. **Note**: Due to the time changing every second, catching a 304 response will be very rare
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/SystemClock/

HEAD /Compute/Devices/{id}/SystemClock/ pings the System Clock resource.

Table 142: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 143: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/SystemClock/

OPTIONS /Compute/Devices/{id}/SystemClock/ returns the allowable HTTP Methods list for this resource.

Table 144: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 145: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.9 /Compute/Devices/{id}/Jobs/

The Collection-level methods and attributes of /Compute/Devices/{id}/Jobs/ return information about the Jobs collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 146: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Jobs/{id}/** ***(page 100)*** for Instance Attribute details

GET /Compute/Devices/{id}/Jobs/

GET /Compute/Devices/{id}/Jobs/ returns the Collection of Jobs.

Table 147: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Jobs/
```

Table 148: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Jobs + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Jobs/

HEAD /Compute/Devices/{id}/Jobs/ pings the Jobs collection.

Table 149: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 150: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Jobs/

OPTIONS /Compute/Devices/{id}/Jobs/ returns the allowable HTTP Methods list for this resource.

Table 151: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 152: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.10 /Compute/Devices/{id}/Jobs/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Jobs/{id}/ return information about a specific Job and enable management operations.

HTTP Methods: GET DELETE HEAD OPTIONS

Table 153: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	"FactoryReset"	The Name of this particular Job
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service" | "Inoperative" | "Write Protected" | "Reboot Needed" | "Activate Needed" | "Activate with reboot needed" | "Activate invalidated by reboot"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0-16 | 65537-65542	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
PercentComplete	Number	Mandatory	No	0-100	The percent complete status of the Job

GET /Compute/Devices/{id}/Jobs/{id}/

GET /Compute/Devices/{id}/Jobs/{id}/ returns information regarding a specific Job based on the {id} in the URI address string.

Table 154: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Jobs/FactoryReset_0123456789/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Jobs/
FactoryReset_0123456789/",
    "ID": "FactoryReset_0123456789",
    "StartTime": "02-Oct-2023 16:49:07",
    "ElapsedTime": "02-Oct-2023 16:49:22",
    "Owner": "http://10.20.30.40/Compute/Devices/0123456789/",
    "PercentComplete": 100,
    "Status": {
        "State": {
            "ID": 8,
            "Name": "Completed"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
```

```
    }
}
```

Table 155: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Job information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Jobs/{id}/

DELETE /Compute/Devices/{id}/Jobs/{id}/ deletes a specific Job.

Note: When deleting firmware-related jobs, the recommended order is to delete FirmwareUpdate before FirmwareActivate.

Note: Jobs cannot be deleted when in-progress (i.e. when PercentComplete is less than 100%).

Table 156: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Storage/Devices/0123456789/Jobs/
FactoryReset_0123456789
```

Table 157: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	If job is still in progress
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Jobs/{id}/

HEAD /Compute/Devices/{id}/Jobs/{id}/ pings this specific Job.

Table 158: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 159: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Jobs/{id}/

OPTIONS /Compute/Devices/{id}/Jobs/{id}/ returns the allowable HTTP Methods list for this resource.

Table 160: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 161: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.11 /Compute/Devices/{id}/Assets/

The Collection-level methods and attributes of /Compute/Devices/{id}/Assets/ return information about the Asset(s) collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 162: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Assets/{id}/** ***(page 108)*** for Instance Attribute details

GET /Compute/Devices/{id}/Assets/

GET /Compute/Devices/{id}/Assets/ returns the Collection of Assets.

Table 163: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Assets/
```

Table 164: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Assets + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Assets/

HEAD /Compute/Devices/{id}/Assets/ pings the Assets Collection.

Table 165: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 166: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Assets/

OPTIONS /Compute/Devices/{id}/Assets/ returns the allowable HTTP Methods list for this resource.

Table 167: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 168: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.12 /Compute/Devices/{id}/Assets/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Assets/{id}/ return information about a specific Asset and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 169: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier this particular Asset
Name	String	Mandatory	No	Free-Form 256 Chars	The name of the Asset. Must be unique to this Asset Collection
Description	String	Optional	No	Free-Form 256 Characters	The description of this Asset
Type. ID	Number	Mandatory	No		
Type. Name	String	Mandatory	No		
Accounts	Object Array	Optional	Yes	BasicAuthentication: {"UserPassword": "<password>"}, "UserID": "<username>"}	Provides a username and password pair to tell Resource Manager Data Center Edition how to connect to a remote/proxied Device (typically the Redfish-based Devices)
SubType. ID	Number	Mandatory	No		
SubType. Name	String	Mandatory	No		
AssetInformation	Object	Mandatory	No	Device GET Full information	Contains one or more discovered Devices' GET /Full information

GET /Compute/Devices/{id}/Assets/{id}/

GET /Compute/Devices/{id}/Assets/{id}/ returns information regarding a specific Asset based on the {id} in the URI address string.

Table 170: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Assets/5000ccab0410a180/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Assets/5000ccab0410a180/",
    "ID": "5000ccab0410a180",
    "Name": "5000ccab0410a180",
    "Accounts": [
        {
            "BasicAuthentication": {
                "UserPassword": " ******* "
            },
            "UserID": "admin"
        }
    ],
    "Description": "Full Device Asset Information",
    "Type": {
        "ID": 2,
        "Name": "Storage"
    },
    "SubType": {
        "ID": 1,
        "Name": "Enclosure"
    },
    "AssetInformation": {
        "Accounts": {...},
        "Adapters": {...},
        "Capabilities": {...},
        "ChassisPackageType": {...},
        "Connectors": {...},
        "Controllers": {...},
        "DurableName": "5000CCAB0410A180",
        "Expanders": {...},
        "Fans": {...},
        "Files": {...},
        "Format": {...},
        "Hostname": "oobm-00-0c-ca-08-39-3c",
        "ID": "5000ccab0410a180",
        "IPAddresses": {...},
        "IndicatorLED": {...},
        "InformationStructure": {...},
        "Jobs": {...},
        "Location": {...},
```

```
        "Manufacturer": "HGST",
        "Media": {...},
        "Model": "H4102-J",
        "Name": "5000ccab0410a180",
        "OperatingSystem": {...},
        "Ports": {...},
        "PowerState": {...},
        "Sanitize": {...},
        "Self": "http://10.20.30.40:80/Storage/Devices/5000ccab0410a180/?
IPAddress=10.20.30.40",
        "Sensors": {...},
        "SerialNumber": "THCLS00220EA0059",
        "SlotNumber": 2,
        "Status": {...},
        "SystemType": {...},
        "TotalCapacity": 150012470231040,
        "ZoneSets": {...},
    }
}
```


Table 171: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Asset information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Assets/{id}/

PUT /Compute/Devices/{id}/Assets/{id}/ modifies an attribute or executes an operation for this Asset.

Table 172: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Assets/123456
    --data '{
        "Accounts": [
            {
                "BasicAuthentication": {
                    "UserPassword": "password",
                    "UserID": "name"
                }
            }
        ]
    }'
```

Table 173: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Asset information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Assets/{id}/

HEAD /Compute/Devices/{id}/Assets/{id}/ pings this Asset.

Table 174: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 175: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body

Response Code	Name	Definition
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Assets/{id}/

OPTIONS /Compute/Devices/{id}/Assets/{id}/ returns the allowable HTTP Methods list for this resource.

Table 176: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 177: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.13 /Compute/Devices/{id}/Policies/

The Collection-level methods and attributes of /Compute/Devices/{id}/Policies/ return information about the Policies collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 178: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Policies/{id}/** ***(page 117)*** for Instance Attribute details

GET /Compute/Devices/{id}/Policies/

GET /Compute/Devices/{id}/Policies/ returns the Collection of Policies.

Table 179: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Policies/
```

Table 180: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Policies + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Policies/

POST /Compute/Devices/{id}/Policies/ creates a new Policy.

Table 181: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/Compute/Devices/01234567890/Policies/
    --data '{
        "Name": "Policy 1",
        "Description": "This is a Policy",
        "Enabled": true,
        "Rules": [
            {
                "Name": "Rule1",
                "Log": "Triggered Check Health not OK on Device ID 0123456789",
                "Description": "Check Health not OK on Device ID 0123456789",
                "Salience": 1,
                "Conditions": [
                    {
                        "Member": "Health_ID",
                        "Operator": "!=",
                        "ValueInt": 5,
                        "IsEnd": false
                    },
                    {
                        "Member": "ID",
                        "Operator": "==",
                        "ValueString": "0123456789",
                        "IsEnd": true
                    }
                ],
                "ActionMethods": [
                    {
                        "Name": "SendMail"
                    }
                ],
                "ConditionsOperator": [
                    "&&"
                ]
```

```
            }
        ]
    }'
```

Example POST Request Body

```
{
    "Name": "Policy3",
    "Description": "Policy to track inventory",
    "Enabled": true|false,
    "Rules": [
        {
            "Name": "CheckHealth",
            "Description": "Check the Health values",
            "Salience": 0 - 100,
            "Conditions": [
                {
                    "Member": "Health_ID",
                    "Operator": "!=",
                    "ValueInt": 5,
                    "ValueString": "",
                    "ValueBool": true|false,
                    "IsEnd": true|false
                }
            ],
            "ConditionsOperator": [
                "=="
            ],
            "ActionMethods": [
                {
                    "Name": "SendMail",
                }
            ],
            "Log": "CheckHealth is performed"
        }
    ]
}
```

Table 182: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Account information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Policies/

HEAD /Compute/Devices/{id}/Policies/ pings this Policies Collection.

Table 183: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 184: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Policies/

OPTIONS /Compute/Devices/{id}/Policies/ returns the allowable HTTP Methods list for this resource.

Table 185: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 186: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.14 /Compute/Devices/{id}/Policies/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Policies/{id}/ return information about a specific Policy and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 187: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The identifier for this particular Policy
Name	String	Mandatory	Yes	The name of the Policy. Must be unique to this Policy Collection	The user-defined Friendly Name for this Resource
Description	String	Optional	Yes	Free-Form 256 Characters	The description of this Policy
CreateDate	String	Mandatory	No	datetime	Displays when this Policy was created
LastModified	String	Mandatory	No	datetime	Displays when this Policy was last modified
Type. ID	Number	Mandatory	No	0 | 1	Type of Policy: User-Defined = 0 System-Defined=1 Default = 0. Only "User-Defined" will be returned when Users create new Policies.
Type. Name	String	Mandatory	No	"User-Defined" | "System-Defined"	Type of Policy: User-Defined = 0 (default). Only "User-Defined" will be returned when Users create new Policies. System-Defined=1
Enabled	Boolean	Mandatory	Yes	true | false	Indicates or Sets the Policy as enabled or disabled

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Rules[]. Name	String	Mandatory	Yes	Free-Form 256 Characters	The name of this Rule. #Must be unique to this collection of Rules
Rules[]. Description	String	Mandatory	Yes	0 | 1 | 2	ConditionListType determines how to apply conditionals. Default = 0
Rules[]. Salience	Number	Mandatory	Yes	0 - 100	Salience determines how focused this rule is to be considered
Rules[]. Conditions[]. Member	String	Mandatory	Yes	Free-Form 256 Characters	Condition Member name or alias
Rules[]. Conditions[]. Operator	String	Mandatory	Yes	"==" | "!=" | ">=" | "<=" | ">" | "<"	Operator type for the Condition of this Rule
Rules[]. Conditions[]. ValueInt	Number	Optional	Yes	-n to n	Indicates an "integer" value to test against
Rules[]. Conditions[]. ValueString	String	Optional	Yes	Free-Form 256 Characters	Indicates a "string" value to test against
Rules[]. Conditions[]. ValueBool	Boolean	Optional	Yes	true | false	Indicates a "boolean" value to test against
Rules[]. Conditions[]. IsEnd	Boolean	Mandatory	Yes	true | false	Indicates if there are more Conditions after this one
Rules[]. ConditionsOperator	String	Mandatory	Yes	"&&" | "||"	Indicates the overall type of operator for this Rule between each Rule Conditions Member entry
Rules[]. ActionsMethods[]. Name	String Array	Mandatory	Yes	"SendMail" | "SendTrap"	Name of the Action; can be one or the other or both as separate "Name" entries
Rules[]. Log	String	Optional	Yes	Free-Form 256 Characters	Log information for this Rule

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Notifications	String	Mandatory	No	Free-Form URI pattern	Link to any Notifications associations. This link is automatically generated

GET /Compute/Devices/{id}/Policies/{id}/

GET /Compute/Devices/{id}/Policies/{id}/ returns information regarding a specific Policy based on the {id} in the URI address string.

Table 188: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/
Policies/9285b6f7-42b6-2a53-9c2e-37767e1b7300
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Policies/9285b6f7-42b6-2a53-9c2e-37767e1b7300/",
    "ID": "9285b6f7-42b6-2a53-9c2e-37767e1b7300",
    "Name": "Policy3",
    "Description": "Policy to track inventory",
    "CreateDate": "12-Feb-2023 22:23:33",
    "LastModified": "17-Feb-2023 06:43:56",
    "Enabled": true,
    "Priority": 0,
    "TimePeriod": "202302120T222333/THISANDFUTURE",
    "Type": {
        "ID": 0,
        "Name": "User-Defined"
    },
    "DecisionStrategy": {
        "ID": 0,
        "Name": "Unknown"
    },
    "Rules": [
        {
            "Name": "CheckHealth",
            "Description": "Check the Health values",
```

```
            "Salience": 10,
            "Conditions": [
                {
                    "Member": "Health_ID",
                    "Operator": "!=",
                    "ValueInt": 5,
                    "IsEnd": false
                },
                {
                    "Member": "Health_Name",
                    "Operator": "!=",
                    "ValueString": "OK",
                    "IsEnd": true
                }
            ],
            "ConditionsOperator": [
                "&&"
            ],
            "ActionMethods": [
                {
                    "Name": "SendMail"
                }
            ],
            "Log": "CheckHealth is performed"
        },
        {
            "Name": "CheckMedia",
            "Description": "Description dummy rule2",
            "Salience": 99,
            "Conditions": [
                {
                    "Member": "PowerState_Name",
                    "Operator": "==",
                    "ValueString": "Off",
                    "IsEnd": false
                },
                {
                    "Member": "IndicatorLED_Name",
                    "Operator": "==",
                    "ValueString": "On",
                    "IsEnd": true
                }
            ],
            "ConditionsOperator": [
                "||"
            ],
            "ActionMethods": [
                {
                    "Name": "SendTrap"
                }
            ],
            "Log": "CheckMedia is performed"
        }
    ],
    "Notifications": "http://10.20.30.40/Compute/Devices/0123456789/Notifications/?
Policies=9285b6f7-42b6-2a53-9c2e-37767e1b7300"
}
```

Table 189: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Policy information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Policies/{id}/

PUT /Compute/Devices/{id}/Policies/{id}/ modifies an attribute or executes an operation for this Policy.

Table 190: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Policies/123456
    --data '{
        "Name": "Policy 123",
        "Description": "Add new rule to this is a Policy",
        "Enabled": true,
        "Rules": [
            {
                "Name": "Rule1",
                "Description": "Check Health not OK on Device ID 0123456789",
                "Salience": 1,
                "Conditions": [
                    {
                        "Member": "Health_ID",
                        "Operator": "!=",
                        "ValueInt": 5,
```

```
                        "IsEnd": false
                    },
                    {
                        "Member": "ID",
                        "Operator": "==",
                        "ValueString": "0123456789",
                        "IsEnd": true
                    }
                ],
                "ConditionsOperator": [
                    "&&"
                ],
                "ActionMethods": [
                    {
                        "Name": "SendMail"
                    }
                ],
                "Log": "Triggered Check Health not OK on Device ID 0123456789"
            },
            {
                "Name": "Rule2",
                "Description": "Check Sensor Temperature",
                "Salience": 1,
                "Conditions": [
                    {
                        "Member": "Type",
                        "Operator": "==",
                        "ValueString": "Sensor",
                        "IsEnd": false
                    },
                    {
                        "Member": "SensorType_ID",
                        "Operator": "==",
                        "ValueInt": 2,
                        "IsEnd": false
                    },
                    {
                        "Member": "CurrentReading",
                        "Operator": ">",
                        "ValueInt": 60,
                        "IsEnd": true
                    }
                ],
                "ActionMethods": [
                    {
                        "Name": "SendTrap"
                    }
                ],
                "Log": "Triggered Check Sensor Temperature above 60 C",
                "ConditionsOperator": [
                    "&&",
                    "&&"
                ]
            }
        ]
    }'
```

Example PUT Request Body

```
{
    "Name": "Policy_newname",
    "Description": "Policy to track inventory",
    "Enabled": true|false,
    "Rules": [
        {
            "Name": "CheckHealth",
            "Description": "Check the Health values",
            "Salience": 0 - 100,
            "Conditions": [
                {
                    "Member": "Health_ID",
                    "Operator": "!=",
                    "ValueInt": 5,
                    "IsEnd": true|false
                }
            ],
            "ConditionsOperator": [
                "=="
            ],
            "ActionMethods": [
                {
                    "Name": "SendMail"
                }
            ],
            "Log": "CheckHealth is performed"
        }
    ]
}
```

Table 191: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Policy information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Policies/{id}/

DELETE /Compute/Devices/{id}/Policies/{id}/ deletes a specific Policy.

Table 192: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Policies/123456
```

Table 193: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	If job is still in progress
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Policies/{id}/

HEAD /Compute/Devices/{id}/Policies/{id}/ pings this Policy.

Table 194: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 195: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Policies/{id}/

OPTIONS /Compute/Devices/{id}/Policies/{id}/ returns the allowable HTTP Methods list for this resource.

Table 196: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 197: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.15 /Compute/Devices/{id}/Notifications/

The Collection-level methods and attributes of /Compute/Devices/{id}/Notifications/ return information about the Notifications collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 198: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Notifications/{id}/** ***(page 131)*** for Instance Attribute details

GET /Compute/Devices/{id}/Notifications/

GET /Compute/Devices/{id}/Notifications/ returns the Collection of Notifications.

Table 199: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Notifications/
```

Table 200: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Notifications + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Notifications/

POST /Compute/Devices/{id}/Notifications/ creates a new Notification.

Table 201: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST "http://10.20.30.40/Compute/Devices/0123456789/Notifications/"
    --form 'SmtpSettings="{
        "Name": "Notification1",
        "Type": {
            "ID": 1,
            "Name": "SMTPAlert"
        },
        "Description": "This is an email notification",
        "SMTPSettings": {
            "Server": "mail-server.com",
            "Subject": "This is a Notification",
            "Port": 25,
            "Username": "admin",
            "Password": "password",
            "From": "This Device"
        },
        "Distributions": [
            {
                "Name": "Admin",
                "Recipients": [
                    "admin@test.com"
                ]
            }
        ]
    }"'
    --form 'Certificate=@"smtpscertificate.pem"'
```

Example POST Request Body – SMTPAlert

```
"SmtpSettings"="{
    "Name" : "notification_name",
    "Description" : "description",
    "Type" : {
        "ID" : 1
```

```
    },
    "SMTPSettings" : {
        "Server" : "email_server",
        "Subject" : "email_subjectline",
        "Port" : 25,
        "Username" : "user",
        "Password" : "password",
        "From" : "From this Device"
    }
    "Distributions" : [
        {
            "Name" : "Admin" | "Operator",
            "Recipients" : [
                {
                    "recip1",
                    "recip2",
                    ...
                    "recipN"
                }
            ]
        }
    ],
}"
Certificate=@"Certificate.pem"
```

Example POST Request Body – SNMP Trap

```
{
    "Name" : "notification_name"
    "Description" : "description",
    "Type" : {
        "ID": 2
    },
    "SNMPSettings" : {
        "TargetIP" : "10.20.30.40",
        "TargetPort" : 162,
        "Version" : "Version1 | Version2c | Version3",
        "Community" : "public"(,)
        (Selecting Version3 additional below)
        "UserName" : "user",
        "EngineID" : 1234,
        "AuthenticationProtocol" : "SHA",
        "AuthenticationPassphrase" : "password",
        "PrivacyProtocol" : "DES",
        "PrivacyPassphrase" : "password"
    }
}
```

Table 202: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Notification information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Notifications/

HEAD /Compute/Devices/{id}/Notifications/ pings the Notifications collection.

Table 203: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 204: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Notifications/

OPTIONS /Compute/Devices/{id}/Notifications/ returns the allowable HTTP Methods list for this resource.

Table 205: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 206: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.16 /Compute/Devices/{id}/Notifications/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Notifications/{id}/ return information about a specific Notification and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 207: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier for this particular Notification
Name	String	Mandatory	Yes	Free-Form 256 Chars	The name of the Notification. Must be unique to this Notifications Collection
Description	String	Optional	Yes	Free-Form 256 Characters	The description of this Notification
CreateDate	String	Mandatory	No	datetime	Displays when the Notification was created
LastModified	String	Mandatory	No	datetime	Displays when the Notification was last modified
Type. ID	Number	Mandatory	Yes	1 | 2	Type of Notification: SMTPAlert = 1 SNMPTrap = 2
Type. Name	String	Mandatory	No	"SMTPAlert" | "SNMPTrap"	Type of Notification SMTPAlert = 1 SNMPTrap = 2

Table 208: SMTP Alert Notification Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
SMTPSettings. Server	String	Mandatory	Yes	Free-Form 256 Characters of an IP Address or Hostname	Indicates the current Email Target Server IP Address or Hostname location. Use if Type is SMTPAlert

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
SMTPSettings. Subject	String	Optional	Yes	Free-Form 256 Characters	The "subject line" for any outgoing email alerts or reports
SMTPSettings. Port	Number	Optional	Yes	Integer value; Default = 25	This is the target email server SMTP Port Number where 25 is the IANA default
SMTPSettings. Username	String	Mandatory	Yes	Username string value	This is the Username entry for the target email server
SMTPSettings. Password	String	Mandatory	Yes	Password format (displays "*******")	This is the Password entry for the target email server
SMTPSettings. From	String	Optional	Yes	Free-Form 64 Characters	This is the "From" entry for the email to be sent
Distributions[]. Name	String Array	Mandatory	Yes	"Admin" | "Operator"	The name of this Distribution List. #There are two choices, either "Admin" or "Operator" distribution list names
Distributions[]. Recipients[]	String Array	Mandatory	Yes	Email Format entries	The Recipients array for this "Name". Format is either IP Addresses, Email Address, or Hostnames depending on the Type

Table 209: SNMP Alert Notification Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
SNMPSettings. TargetIP	String	Mandatory	Yes	IPv4 Formatted entry or Hostname formatted entry	This is the target or destination server IP Address or Hostname to send the trap
SNMPSettings. TargetPort	Number	Mandatory	Yes	Integer Port number; Default is IANA Port 162	This is the target or destination server Port Number to send the trap

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
SNMPSettings. Version	String	Mandatory	Yes	Version1, Version2c, Version3	This is the SNMP Version to send the trap. Version 3 requires more attribute entries
SNMPSettings. Community	String	Mandatory	Yes	"public" (default)	This is the "Community" for the SNMP Trap information

Table 210: Additional SNMP Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
SNMPSettings. UserName	String	Mandatory	Yes	Username value	Username for the secure SNMP v3 entry
SNMPSettings. EngineID	Number	Mandatory	Yes	Integer number	Engine identifier located at the target or destination SNMP Listener server
SNMPSettings. AuthenticationProtocol	String	Mandatory	Yes	"SHA" (default)	Authentication protocol type for the SNMP v3 transfer
SNMPSettings. AuthenticationPassphrase	String	Mandatory	Yes	Password format (displays "*******")	Authentication passphrase for the SNMP v3 transfer
SNMPSettings. PrivacyProtocol	String	Mandatory	Yes	"DES" (default)	Privacy protocol type for the SNMP v3 transfer
SNMPSettings. PrivacyPassphrase	String	Mandatory	Yes	Password format (displays "*******")	Privacy passphrase for the SNMP v3 transfer

GET /Compute/Devices/{id}/Notifications/{id}/

GET /Compute/Devices/{id}/Notifications/{id}/ returns information regarding a specific Notification based on the {id} in the URI address string.

Table 211: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Notifications/6202d750
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Notifications/6202d750/",
    "ID": "6202d750-fefc-dff5-68ec-d8cc7fc9fb79",
    "Name": "Email Alert 1",
    "Description": "Test",
    "CreateDate": "07-Sep-2023 12:49:08",
    "LastModified": "07-Sep-2023 19:09:56",
    "Type": {
        "ID": 1,
        "Name": "SMTPAlert"
    },
    "SMTPSettings": {
        "Server": "mail@server.com",
        "Subject": "This is an Email Alert",
        "Port": 25,
        "Username": "admin",
        "Password": "*******",
        "From": "Device 12345"
    },
    "SNMPSettings": {},
    "Distributions": [
        {
            "Name": "Admin",
            "Recipients": [
                "bill@wdc.com"
            ]
        },
        {
            "Name": "Operator",
            "Recipients": [
                "ted@wd.com"
            ]
        }
    ]
}
```


Table 212: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Notification information in the response body + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Notifications/{id}/

PUT /Compute/Devices/{id}/Notifications/{id}/ modifies an attribute or executes an operation for this Notification.

Table 213: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -X PUT "http://10.20.30.40/Compute/Devices/0123456789/Notifications/123456"
    -H "If-Match: 87f24742e6308fa877b8453281f7bca5"
    --form 'SmtpSettings="{
        "Name": "Notification1",
        "Type": {
            "ID": 1,
            "Name": "SMTPAlert"
        },
        "Description": "This is an email notification",
        "SMTPSettings": {
            "Server": "mail-server.com",
            "Subject": "This is a Notification",
            "Port": 25,
            "Username": "admin",
            "Password": "password",
            "From": "This Device"
        },
        "Distributions": [
            {
                "Name": "Admin",
                "Recipients": [
                    "admin@test.com"
                ]
            }
        ]
    }"'
    --form 'Certificate=@"smtpscertificate.pem"'
```

Example PUT Request Body – SMTPAlert

```
'SmtpSettings"="{
    "Name": "notification_name"
    "Description": "description",
    "SMTPSettings": {
        "Server": "email_server",
        "Subject": "email_subjectline",
        "Port": 25,
        "Username": "user",
        "Password": "password",
        "From": "From this Device"
    }
    "Distributions" : [
        {
            "Name": "Admin" | "Operator",
            "Recipients": [
                {
                    "recip1", "recip2", ... "recipN"
                }
            ]
        }
```

```
    ],
}'
Certificate=@"Certificate.pem"
```

Example PUT Request Body – SNMPTrap

```
{
    "Name": "notification_name"
    "Description": "description",
    "SNMPSettings": {
        "TargetIP": "10.20.30.40",
        "TargetPort": 162,
        "Version": "Version1 | Version2c | Version3",
        "Community": "public",
        "UserName": "user",
        "EngineID": 1234,
        "AuthenticationProtocol": "SHA",
        "AuthenticationPassphrase": "password",
        "PrivacyProtocol": "DES",
        "PrivacyPassphrase": "password"
    }
}
```

Table 214: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Notification information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Notifications/{id}/

DELETE /Compute/Devices/{id}/Notifications/{id}/ deletes a specific Notification.

Table 215: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Notifications/123456
```

Table 216: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one notification, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Notifications/{id}/

HEAD /Compute/Devices/{id}/Notifications/{id}/ pings this Notification.

Table 217: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 218: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Notifications/{id}/

OPTIONS /Compute/Devices/{id}/Notifications/{id}/ returns the allowable HTTP Methods list for this resource.

Table 219: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 220: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.17 /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/

The Device-level methods of /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/ return information about the SMTPS certificate associated with a specific Notification.

HTTP Methods: GET DELETE HEAD OPTIONS

GET /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/

GET /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/ returns the certificate for this Notification ID.

Table 221: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 222: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Notification information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/

DELETE /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/ deletes the Certificate for this Notification ID.

Table 223: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Table 224: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one notification, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/

HEAD /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/ pings this Notification.

Table 225: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 226: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/

OPTIONS /Compute/Devices/{id}/Notifications/{id}/SMTPSCert/ returns the allowable HTTP Methods list for this resource.

Table 227: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text

Header	Mandatory or Optional	Description
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 228: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.18 /Compute/Devices/{id}/Analytics/

The Collection-level methods and attributes of /Compute/Devices/{id}/Analytics/ return information about the Analytics collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 229: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Analytics/{id}/** *(page 147)* for Instance Attribute details

GET /Compute/Devices/{id}/Analytics/

GET /Compute/Devices/{id}/Analytics/ returns the collection of Analytics.

Note: This will not return the "AnalyticInformation" but only the collection of Analytics and their attributes, unless the query parameter is sent. The query parameter `?Analytics=true` will add the "AnalyticInformation" objects to all the Analytics Instances in the Collection.

Table 230: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Analytics/
```

Table 231: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Analytics + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Analytics/

POST /Compute/Devices/{id}/Analytics/ creates a new Analytic.

Table 232: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Analytics/
    --data '{
        "Name": "Enclosure Controller B Health Status",
        "Description": "Show Enclosure Controller B Health status of the Device",
        "AnalyticAttributes": {
            "TargetEnclosureID": "04105300",
            "TargetComponent": "Controllers",
            "TargetComponentID": "IOModuleBFRU",
            "TargetAttribute": "Health"
        }
    }'
```

Example POST Request Body

```
{
    "Name": "name",
    "Description": "description",
    "AnalyticAttributes": {
        "TargetEnclosureID": "5000ccab04105300",
        "TargetComponent": "Controllers",
        "TargetComponentID": "IOModuleBFRU",
        "TargetAttribute": "Health",
        "StartTimestamp": "2023-11-20T12:00:00Z",
        "EndTimestamp": "2023-11-29T12:00:00Z"
    }
}
```

Table 233: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Analytic information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Analytics/

HEAD /Compute/Devices/{id}/Analytics/ pings the Analytics collection.

Table 234: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 235: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Analytics/

OPTIONS /Compute/Devices/{id}/Analytics/ returns the allowable HTTP Methods list for this resource.

Table 236: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 237: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.19 /Compute/Devices/{id}/Analytics/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Analytics/{id}/ return information about a specific Analytic and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 238: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier for this particular Analytic
Name	String	Mandatory	Yes	Free-Form 64 Chars	The name of the Analytic. Must be unique to this Analytics Collection
Description	String	Optional	Yes	Free-Form 256 Characters	The description of this Analytic
CreateDate	String	Mandatory	No	datetime	Displays when the Analytic was created
LastModified	String	Mandatory	No	datetime	Displays when the Analytic was last modified
Type. ID	Number	Mandatory	Yes	1 - 4	Type of Analytic (see Type.Name): 1=Descriptive 2=Diagnostic 3=Predictive (future) 4=Prescriptive (future)
Type. Name	String	Mandatory	No	"Descriptive Data Analytics" | "Diagnostic Data Analytics" | "Predictive Data Analytics" | "Prescriptive Data Analytics"	Descriptive=Trending Diagnostic=Anomaly detection Predictive=Correlation of anomalies and trending to predict future issues Prescriptive=Provides solutions based on predictions

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
AnalyticAttributes. TargetEnclosureID	String	Mandatory	Yes	UUID or other string format	Apply Analytic to this Device based on its ID
AnalyticAttributes. TargetEnclosureModel	String	Optional	Yes	"All" | "H4102-J" | "UD102" | "H4060-J" | "UD60" | "OpenFlex Data24" | "OpenFlex Data24 3xxx" | "OpenFlex Data24 4xxx"	Enclosure models, needed only in case of Diagnostic data Analytics when TargetEnclosureID is absent
AnalyticAttributes. TargetComponent	String	Mandatory	Yes	"Enclosure" | "Connectors" | "Controllers" | "Expanders" | "Fans" | "Media" | "Performance" | "Ports" | "PowerSupplies" | "Processors" | "Sensors" | "Slots" | "Volumes"	Apply Analytic to this Device's supported Component Type **NOTE**: Not all Target Components have all the Target Attributes but will return the appropriate Attributes within the AnalyticInformation structure for a given Component. This is an exclusive list of Target Components, but all are not supported for Diagnostic Data Analytics.
AnalyticAttributes. TargetComponentID	String	Mandatory	Yes	UUID or other string format	Apply Analytic to this Device's Component ID
AnalyticAttributes. TargetAttribute	String	Mandatory	Yes	"Health" | "TotalCapacity" | "RemainingCapacity" | "CurrentReading" | "CurrentSpeed" | "DiskStats" | "IOStats" | "LoadPercentage" | "Memory" | "PowerState" | "Speed"	Supported Attribute type to focus Analytic data. **NOTE**: Not all Target Attributes are member properties of all Target Components, but will return the appropriate Attributes within the AnalyticInformation[] structure for a given Component. This is an exclusive list of Target Attributes, but all are not supported for Diagnostic Data Analytics.

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
AnalyticAttributes. StartTimestamp	String	Optional	No	datetime	Starting time for the range of data for this Analytic
AnalyticAttributes. EndTimestamp	String	Optional	No	datetime	End time for the range of data for this Analytic from the Persistent Storage
Associations	Strings Array	Mandatory	No	Analytics ID | Reports ID(s)	Shows association between the Analytic and any Report(s) that it might be used in. **NOTE:** The case of a second Analytic ID in the Associations indicates an associative relationship to a second Analytic that is the exact set of Attributes but is of the other Type, for ease of correlation between a Descriptive and Diagnostic Analytic.
AnalyticInformation[]	String Array	Mandatory (Instance) | Optional (Collection)	No	ID | Name | Status (Health and State) | Timestamp | Speed (b/s) | Capacity (Bytes) | CurrentReading (Temp, Volts, Current, RPM) | CurrentSpeed (b/s)	Analytic Data array containing Target attribute information across the time range
Reports. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Reports Resource(s) regarding this Analytic

GET /Compute/Devices/{id}/Analytics/{id}/

GET /Compute/Devices/{id}/Analytics/{id}/ returns information regarding a specific Analytic based on the {id} in the URI address string.

Note: This will trigger returning the actual "AnalyticInformation" data from the Persistent Storage based on the AnalyticAttributes.

Table 239: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Analytics/72badd62eaca
```

Example GET Response Body of Descriptive Data Analytics


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Analytics/72badd62eaca/",
    "ID": "72badd62eaca",
    "Name": "Analytics3",
    "Description": "Server1: Processors: CPU1: Load Percentage",
    "CreateDate": "08-May-2024 02:28:25 UTC",
    "LastModified": "08-May-2024 02:28:25 UTC",
    "Type": {
        "ID": 1,
        "Name": "Descriptive Data Analytics"
    },
    "Associations": {
        "Analytics": [
            "72badd62eaca"
        ],
        "Reports": [
            "a3e340b25534",
            "89de5ac92d93"
        ]
    },
    "Reports": "http://10.20.30.40/Compute/Devices/0123456789/Reports/?
Analytics=72badd62eaca",
    "AnalyticAttributes": {
        "TargetEnclosureID": "1ace035696544fe38b1149ccd18d3ca1",
        "TargetComponent": "Processors",
        "TargetComponentID": "CPU1",
        "TargetAttribute": "LoadPercentage",
        "StartTimestamp": "2024-04-07T21:00:11.418469852-06:00",
        "EndTimestamp": "2024-05-07T21:00:11.418469852-06:00"
    },
    "AnalyticInformation": [
        {
            "Architecture": "x86_64",
            "Cores": 10,
            "ID": "CPU1",
            "LoadPercentage": 0.1,
            "LogicalProcessors": 20,
            "Name": "CPU1",
            "ProcessorID": "BFEBFBFF000306F2",
            "Timestamp": "2024-04-11T14:04:08.156343903Z"
```

```
        },
        {
            "Architecture": "x86_64",
            "Cores": 10,
            "ID": "CPU1",
            "LoadPercentage": 0.03,
            "LogicalProcessors": 20,
            "Name": "CPU1",
            "ProcessorID": "BFEBFBFF000306F2",
            "Timestamp": "2024-04-11T14:19:01.700714765Z"
        },
        {
            "Architecture": "x86_64",
            "Cores": 10,
            "ID": "CPU1",
            "LoadPercentage": 0.1,
            "LogicalProcessors": 20,
            "Name": "CPU1",
            "ProcessorID": "BFEBFBFF000306F2",
            "Timestamp": "2024-04-11T14:30:38.5508993Z"
        }
    ]
}
```


Example GET Response Body of Diagnostic Data Analytics


```
{
  "Self": "http://10.20.30.40:80/Compute/Devices/ac4cb9a3c4014f55b2df2a81fd8cbd76/
Analytics/fb938d06-774b-57e5-8034-01ff1791387f/",
  "ID": "fb938d06-774b-57e5-8034-01ff1791387f",
  "Name": "Single enclosure 5000ccab040e5c80",
  "Description": "Enclosure 5000ccab040e5c80",
  "CreateDate": "22-Apr-2025 07:44:07 UTC",
  "LastModified": "22-Apr-2025 07:44:07 UTC",
  "Type": {
    "ID": 2,
    "Name": "Diagnostic Data Analytics"
  },
  "Associations": {
    "Analytics": [
      "fb938d06-774b-57e5-8034-01ff1791387f"
    ]
  },
  "Reports": "http://10.20.30.40:80/Compute/Devices/ac4cb9a3c4014f55b2df2a81fd8cbd76/
Reports/?Analytics=fb938d06-774b-57e5-8034-01ff1791387f",
  "AnomalyDetected": true,
  "AnalyticAttributes": {
    "TargetEnclosureID": "5000ccab040e5c80",
    "TargetComponent": "Enclosure",
    "StartTimestamp": "2025-03-24T12:03:27.55181399Z",
    "EndTimestamp": "2025-04-23T12:03:27.55181399Z"
  },
  "Files": {
    "Self": "http://10.20.30.40:80/Compute/Devices/ac4cb9a3c4014f55b2df2a81fd8cbd76/
Analytics/fb938d06-774b-57e5-8034-01ff1791387f/Files/"
  },
  "AnalyticInformation": [
    {
      "AnomalyDetails": {
```

```
        "AnomalyScore": 0.5,
        "MaxNormal": 0.49
      },
      "ID": "5000ccab040e5c80",
      "Media": {
        "Members": [
          {
            "ID": "1",
            "Name": "SLOT 00"
          },
     ...
          {
            "ID": "60",
            "Name": "SLOT 59"
          }
        ]
      },
      "Model": "H4060-J",
      "Timestamp": "2025-03-25T12:46:17.729482019Z",
      "TotalCapacity": 92006789414912
    },
    {
    ...
    }
  ]
}
```


Table 240: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Analytic information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Analytics/{id}/

PUT /Compute/Devices/{id}/Analytics/{id}/ modifies an attribute or executes an operation for this Analytic.

Table 241: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Analytics/
    --data '{
        "Name":"DEVICE 5000ccab04105300 Controller B Health Status",
        "Description":"Show DEVICE 04105300 Controller B Health status of the Device"
    }'
```

Example PUT Request Body

```
{
    "Name": "new_name",
    "Description": "new_description",
    "AnalyticAttributes": {
        "TargetEnclosureID": "5000ccab04105301",
        "TargetComponent": "PowerSupplies",
        "TargetComponentID": "PS A",
        "TargetAttribute": "Health"
    }
}
```

Table 242: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Analytic information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Analytics/{id}/

DELETE /Compute/Devices/{id}/Analytics/{id}/ deletes a specific Analytic.

Table 243: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Analytics/123456
```

Table 244: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one notification, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Analytics/{id}/

HEAD /Compute/Devices/{id}/Analytics/{id}/ pings a specific Analytic.

Table 245: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 246: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body

Response Code	Name	Definition
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Analytics/{id}/

OPTIONS /Compute/Devices/{id}/Analytics/{id}/ returns the allowable HTTP Methods list for this resource.

Table 247: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 248: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.20 /Compute/Devices/{id}/Analytics/{id}/Files/

The Collection-level methods and attributes of /Compute/Devices/{id}/Analytics/{id}/Files/ return information about the Analytic files collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 249: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
AnalyticFiles	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of URLs for each Analytic File for an Analytic

GET /Compute/Devices/{id}/Analytics/{id}/Files/

GET /Compute/Devices/{id}/Analytics/{id}/Files/ returns the collection or the list of files for the Analytic.

Note: This will not return the contents of each file.

Table 250: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40:80/Compute/Devices/0123456789/Analytics/8d8ea16353c8/
Files/
```

Table 251: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of files + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Analytics/{id}/Files/

POST /Compute/Devices/{id}/Analytics/{id}/Files/ creates a new Analytic file.

Note: Data analysis for generating the file happens asynchronously.

Table 252: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40:80/Compute/Devices/0123456789/Analytics/8d8ea16353c8/
Files/
```

Note: POST commands to this resource do not require a data parameter. If accepted, analytic data collection will occur in the background.

Table 253: POST Response Codes

Response Code	Name	Definition
202	Accepted	New URI in Location Header to get the File
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Analytics/{id}/Files/

HEAD /Compute/Devices/{id}/Analytics/{id}/Files/ pings the Analytic files collection.

Table 254: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 255: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Analytics/{id}/Files/

OPTIONS /Compute/Devices/{id}/Analytics/{id}/Files/ returns the allowable HTTP Methods list for this resource.

Table 256: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 257: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.21 /Compute/Devices/{id}/Analytics/{id}/Files/{id}

The Device-level methods and attributes of /Compute/Devices/{id}/Analytics/{id}/Files/{id} return information about a specific Analytic file and enable management operations.

HTTP Methods: GET DELETE HEAD OPTIONS

Table 258: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier for this particular Analytic
Name	String	Mandatory	Yes	Free-Form 64 Chars	The name of the Analytic. Must be unique to this Analytics Collection
Description	String	Optional	Yes	Free-Form 256 Characters	The description of this Analytic
CreateDate	String	Mandatory	No	datetime	Displays when the Analytic was created
LastModified	String	Mandatory	No	datetime	Displays when the Analytic was last modified
Type. ID	Number	Mandatory	Yes	1 - 4	Type of Analytic (see Type.Name): 1=Descriptive 2=Diagnostic 3=Predictive (future) 4=Prescriptive (future)
Type. Name	String	Mandatory	No	"Descriptive Data Analytics" | "Diagnostic Data Analytics" | "Predictive Data Analytics" | "Prescriptive Data Analytics"	Descriptive=Trending Diagnostic=Anomaly detection Predictive=Correlation of anomalies and trending to predict future issues Prescriptive=Provides solutions based on predictions

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
AnalyticAttributes. TargetEnclosureID	String	Mandatory	Yes	UUID or other string format	Apply Analytic to this Device based on its ID
AnalyticAttributes. TargetEnclosureModel	String	Optional	Yes	"All" | "H4102-J" | "UD102" | "H4060-J" | "UD60" | "OpenFlex Data24" | "OpenFlex Data24 3xxx" | "OpenFlex Data24 4xxx"	Enclosure models, needed only in case of Diagnostic data Analytics when TargetEnclosureID is absent
AnalyticAttributes. TargetComponent	String	Mandatory	Yes	"Enclosure" | "Connectors" | "Controllers" | "Expanders" | "Fans" | "Media" | "Performance" | "Ports" | "PowerSupplies" | "Processors" | "Sensors" | "Slots" | "Volumes"	Apply Analytic to this Device's supported Component Type **NOTE**: Not all Target Components have all the Target Attributes but will return the appropriate Attributes within the AnalyticInformation structure for a given Component. This is an exclusive list of Target Components, but all are not supported for Diagnostic Data Analytics.
AnalyticAttributes. TargetComponentID	String	Mandatory	Yes	UUID or other string format	Apply Analytic to this Device's Component ID
AnalyticAttributes. TargetAttribute	String	Mandatory	Yes	"Health" | "TotalCapacity" | "RemainingCapacity" | "CurrentReading" | "CurrentSpeed" | "DiskStats" | "IOStats" | "LoadPercentage" | "Memory" | "PowerState" | "Speed"	Supported Attribute type to focus Analytic data. **NOTE**: Not all Target Attributes are member properties of all Target Components, but will return the appropriate Attributes within the AnalyticInformation[] structure for a given Component. This is an exclusive list of Target Attributes, but all are not supported for Diagnostic Data Analytics.

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
AnalyticAttributes. StartTimestamp	String	Optional	No	datetime	Starting time for the range of data for this Analytic
AnalyticAttributes. EndTimestamp	String	Optional	No	datetime	End time for the range of data for this Analytic from the Persistent Storage
Associations	Strings Array	Mandatory	No	Analytics ID | Reports ID(s)	Shows association between the Analytic and any Report(s) that it might be used in. **NOTE:** The case of a second Analytic ID in the Associations indicates an associative relationship to a second Analytic that is the exact set of Attributes but is of the other Type, for ease of correlation between a Descriptive and Diagnostic Analytic.
AnalyticInformation[]	String Array	Mandatory (Instance) | Optional (Collection)	No	ID | Name | Status (Health and State) | Timestamp | Speed (b/s) | Capacity (Bytes) | CurrentReading (Temp, Volts, Current, RPM) | CurrentSpeed (b/s)	Analytic Data array containing Target attribute information across the time range
Reports. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Reports Resource(s) regarding this Analytic

GET /Compute/Devices/{id}/Analytics/{id}/Files/{id}

GET /Compute/Devices/{id}/Analytics/{id}/Files/{id} returns information regarding a specific Analytic file based on the {id} in the URI address string.

Note: This will trigger returning the actual "AnalyticInformation" data from the File.

Table 259: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40:80/Compute/Devices/0123456789/Analytics/8d8ea16353c8/
Files/8d8ea16353c8_1745498914
```

Table 260: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Analytic information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Analytics/{id}/Files/{id}

DELETE /Compute/Devices/{id}/Analytics/{id}/Files/{id} deletes a specific Analytic file.

Table 261: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: b95271160faf6c055f174a87b7ed4256"
    -X DELETE http://10.20.30.40:80/Compute/Devices/0123456789/Analytics/8d8ea16353c8/
Files/8d8ea16353c8_1745498914
```

Table 262: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one notification, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Analytics/{id}/Files/{id}

HEAD /Compute/Devices/{id}/Analytics/{id}/Files/{id} pings a specific Analytic file.

Table 263: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 264: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Analytics/{id}/Files/{id}

OPTIONS /Compute/Devices/{id}/Analytics/{id}/Files/{id} returns the allowable HTTP Methods list for this resource.

Table 265: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 266: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.22 /Compute/Devices/{id}/Reports/

The Collection-level methods and attributes of /Compute/Devices/{id}/Reports/ return information about the Reports collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 267: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Reports/{id}/** *(page 169)* for Instance Attribute details.

GET /Compute/Devices/{id}/Reports/

GET /Compute/Devices/{id}/Reports/ returns the collection of Reports.

Note: This will not return the "ReportInformation" but only the collection of Reports and their attributes unless the Query Parameter is sent.

Table 268: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Reports/
```

Table 269: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Reports + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Reports/

POST /Compute/Devices/{id}/Reports/ creates a new Report.

Table 270: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Reports/
    --data '{
        "Name": "Report1",
        "Description": "Included Analytics: Analytics1, Analytics2",
        "ReportAttributes": {
            "Capability": {
                "ID": 2
            },
            "Type": {
                "ID": 1
            }
        },
        "Resources": [
            {
                "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Analytics/8382835b7f22/"
            },
            {
                "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Analytics/72badd62eaca/"
            }
        ],
        "StartTimestamp": "2024-05-08T02:44:57Z",
        "EndTimestamp": "2024-04-08T02:44:57Z"
    }'
```

Example POST Request Body

```
{
    "Name": "name",
```

```
    "Description": "description",
    "ReportAttributes": {
        "Type": {
            "ID", 1|2
        },
        "Capability": {
            "ID": 1|2
        },
    }
    "StartTimestamp": "2023-11-20T12:00:00Z",
    "EndTimestamp": "2023-11-29T12:00:00Z",
    "Resources" : [
        {
            "Self":"analytic_uri1"
        },
        {
            "Self":"analytic_uri2"
        },
        {
        ...
        }
    ]
}
```

Table 271: POST Response Codes

Response Code	Name	Definition
202	Created	Created & Report information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Reports/

HEAD /Compute/Devices/{id}/Reports/ pings the Reports collection.

Table 272: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 273: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Reports/

OPTIONS /Compute/Devices/{id}/Reports/ returns the allowable HTTP Methods list for this resource.

Table 274: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 275: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.23 /Compute/Devices/{id}/Reports/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Reports/{id}/ return information about a specific Report and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 276: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier for this particular Report
Name	String	Mandatory	Yes	Free-Form 64 Chars	The name of the Report. Must be unique to this Reports Collection
Description	String	Optional	Yes	Free-Form 768 Characters	The description of this Report
CreateDate	String	Mandatory	No	datetime	Displays when the Report was created
LastModified	String	Mandatory	No	datetime	Displays when the Report was last modified
ReportAttributes. Type.ID	Number	Mandatory	Yes	1 | 2	Type of Report
ReportAttributes. Type.Name	String	Mandatory	No	"JSON" | "CSV"	JSON Formatted Data, or CSV Formatted Data (within the JSON response wrapper)
ReportAttributes. Capability.ID	Number	Mandatory	Yes	1 | 2	Capability ID representing the kind of Report where Dynamic is a sliding timestamp window and the Snapshot is a fixed and saved timestamp range. Use this attribute to create or modify the Report Capability

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
ReportAttributes.Capability.Name	String	Mandatory	No	"Dynamic" | "Snapshot"	Capability Name representing the kind of Report where Dynamic is a sliding timestamp window and the Snapshot is a fixed and saved timestamp range
StartTimestamp	String	Optional	Yes	datetime	Starting time for the range of data for this Report
EndTimestamp	String	Optional	Yes	datetime	Ending time for the range of data for this Report
Resources	String Array	Mandatory	Yes	Analytic Self Entries	Array of one or more Analytic entries for the Report
Associations	String Array	Mandatory	No	Reports ID | Analytics ID(s)	Shows association between the Report and any Analytic(s) that it might be used in
Analytic.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Analytic Resource(s) regarding this Report
ReportInformation[]	String Array	Optional (Collection) | Mandatory (Instance)	No	Analytic Data Entries	Report Data array containing Analytics Data information across the time range

GET /Compute/Devices/{id}/Reports/{id}/

GET /Compute/Devices/{id}/Reports/{id}/ returns information regarding a specific Report based on the {id} in the URI address string.

Note: This will trigger returning the actual "ReportInformation" data from the Persistent Storage based on the ReportAttributes.

Table 277: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Reports/89de5ac92d93
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Reports/89de5ac92d93/",
    "ID": "89de5ac92d93",
    "Name": "Report1",
    "Description": "Included Analytics - Analytics2, Analytics3",
    "CreateDate": "08-May-2024 02:56:06 UTC",
    "LastModified": "08-May-2024 02:56:06 UTC",
    "ReportAttributes": {
        "Type": {
            "ID": 1,
            "Name": "JSON"
        },
        "Capability": {
            "ID": 1,
            "Name": "Dynamic"
        },
        "StartTimestamp": "2024-04-07T21:12:45.995834841-06:00",
        "EndTimestamp": "2024-05-07T21:12:45.995834841-06:00"
    },
    "Resources": [
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Analytics/8382835b7f22/"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Analytics/72badd62eaca/"
        }
    ],
    "Associations": {
        "Analytics": [
            "8382835b7f22",
            "72badd62eaca"
        ],
        "Reports": [
            "89de5ac92d93"
        ]
    },
    "Analytics": "http://10.20.30.40/Compute/Devices/0123456789/Analytics/?
Reports=89de5ac92d93",
    "ReportInformation": [
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Reports/89de5ac92d93/Analytics/8382835b7f22/",
```

```
            "ID": "8382835b7f22",
            "Name": "Analytics2",
            "Description": "Server4: Processors: CPU1: Load Percentage",
            "CreateDate": "08-May-2024 02:26:50 UTC",
            "LastModified": "08-May-2024 02:26:50 UTC",
            "Type": {
                "ID": 1,
                "Name": "Descriptive Data Analytics"
            },
            "Associations": {
                "Analytics": [
                    "60e12334-f111-a460-2925-8382835b7f22"
                ],
                "Reports": [
                    "e6c691f1-378e-738d-350c-a3e340b25534",
                    "6b692fa0-14f9-75a3-6cf7-89de5ac92d93"
                ]
            },
            "Reports": "http://10.20.30.40/Compute/Devices/0123456789/
Reports/89de5ac92d93/Reports/?Analytics=8382835b7f22",
            "AnalyticAttributes": {
                "TargetEnclosureID": "3537972c02794b8d968e6bcb22262286",
                "TargetComponent": "Processors",
                "TargetComponentID": "CPU1",
                "TargetAttribute": "LoadPercentage",
                "StartTimestamp": "2024-04-07T21:12:45.995834841-06:00",
                "EndTimestamp": "2024-05-07T21:12:45.995834841-06:00"
            },
            "AnalyticInformation": [
                {
                    "Architecture": "x86_64",
                    "Cores": 10,
                    "ID": "CPU1",
                    "LoadPercentage": 0,
                    "LogicalProcessors": 20,
                    "Name": "CPU1",
                    "ProcessorID": "BFEBFBFF000306F2",
                    "Timestamp": "2024-04-11T14:04:56.768616776Z"
                },
                {
                    "Architecture": "x86_64",
                    "Cores": 10,
                    "ID": "CPU1",
                    "LoadPercentage": 10.16,
                    "LogicalProcessors": 20,
                    "Name": "CPU1",
                    "ProcessorID": "BFEBFBFF000306F2",
                    "Timestamp": "2024-04-11T14:21:04.850948626Z"
                }
            ]
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/
Reports/89de5ac92d93/Analytics/72badd62eaca/",
            "ID": "72badd62eaca",
            "Name": "Analytics3",
            "Description": "Server5: Processors: CPU1: Load Percentage",
            "CreateDate": "08-May-2024 02:28:25 UTC",
            "LastModified": "08-May-2024 02:28:25 UTC",
```

```
            "Type": {
                "ID": 1,
                "Name": "Descriptive Data Analytics"
            },
            "Associations": {
                "Analytics": [
                    "72badd62eaca"
                ],
                "Reports": [
                    "a3e340b25534",
                    "89de5ac92d93"
                ]
            },
            "Reports": "http://10.20.30.40/Compute/Devices/0123456789/
Reports/89de5ac92d93/Reports/?Analytics=72badd62eaca",
            "AnalyticAttributes": {
                "TargetEnclosureID": "1ace035696544fe38b1149ccd18d3ca1",
                "TargetComponent": "Processors",
                "TargetComponentID": "CPU1",
                "TargetAttribute": "LoadPercentage",
                "StartTimestamp": "2024-04-07T21:12:45.995834841-06:00",
                "EndTimestamp": "2024-05-07T21:12:45.995834841-06:00"
            },
            "AnalyticInformation": [
                {
                    "Architecture": "x86_64",
                    "Cores": 10,
                    "ID": "CPU1",
                    "LoadPercentage": 0.1,
                    "LogicalProcessors": 20,
                    "Name": "CPU1",
                    "ProcessorID": "BFEBFBFF000306F2",
                    "Timestamp": "2024-04-11T14:04:08.156343903Z"
                },
                {
                    "Architecture": "x86_64",
                    "Cores": 10,
                    "ID": "CPU1",
                    "LoadPercentage": 0.03,
                    "LogicalProcessors": 20,
                    "Name": "CPU1",
                    "ProcessorID": "BFEBFBFF000306F2",
                    "Timestamp": "2024-04-11T14:19:01.700714765Z"
                },
                {
                    "Architecture": "x86_64",
                    "Cores": 10,
                    "ID": "CPU1",
                    "LoadPercentage": 0.1,
                    "LogicalProcessors": 20,
                    "Name": "CPU1",
                    "ProcessorID": "BFEBFBFF000306F2",
                    "Timestamp": "2024-04-11T14:30:38.5508993Z"
                }
            ]
        }
    ]
}
```

Table 278: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Analytic information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Reports/{id}/

PUT /Compute/Devices/{id}/Reports/{id}/ modifies an attribute or executes an operation for this Report.

Table 279: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Reports/123456
    --data '{
        "Name": "new_name"
        "Description": "new_description"
    }'
```

Example PUT Request Body

```
{
    "Name": "name",
    "Description": "description",
    "ReportAttributes": {
        "Type": {
            "ID", 1|2
        },
        "Capability": {
            "ID": 1|2
        }
    }
    "StartTimestamp": "2023-11-20T12:00:00Z",
    "EndTimestamp": "2023-11-29T12:00:00Z",
    "Resources" : [
        {
            "Self":"analytic_uri1"
        },{
            "Self":"analytic_uri2"
        },{
            ...
        }
    ]
}
```

Table 280: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Report information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Reports/{id}/

DELETE /Compute/Devices/{id}/Reports/{id}/ deletes a specific Report.

Table 281: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Reports/123456
```

Table 282: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one notification, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Reports/{id}/

HEAD /Compute/Devices/{id}/Reports/{id}/ pings a specific Report.

Table 283: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 284: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Reports/{id}/

OPTIONS /Compute/Devices/{id}/Reports/{id}/ returns the allowable HTTP Methods list for this resource.

Table 285: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 286: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.24 /Compute/Devices/{id}/Reports/{id}/Files/

The Collection-level methods of /Compute/Devices/{id}/Reports/{id}/Files/ return information about the Report Files collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 287: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
ReportFiles	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of URLs for each Report File for a Report

GET /Compute/Devices/{id}/Reports/{id}/Files/

GET /Compute/Devices/{id}/Reports/{id}/Files/ returns the Collection or list of Files for a Report.

Note: This will not return the contents of each file.

Table 288: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40:80/Compute/Devices/0123456789/Reports/8d8ea16353c8/Files/
```

Table 289: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Reports + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Reports/{id}/Files/

POST /Compute/Devices/{id}/Reports/{id}/Files/ creates a new Report file. Data analysis for generating the file happens asynchronously only when Report Capability is Dynamic.

Note: No file is created if Report Capability is Snapshot, as with initial POST on Reports, data is already captured and saved in Persistent Storage.

Table 290: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Reports/8d8ea16353c8/Files/
```

Note: POST commands to this resource do not require a data parameter. If accepted, report data collection will occur in the background.

Table 291: POST Response Codes

Response Code	Name	Definition
202	Created	New URI in Location Header to get the File
302	Found	If Report Capability is Snapshot, with initial POST request on Reports, data is already captured and saved
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Reports/{id}/Files/

HEAD /Compute/Devices/{id}/Reports/{id}/Files/ pings a Reports Files collection.

Table 292: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 293: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Reports/{id}/Files/

OPTIONS /Compute/Devices/{id}/Reports/{id}/Files/ returns the allowable HTTP Methods list for this resource.

Table 294: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 295: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.25 /Compute/Devices/{id}/Reports/{id}/Files/{id}/

The Device-level methods of /Compute/Devices/{id}/Reports/{id}/Files/{id}/ return information about a specific Report File and enable management operations.

HTTP Methods: GET DELETE HEAD OPTIONS

Table 296: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	UUID Format	The Identifier for this particular Report
Name	String	Mandatory	Yes	Free-Form 64 Chars	The name of the Report. Must be unique to this Reports Collection
Description	String	Optional	Yes	Free-Form 768 Characters	The description of this Report
CreateDate	String	Mandatory	No	datetime	Displays when the Report was created
LastModified	String	Mandatory	No	datetime	Displays when the Report was last modified
ReportAttributes. Type.ID	Number	Mandatory	Yes	1 | 2	Type of Report
ReportAttributes. Type.Name	String	Mandatory	No	"JSON" | "CSV"	JSON Formatted Data, or CSV Formatted Data (within the JSON response wrapper)
ReportAttributes. Capability.ID	Number	Mandatory	Yes	1 | 2	Capability ID representing the kind of Report where Dynamic is a sliding timestamp window and the Snapshot is a fixed and saved timestamp range. Use this attribute to create or modify the Report Capability

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
ReportAttributes.Capability.Name	String	Mandatory	No	"Dynamic" | "Snapshot"	Capability Name representing the kind of Report where Dynamic is a sliding timestamp window and the Snapshot is a fixed and saved timestamp range
StartTimestamp	String	Optional	Yes	datetime	Starting time for the range of data for this Report
EndTimestamp	String	Optional	Yes	datetime	Ending time for the range of data for this Report
Resources	String Array	Mandatory	Yes	Analytic Self Entries	Array of one or more Analytic entries for the Report
Associations	String Array	Mandatory	No	Reports ID | Analytics ID(s)	Shows association between the Report and any Analytic(s) that it might be used in
Analytic.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Analytic Resource(s) regarding this Report
ReportInformation[]	String Array	Optional (Collection) | Mandatory (Instance)	No	Analytic Data Entries	Report Data array containing Analytics Data information across the time range

GET /Compute/Devices/{id}/Reports/{id}/Files/{id}/

GET /Compute/Devices/{id}/Reports/{id}/Files/{id}/ returns information regarding a specific Report File based on the {id} in the URI address string.

Note: This will trigger returning the actual "ReportInformation" data from the File.

Table 297: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40:80/Compute/Devices/0123456789/Reports/8d8ea16353c8/
Files/8d8ea16353c8_1745498914
```

Table 298: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Analytic information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Reports/{id}/Files/{id}/

DELETE /Compute/Devices/{id}/Reports/{id}/Files/{id}/ deletes a specific Report File.

Table 299: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: b95271160faf6c055f174a87b7ed4256"
    -X DELETE http://10.20.30.40:80/Compute/Devices/0123456789/Reports/8d8ea16353c8/
Files/8d8ea16353c8_1745498914
```

Table 300: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one notification, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Reports/{id}/Files/{id}/

HEAD /Compute/Devices/{id}/Reports/{id}/Files/{id}/ pings a specific Report File.

Table 301: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 302: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Reports/{id}/Files/{id}/

OPTIONS /Compute/Devices/{id}/Reports/{id}/Files/{id}/ returns the allowable HTTP Methods list for this resource.

Table 303: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 304: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

3.3.26 /Compute/Devices/{id}/Telemetry/

The methods and attributes of /Compute/Devices/{id}/Telemetry/ return information about the Telemetry resource and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 305: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
LastAccessed	String	Mandatory	No	datetime based on UTC	The UTC time of the request
Members[]	String Array	Mandatory	No	Member Entry	This is the instantaneous, fully expanded metadata from this Device

GET /Compute/Devices/{id}/Telemetry/

GET /Compute/Devices/{id}/Telemetry/ returns the information of the Telemetry resource.

Table 306: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.41.100.60:80/Compute/Devices/A0A9CB7F-7D63-4075-8826-48E0ADF3F2C0/
Telemetry/
```

Example GET Response Body


```
{
    "Self": "http://10.41.100.60:80/Compute/Devices/
A0A9CB7F-7D63-4075-8826-48E0ADF3F2C0/Telemetry/",
    "LastAccessed": "05-Oct-2023 18:47:33 UTC",
    "Members": [
        {
            "Self": "http://10.41.100.60:80/Compute/Devices/
A0A9CB7F-7D63-4075-8826-48E0ADF3F2C0/",
            "SystemType": {...},
            "Name": "WDAP-LGWPTOMIDK",
            "ID": "A0A9CB7F-7D63-4075-8826-48E0ADF3F2C0",
            "OperatingSystem": {...},
            "Model": "Resource Manager Data Center",
            "Manufacturer": "WDC",
            "Capabilities": {...},
            "Status": {...},
            "IPAddresses": {...},
            "InformationStructure": {...},
            "ApplicationSettings": {...},
            "Location": {...},
            "Accounts": {...},
            "Files": {...},
            "Jobs": {...},
            "SystemClock": {...},
            "Assets": {...},
            "Analytics": {...},
            "Notifications": {...},
            "Policies": {...},
            "Reports": {...},
            "Telemetry": {...}
        }
    ]
}
```


Table 307: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Telemetry resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Telemetry/

HEAD /Compute/Devices/{id}/Telemetry/ pings the Telemetry resource.

Table 308: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 309: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Telemetry/

OPTIONS /Compute/Devices/{id}/Telemetry/ returns the allowable HTTP Methods list for this resource.

Table 310: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 311: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

Compute Service

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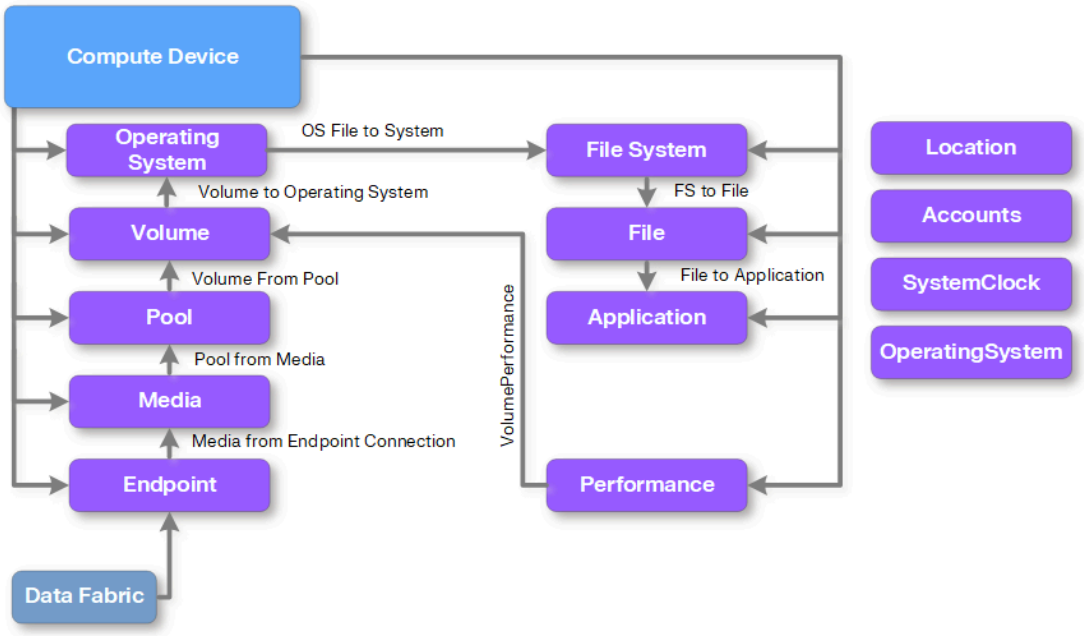
4.1 Modeling Compute Device Resources

The Compute Device modeling includes both physical and logical/virtual resources. The physical resource model includes the overall Compute Device that contains a collection of processors, power supplies and fans, and the attached Storage Media (Storage and/or Memory). A Chassis Device may provide the power and cooling resources separately if the Compute Device is separate from the enclosure services. These physical resources are represented in the figure below.

The logical and virtual resources model include those resources that are logical or virtual in nature. These include Storage Pools, Storage Volumes, Storage Endpoints, File System, and Performance.

Figure 2: Compute Resource Physical and Virtual/Logical Model



4.2 Physical Devices

The OCAPI's physical resource model includes the overall Compute Proxy Device. The topics in this section provide information for accessing and managing the features of these physical resources.

4.2.1 /Compute/Devices/

The Collection-level methods and attributes of /Compute/Devices/ return information about the Devices collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 312: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/** *(page 193)* for Instance Attribute details

GET /Compute/Devices/

GET /Compute/Devices/ returns the Collection of Devices.

Table 313: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/
```

Table 314: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Devices + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/

HEAD /Compute/Devices/ pings the Devices collection.

Table 315: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 316: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/

OPTIONS /Compute/Devices/ returns the allowable HTTP Methods list for this resource.

Table 317: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 318: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.2.2 /Compute/Devices/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/ return information about a specific Device and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 319: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
SystemType. Name	String	Mandatory	No	"Compute"	The domain type string indicating the type of Device Resource
SystemType. ID	Number	Mandatory	No	1	The domain type number indicating the type of Device Resource
Name	String	Mandatory	Yes	Free-Form 256 Chars	The user-defined Friendly Name for this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
OperatingSystem. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Operating System Resource
OperatingSystem. Name	String	Mandatory	No	Free-Form URI	Resource Manager Data Center Edition Application Name
OperatingSystem. OSType. Name	String	Mandatory	No	"Winnt" | "Linux"	Resource Manager Data Center Edition running on this OS Type string value
OperatingSystem. OSType. ID	Number	Mandatory	No	18 | 36	Resource Manager Data Center Edition running on this OS Type number value
OperatingSystem. Version	String	Mandatory	No	Free-Form URI	Resource Manager Data Center Edition Application Version
Serial Number	String	Mandatory	No	Free-Form 64 Chars	The Compute Node Serial Number

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Model	String	Mandatory	No	"Resource Manager Data Center"	Compute Node application model
Manufacturer	String	Mandatory	No	"WDC"	Compute Node manufacturer
Capabilities. Members[]. Name	String	Mandatory	No	"Management" | "File Server", | "Server", | "Central Processor Device"	The capabilities information value(s) regarding the Resource
Capabilities. Members[]. ID	Number	Mandatory	No	14 | 16 | 39 | 47101	The capabilities information value map(s) regarding the Resource
Capabilities. Members[]. CapabilitiesDescription	String	Mandatory	No	Free-form characters	Entry describing the capability
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries.
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536 - 65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
IPAddresses. Members[]. IPAddress	String Array	Mandatory	No	IPv4 or IPv6 pattern	The IP Address list for this Resource
Architecture	String	Optional	No	Free-form characters	The Compute Architecture type
Bios.Manufacturer	String	Mandatory	No	Free-form characters	The BIOS Manufacturer information
Bios.Version	String	Mandatory	No	Free-form characters	The BIOS Version information
Bios.ReleaseDate	String	Mandatory	No	Free-form characters	The BIOS Release Date
ChassisPackageType. Name	String	Optional	No	"Other" | "Unknown" | "Desktop" | "Low Profile Desktop" | "Pizza Box" | "Mini Tower" | "Tower" | "Portable" | "Laptop" | "Notebook" | "Hand Held" | "Docking Station" | "All in One" | "Sub Notebook" | "Space-Saving" | "Lunch Box" | "Main System Chassis" | "Expansion Chassis" | "SubChassis" | "Bus Expansion Chassis" | "Peripheral Chassis" | "Storage Chassis" | "Rack Mount Chassis" | "Sealed-Case PC" | "Multi-system Chassis"	The Compute Node chassis package value information
ChassisPackageType. ID	Number	Optional	No	1 - 25	The Compute Node chassis package value map
Domain	String	Mandatory	No	Free-Form 64 Chars	The Domain that this Compute Node is a member of
Memory	Number	Mandatory	No	Integer in Bytes	The total memory on this Compute Node in Bytes
RemainingMemory	Number	Mandatory	No	Integer in Bytes	The remaining (not used) memory on this Compute Node in Bytes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Rackmountable. Name				"Unknown" | "False" | "True"	The Rackmountable value based on ChassisType
Rackmountable. ID	Number	Optional	No	1 - 25	The Rackmountable value map based on ChassisType value map
InformationStructure. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Information Structure Resource
Location. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Location Resource
Accounts. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Accounts Resource
Files. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Files Resource
SystemClock. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the System Clock Resource
Adapters. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Adapters Resource collection
Media. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Media Resource
Performance. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Performance Resource
Ports. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Ports Resource
PowerSupplies. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Power Supplies Resource
Processors. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Proessors Resource
Volumes. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Volumes Resource

GET /Compute/Devices/{id}/

GET /Compute/Devices/{id}/ returns information regarding a specific Device based on the {id} in the URI address string.

Table 320: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/",
    "SystemType": {
        "ID": 1,
        "Name": "Compute"
    },
    "Name": "cos-r16-s1",
    "ID": "0123456789",
    "OperatingSystem": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/OperatingSystem/",
        "Name": "Linux",
        "Architecture": "x86_64",
        "OSDescription": "Ubuntu 22.04 (Jammy Jellyfish) 5.15.0-69-generic",
        "SerialNumber": "0123456789",
        "OSType": {
            "ID": 36,
            "Name": "Linux"
        },
        "Version": "Ubuntu 22.04 (Jammy Jellyfish) 5.15.0-69-generic",
        "Status": {
            "State": {
                "ID": 16,
                "Name": "In service"
            },
            "Health": [
                {
                    "ID": 5,
                    "Name": "OK"
                }
            ]
        }
    },
    "SerialNumber": "S199310X5A12689",
    "Model": "SSG-5028R-E1CR12L-CE010",
    "Manufacturer": "Supermicro",
    "Capabilities": {
        "Members": [
            {
                "ID": 14,
                "Name": "Management",
```

```
                "CapabilityDescription": "This device provides software system
 management capabilities."
            },
            {
                "ID": 16,
                "Name": "File Server",
                "CapabilityDescription": "This device is a file server."
            },
            {
                "ID": 39,
                "Name": "Server",
                "CapabilityDescription": "This device is a compute server."
            },
            {
                "ID": 47101,
                "Name": "Central Processor Device",
                "CapabilityDescription": "This device provides general purpose CPU-
based compute processors."
            }
        ]
    },
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "None"
        ]
    },
    "IPAddresses": {
        "Members": [
            {
                "IPAddress": "10.202.239.211"
            }
        ]
    },
    "Architecture": "x86_64",
    "Bios": {
        "Manufacturer": "American Megatrends Inc.",
        "Version": "2.0b",
        "ReleaseDate": "04/14/2017"
    },
    "ChassisPackageType": {
        "ID": 1,
        "Name": "Other"
    },
    "InformationStructure": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/InformationStructure/"
    },
    "Domain": "",
    "Memory": 134961729536,
    "RemainingMemory": 128453312512,
```

```
    "Rackmountable": {
        "ID": 1,
        "Name": "Unknown"
    },
    "Location": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Location/"
    },
    "Accounts": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Accounts/"
    },
    "Adapters": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Adapters/"
    },
    "Files": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/"
    },
    "Media": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Media/"
    },
    "Performance": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Performance/"
    },
    "Ports": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Ports/"
    },
    "PowerSupplies": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/PowerSupplies/"
    },
    "Processors": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Processors/"
    },
    "SystemClock": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/SystemClock/"
    },
    "Volumes": {
        "Self": "http://10.20.30.40/Compute/Devices/0123456789/Volumes/"
    }
}
```


Table 321: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Device information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/

HEAD /Compute/Devices/{id}/ pings this Storage Device.

Table 322: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 323: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/

OPTIONS /Compute/Devices/{id}/ returns the allowable HTTP Methods list for this resource.

Note: There may be some attributes returned in the OPTIONS response schema that may not be directly supported by this resource.

Table 324: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 325: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.2.3 /Compute/Devices/{id}/Processors/

The Collection-level methods and attributes of /Compute/Devices/{id}/Processors/ return information about the Processors collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 326: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Processors/{id}/** *(page 204)* for Instance Attribute details

GET /Compute/Devices/{id}/Processors/

GET /Compute/Devices/{id}/Processors/ returns the Collection of Controllers.

Table 327: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Processors/
```

Table 328: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Controllers + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Processors/

HEAD /Compute/Devices/{id}/Processors/ pings the Controllers collection.

Table 329: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 330: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Processors/

OPTIONS /Compute/Devices/{id}/Processors/ returns the allowable HTTP Methods list for this resource.

Table 331: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

4.2.4 /Compute/Devices/{id}/Processors/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Processors/{id}/ return information about a specific Processor and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 332: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	Yes	Free-Form 256 Chars	The user-defined Friendly Name for this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Architecture	String	Mandatory	No	Free-form string indicating the architecture	The architecture type for the processor
Cores	Number	Mandatory	No	Unsigned Integer	The number of processor cores
DataWidth	String	Mandatory	No	Free-form string indicating the data width in bits	The number of processor cores
ProcessorDescription	String	Mandatory	No	Free-form 256 Chars	The number of processor cores
Family	Number	Mandatory	No	Unsigned Integer	The processor family value
LogicalProcessors	Number	Mandatory	No	Unsigned Integer	The number of logical processors
Manufacturer	String	Mandatory	No	Free-form 64 Chars	The processor manufacturer
Model	Number	Mandatory	No	Unsigned Integer	The processor model name or number
ProcessorID	String	Mandatory	No	Hex String	The processor unique identifier
ProcessorSpeed. BaseUnits	String	Mandatory	No	MHz	The base units for the processor speed
ProcessorSpeed. CurrentClockSpeed	Number	Mandatory	No	0 - n	The current speed of the processor
ProcessorSpeed. MaxClockSpeed	Number	Mandatory	No	0 - n	The maximum speed of the processor
Stepping	Number	Mandatory	No	Unsigned Integer	The processor stepping value
Revision	String	Mandatory	No	Free-form 64 Chars	The processor version
LoadPercentage	Number	Mandatory	No	0 - 100	The processor stepping value
Role	String	Mandatory	No	"Central Processor" | "Math Processor" | "Graphics Processing Unit" | "Field-Programmable Gate Array"	The processor role

GET /Compute/Devices/{id}/Processors/{id}/

GET /Compute/Devices/{id}/Processors/{id}/ returns information regarding a specific Controller based on the {id} in the URI address string.

Table 333: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Processors/CPU1
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Processors/CPU1/",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "None"
        ]
    },
    "Architecture": "x86_64",
    "Cores": 10,
    "DataWidth": "64-bit",
    "ProcessorDescription": "Type 0, Family 6, Model 63, Stepping 2",
    "Family": 6,
    "ID": "CPU1",
    "LogicalProcessors": 20,
    "Manufacturer": "Intel",
    "Model": 63,
    "Name": "Intel(R) Xeon(R) CPU E5-2650 v3 @ 2.30GHz",
    "ProcessorID": "BFEBFBFF000306F2",
    "ProcessorSpeed": {
        "BaseUnits": "MHz",
        "CurrentClockSpeed": 2300,
        "MaxClockSpeed": 4000
    },
    "Stepping": 2,
```

```
    "Revision": "Intel(R) Xeon(R) CPU E5-2650 v3 @ 2.30GHz",
    "LoadPercentage": 0,
    "Role": "Central Processor"
}
```

Table 334: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Controller information in the response body + ETag Header
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Processors/{id}/

HEAD /Compute/Devices/{id}/Processors/{id}/ pings this Controller.

Table 335: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 336: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Processors/{id}/

OPTIONS /Compute/Devices/{id}/Processors/{id}/ returns the allowable HTTP Methods list for this resource.

Table 337: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

4.2.5 /Compute/Devices/{id}/Adapters/

The Collection-level methods and attributes of /Compute/Devices/{id}/Adapters/ return information about the Adapters collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 338: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Adapters/{id}/** *(page 210)* for Instance Attribute details

GET /Compute/Devices/{id}/Adapters/

GET /Compute/Devices/{id}/Adapters/ returns the Collection of Adapters.

Table 339: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Adapters
```

Table 340: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Adapters + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Adapters/

HEAD /Compute/Devices/{id}/Adapters/ pings the Adapters collection.

Table 341: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 342: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Adapters/

OPTIONS /Compute/Devices/{id}/Adapters/ returns the allowable HTTP Methods list for this resource.

Table 343: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

4.2.6 /Compute/Devices/{id}/Adapters/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Adapters/{id}/ return information about a specific Adapter and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 344: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Associations	String Array	Mandatory	No	Adapter | Ports	The associations among resources based on resource IDs

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Ports.Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Ports associated with this Adapter

GET /Compute/Devices/{id}/Adapters/{id}/

GET /Compute/Devices/{id}/Adapters/{id}/ returns information regarding a specific Adapter based on the {id} in the URI address string.

Table 345: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Adapters/eno1/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Adapters/eno1/",
    "ID": "eno1",
    "Status": {
        "State": {
            "ID": 0,
            "Name": "Unknown"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "Associations": {
        "Adapters": [
            "eno1"
        ],
        "Ports": [
            "0c_c4_7a_68_e5_60_eno1_inet",
            "0c_c4_7a_68_e5_60_eno1_inet6ll"
        ]
    },
    "Ports": "http://10.20.30.40/Compute/Devices/0123456789/Ports/?Adapters=eno1"
```

```
}
```

Table 346: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Adapter information in the response body + ETag Header
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Adapters/{id}/

HEAD /Compute/Devices/{id}/Adapters/{id}/ pings this Adapter.

Table 347: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 348: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Adapters/{id}/

OPTIONS /Compute/Devices/{id}/Adapters/{id}/ returns the allowable HTTP Methods list for this resource.

Table 349: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

4.2.7 /Compute/Devices/{id}/Adapters/{id}/Ports/

The Collection-level methods and attributes of /Compute/Devices/{id}/Adapters/{id}/Ports/ return information about the Ports collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 350: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Adapters/{id}/Ports/{id}/** ***(page 215)*** for Instance Attribute details

GET /Compute/Devices/{id}/Adapters/{id}/Ports/

GET /Compute/Devices/{id}/Adapters/{id}/Ports/ returns the Collection of Ports.

Table 351: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Adapters/management/Ports
```

Table 352: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Ports + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Adapters/{id}/Ports/

HEAD /Compute/Devices/{id}/Adapters/{id}/Ports/ pings the Ports collection.

Table 353: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 354: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Adapters/{id}/Ports/

OPTIONS /Compute/Devices/{id}/Adapters/{id}/Ports/ returns the allowable HTTP Methods list for this resource.

Table 355: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

4.2.8 /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/ return information about a specific Port and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 356: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
AddressOrigin. Name	String	Mandatory	Yes	"UNKNOWN" | "OTHER" | "NOT APPLICABLE" | "STATIC" | "DHCP"	Information value as to where the address is supplied

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
AddressOrigin.ID	Number	Mandatory	Yes	0 - 4	Information value map as to where the address is supplied
IPv4Address	String	Mandatory	Yes	IPv4 4-Octet pattern + CIDR mask	The IPv4 Network IP Address for this Device. **Note**: 0.0.0.0/8 & 127.0.0.0/8 are reserved and rejected.
IPGateway	String	Optional	Yes	IPv4 4-Octet pattern	The Network IP Gateway address setting
MACAddress	String	Mandatory	No	6-hex digit HW address	
NetworkType.Name	String	Mandatory	No	"LAN" | "WLAN" | "WAN" | "MAN" | "SAN" | "VLAN" | "Ethernet Network" | "IP Network" | "IPv4 Network" | "IPv6 Network"	The network type value
NetworkType.ID	Number	Mandatory	No	0 - 8	The network type value map
MTUBytes	Number	Mandatory	Yes	Range: 46 - 9216 (default: 1500)	The current Maximum Transfer Bytes value
DNS	String Array	Optional	Yes	IPv4 4-Octet pattern	List of DNS server addresses

GET /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/

GET /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/ returns information regarding a specific Port based on the {id} in the URI address string.

Table 357: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Adapters/management/
Ports/01_0a_f7_95_79_3c
```

Example GET Response Body

```
{
```

```
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Adapters/management/
Ports/01_0a_f7_95_79_3c/",
    "ID": "01_0a_f7_95_79_3c",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "AddressOrigin": {
        "ID": 4,
        "Name": "DHCP"
    },
    "IPv4Address": "10.20.30.40/22",
    "IPGateway": "",
    "MACAddress": "01:0a:f7:95:79:3c",
    "NetworkType": {
        "ID": 8,
        "Name": "IPv4 Network"
    },
    "MTUBytes": 1500
}
```


Table 358: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Adapter information in the response body + ETag Header
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

PUT /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/

PUT /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/ modifies an attribute or executes an operation for this Port.

Table 359: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT Response Body

```
{
    "AddressOrigin": {
        "ID": number,
        "Name": "DHCP | STATIC"
    },
    "IPv4Address": "ipv4_address/cidr",
    "IPGateway": "ipv4_gateway",
    "MTUBytes": 1500,
    "DNS": [
        "ipv4_address"
    ]
}
```

Table 360: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Port information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an Etag value in the Request Header
500	Internal Server Error	System could not process the request

HEAD /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/

HEAD /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/ pings this Port.

Table 361: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 362: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/

OPTIONS /Compute/Devices/{id}/Adapters/{id}/Ports/{id}/ returns the allowable HTTP Methods list for this resource.

Table 363: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

4.2.9 /Compute/Devices/{id}/PowerSupplies/

The Collection-level methods and attributes of /Compute/Devices/{id}/PowerSupplies/ return information about the Power Supplies collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 364: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/PowerSupplies/{id}/** *(page 222)* for Instance Attribute details

GET /Compute/Devices/{id}/PowerSupplies/

GET /Compute/Devices/{id}/PowerSupplies/ returns the Collection of Power Supplies.

Table 365: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies
```

Table 366: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Power Supplies + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/PowerSupplies/

HEAD /Compute/Devices/{id}/PowerSupplies/ pings the Power Supplies collection.

Table 367: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 368: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/PowerSupplies/

OPTIONS /Compute/Devices/{id}/PowerSupplies/ returns the allowable HTTP Methods list for this resource.

Table 369: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.2.10 /Compute/Devices/{id}/PowerSupplies/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/PowerSupplies/{id}/ return information about a specific Power Supply and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 370: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Name	String	Mandatory	No	Free-Form 64 Chars	
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Power Supply
PartNumber	String	Mandatory	No	Free-Form 32 Characters	The Part Number for this Power Supply
SerialNumber	String	Mandatory	No	Free-Form 32 Characters	The Serial Number of this Power Supply
Manufacturer	String	Mandatory	No	Free-Form 32 Characters	The Manufacturer of this Power Supply
TotalOutputPower	Number	Optional	No	Number in Watts	The total output power in watts
BaseUnits. ID	Number	Optional	No	7	The Base Units value map
BaseUnits. Name	String	Optional	No	Watts	The Base Units value as Watts
RateUnits. ID	Number	Optional	No	0	The Rate Units value map
RateUnits. Name	String	Optional	No	None	The Rate Units value
UnitModifier	Number	Optional	No	0	The Unit Modifier to determine order of magnitude to multiply the Total Power Output as Base 10 * Unit Modifier

GET /Compute/Devices/{id}/PowerSupplies/{id}/

GET /Compute/Devices/{id}/PowerSupplies/{id}/ returns information regarding a specific Power Supply based on the {id} in the URI address string.

Table 371: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/1
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/PowerSupplies/1/",
    "ID": "1",
    "Name": "PowerSupply 1",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "PartNumber": " PWS-920P-SQ",
    "SerialNumber": " P9212CF32AT4462",
    "Manufacturer": " SUPERMICRO",
    "TotalOutputPower": 920,
    "BaseUnits": {
        "ID": 7,
        "Name": "Watts"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
    },
    "UnitModifier": 0
}
```


Table 372: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Power Supply information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned

Response Code	Name	Definition
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/PowerSupplies/{id}/

PUT /Compute/Devices/{id}/PowerSupplies/{id}/ modifies an attribute or executes an operation for this Power Supply.

Table 373: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/
PowerSupplyAFRU/?IPAddress=10.20.30.41
    --data '{
        "IndicatorLED": {
            "ID": 2
        }
    }'
```

Table 374: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Power Supply information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed

Response Code	Name	Definition
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/PowerSupplies/{id}/

HEAD /Compute/Devices/{id}/PowerSupplies/{id}/ pings this Power Supply.

Table 375: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 376: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/PowerSupplies/{id}/

OPTIONS /Compute/Devices/{id}/PowerSupplies/{id}/ returns the allowable HTTP Methods list for this resource.

Table 377: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 378: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.2.11 /Compute/Devices/{id}/Media/

The Collection-level methods and attributes of /Compute/Devices/{id}/Media/ return information about the Media collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 379: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Media/{id}/** ***(page 230)*** for Instance Attribute details

GET /Compute/Devices/{id}/Media/

GET /Compute/Devices/{id}/Media/ returns the Collection of Media.

Table 380: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 381: GET Query Parameters

Query Parameter	Description
?SmartLog=true	Adds the "SmartLog" JSON object attributes based on the particular protocol type (e.g, ATA, SCSI, and NVMe) for all the Media Instances

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Media/
```

Table 382: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Media + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Media/

HEAD /Compute/Devices/{id}/Media/ pings the Media collection.

Table 383: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 384: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Media/

OPTIONS /Compute/Devices/{id}/Media/ returns the allowable HTTP Methods list for this resource.

Table 385: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

4.2.12 /Compute/Devices/{id}/Media/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Media/{id}/ return information about a specific Medium and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 386: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
SerialNumber	String	Mandatory	No	Free-Form 64 Chars	The serial number for this Resource
Model	String	Mandatory	No	Free-Form 64 Chars	The vendor name for this Resource
Version	String	Mandatory	No	Free-Form 32 Chars	The Operating System or Firmware Version
Manufacturer	String	Mandatory	No	Free-Form 64 Chars	The vendor name for this Resource
Protocol. Name	String	Mandatory	No	"Universal Serial Bus" | "Serial ATA" | "SAS" | "Nvme" | "NvmeoF"	The Media Protocol Name
Protocol. ID	Number	Mandatory	No	15 | 47 | 65536-65538	The Media Protocol ID value map
DiskType. Name	String	Mandatory	No	"Unknown" | "Other" | "HDD" | "SSD" | "Hybrid" | "CD-DVD"	The Media Disk Type Name
DiskType. ID	Number	Mandatory	No	0 - 5	The Media Disk Type ID value map

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Status. Details[]	String Array	Mandatory	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Device
Capacity	Number	Mandatory	No	Integer value in Bytes	The total capacity of the Medium Resource in Bytes
DeviceHandle	String	Optional	No	Free-Form Characters	The current Device Handle name attached to this Medium Resource
MultiplePathsDetected	Boolean	Optional	No	true | false	Indicates if this Media Resource has multiple paths detected
DurableName	String	Optional	No	WWN or UUID Format	The in-band data path identifier

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Associations	String Array	Mandatory	No	Media | Volumes	Indicates the associations among linked resources based on ID
Volumes. Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Volumes associated with this Medium resource

GET /Compute/Devices/{id}/Media/{id}/

GET /Compute/Devices/{id}/Media/{id}/ returns information regarding a specific Medium based on the {id} in the URI address string.

Table 387: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 388: GET Query Parameters

Query Parameter	Description
?SmartLog=true	Adds the "SmartLog" JSON object attributes based on the particular protocol type (e.g, ATA, SCSI, and NVMe) for all the Media Instances

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Media/BTWA53140407240AGN
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Media/BTWA53140407240AGN/",
    "ID": "BTWA53140407240AGN",
    "Name": "",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
```

```
                "Name": "OK"
            }
        ]
    },
    "Associations": {
        "Media": [
            "BTWA53140407240AGN"
        ],
        "Volumes": [
            "4D4B-2427",
            "e5a11a79-bd5e-49de-ba1b-7ba7977878bf"
        ]
    },
    "Capacity": 240057409536,
    "AlternateDeviceHandle": "/dev/sda",
    "DeviceHandle": "/dev/sg0",
    "Manufacturer": "INTEL",
    "Model": "SSDSC2BB24",
    "MultiplePathsDetected": false,
    "Protocol": {
        "ID": 47,
        "Name": "Serial ATA"
    },
    "DiskType": {
        "ID": 2,
        "Name": "HDD"
    },
    "Version": "0130",
    "SerialNumber": "BTWA53140407240AGN",
    "DurableName": "",
    "Volumes": "http://10.20.30.40/Compute/Devices/0123456789/Volumes/?
Media=BTWA53140407240AGN"
}
```


Table 389: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Media information in the response body + ETag Header
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Media/{id}/

HEAD /Compute/Devices/{id}/Media/{id}/ pings this Medium.

Table 390: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 391: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Media/{id}/

OPTIONS /Compute/Devices/{id}/Media/{id}/ returns the allowable HTTP Methods list for this resource.

Table 392: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

4.3 Logical Devices

The OCAPI's logical and virtual resources model include those resources that are logical or virtual in nature. The topics in this section provide information for accessing and managing the features of these logical and virtual resources.

4.3.1 /Compute/Devices/{id}/Volumes/

The Collection-level methods and attributes of /Compute/Devices/{id}/Volumes/ return information about the Volumes collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 393: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Volumes/{id}/** *(page 238)* for Instance Attribute details

GET /Compute/Devices/{id}/Volumes/

GET /Compute/Devices/{id}/Volumes/ returns the Collection of Volumes.

Table 394: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Volumes/
```

Table 395: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Volume + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

POST /Compute/Devices/{id}/Volumes/

POST /Compute/Devices/{id}/Volumes/ creates a new Volume.

Table 396: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST Request Body

```
{
    "Name": "volume_name",
    "Capacity" : bytes,
    "Description": "desc",
    "PoolID": pool_id
}
```

Table 397: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Volume information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
500	Internal Server Error	System could not process the request

HEAD /Compute/Devices/{id}/Volumes/

HEAD /Compute/Devices/{id}/Volumes/ pings the Volume collection.

Table 398: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 399: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Volumes/

OPTIONS /Compute/Devices/{id}/Volumes/ returns the allowable HTTP Methods list for this resource.

Table 400: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

4.3.2 /Compute/Devices/{id}/Volumes/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Volumes/{id}/ return information about a specific Volume and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 401: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	Yes	Free-Form 256 Chars	The user-defined name of this Volume
UUID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Capacity	Number	Mandatory	No	Integer Bytes	The total number of raw bytes for this Volume from the FS point of view
RemainingCapacity	Number	Mandatory	No	Integer Bytes	The total number of raw remaining bytes for this Volume from the FS point of view
BlockSize	Number	Optional	No	Integer Bytes	The block size for this Volume
DeviceHandle	String	Optioal	No	Free-Form 64 Chars	The operating system connection to this Volume
FsType	String	Mandatory	No	"ext4" | "NTFS"	The Filesystem Type formatted onto this Volume
MountPoint	String	Mandatory	No	Free-Form 64 Chars	The Volume mount point name on this system
Associations	String Array	Mandatory	No	Media | Volumes	The associations among resources based on ID

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. State. Name	String Array	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Media. Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link to the Media for this Volume

GET /Compute/Devices/{id}/Volumes/{id}/

GET /Compute/Devices/{id}/Volumes/{id}/ returns information regarding a specific Volume based on the {id} in the URI address string.

Table 402: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Volumes/4D4B-2427
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Volumes/4D4B-2427/",
    "ID": "4D4B-2427",
    "Status": {
        "State": {
            "ID": 0,
            "Name": "Unknown"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "BlockSize": 512,
    "Capacity": 1124995072,
    "RemainingCapacity": 1118629888,
    "DeviceHandle": "/dev/sda1",
    "FsType": "vfat",
    "MountPoint": "/boot/efi",
    "UUID": "4D4B-2427",
    "Associations": {
        "Media": [
            "BTWA53140407240AGN"
        ],
        "Volumes": [
            "4D4B-2427"
        ]
    },
    "Media": "http://10.20.30.40/Compute/Devices/0123456789/Media/?Volumes=4D4B-2427"
}
```


Table 403: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Volume information in the response body + ETag Header
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

PUT /Compute/Devices/{id}/Volumes/{id}/

PUT /Compute/Devices/{id}/Volumes/{id}/ modifies an attribute or executes an operation for this Volume.

Table 404: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Volumes/4D4B-2427
    --data '{
        "Name": "volume_name",
        "Description": "desc",
    }'
```

Table 405: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Volume information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	A new parameter value already exists (e.g., Name)
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
500	Internal Server Error	System could not process the request

DELETE /Compute/Devices/{id}/Volumes/{id}/

DELETE /Compute/Devices/{id}/Volumes/{id}/ deletes this Volume.

Table 406: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Volumes/4D4B-2427
```

Note: The last remaining "Admin" role account cannot be deleted to prevent loss of communication with the Device.

Table 407: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
500	Internal Server Error	System could not process the request

HEAD /Compute/Devices/{id}/Volumes/{id}/

HEAD /Compute/Devices/{id}/Volumes/{id}/ pings this Volume.

Table 408: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 409: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Volumes/{id}/

OPTIONS /Compute/Devices/{id}/Volumes/{id}/ returns the allowable HTTP Methods list for this resource.

Table 410: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

4.3.3 /Compute/Devices/{id}/Accounts/

The Collection-level methods and attributes of /Compute/Devices/{id}/Accounts/ return information about the Accounts collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 411: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Accounts/{id}/** *(page 248)* for Instance Attribute details

GET /Compute/Devices/{id}/Accounts/

GET /Compute/Devices/{id}/Accounts/ returns the Collection of Accounts.

Table 412: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/ComputeDevices/0123456789/Accounts/
```

Table 413: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Accounts + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Accounts/

POST /Compute/Devices/{id}/Accounts/ creates a new Account.

Table 414: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/Compute/Devices/01234567890/Accounts/
    --data '{
        "UserID": "user_id",
        "BasicAuthentication": {
            "UserPassword":"password"
        },
        "DigestAuthentication": {
            "UserPassword":"password"
        },
        "JWTAuthentication": {
            "UserPassword":"password"
        },
        "Role": {
            "ID": 0 | 1
        }
    }'
```

Table 415: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Account information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Accounts/

HEAD /Compute/Devices/{id}/Accounts/ pings the Accounts collection.

Table 416: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 417: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Accounts/

OPTIONS /Compute/Devices/{id}/Accounts/ returns the allowable HTTP Methods list for this resource.

Table 418: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 419: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.4 /Compute/Devices/{id}/Accounts/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Accounts/{id}/ return information about a specific Account and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 420: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 64 Chars Valid characters include: "a-z A-Z 0-9 _ - "	The Identifier this particular Account
BasicAuthentication. UserPassword SHA1Checksum	String	Mandatory	No	Free-Form 128 Characters	The basic authentication checksum
BasicAuthentication. UserPassword	String	Optional	Yes	Free-Form 128 Characters	The user password used to create the checksum (not shown in the GET response)
DigestAuthentication. UserPassword MD5Checksum	String	Mandatory	No	Free-Form 128 Characters	The digest authentication checksum
DigestAuthentication. UserPassword	String	Optional	Yes	Free-Form 128 Characters	The user password used to create the checksum (not shown in the GET response)
DigestAuthentication. Realm	String	Optional	No	Free-Form 128 Characters	The digest authentication realm
Role. Name	String	Mandatory	No	"Admin"	The Role type string for this Account
Role. ID	Number	Mandatory	No	0	The Role type value for this Account
UserID	String	Mandatory	Yes	Free-Form 64 Chars	The user identifier for this Account

GET /Compute/Devices/{id}/Accounts/{id}/

GET /Compute/Devices/{id}/Accounts/{id}/ returns information regarding a specific Account based on the {id} in the URI address string.

Table 421: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Accounts/admin
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Accounts/admin/",
    "ID": "admin",
    "BasicAuthentication": {
        "UserPasswordSHA1Checksum": "{SHA}0DPiKuNIrrVmD8IUCuw1hQxNqZc="
    },
    "DigestAuthentication": {
        "UserPasswordMD5Checksum": "97ebad852d0dabfd6b71ae26fff61fa3",
        "Realm": "Western Digital Corporation"
    },
    "JWTAuthentication": {
        "UserPasswordMD5Checksum": "b80648545a30e0999892b517127c0e17"
    },
    "Role": {
        "ID": 0,
        "Name": "Admin"
    },
    "UserID": "admin"
}
```


Table 422: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Accounts/{id}/

PUT /Compute/Devices/{id}/Accounts/{id}/ modifies an attribute or executes an operation for this Account.

Table 423: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Accounts/test
    --data '{
        "UserID": "test",
        "BasicAuthentication": {
            "UserPassword": "new_password"
        },
        "Role": {
            "ID": 1
        }
    }'
```

Table 424: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Compute/Devices/{id}/Accounts/{id}/

DELETE /Compute/Devices/{id}/Accounts/{id}/ deletes this Account.

Table 425: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Accounts/test
```

Note: The last remaining "Admin" role account cannot be deleted to prevent loss of communication with the Device.

Table 426: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	In the case of only one account, that last one cannot be deleted
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is Busy, client should try again later

HEAD /Compute/Devices/{id}/Accounts/{id}/

HEAD /Compute/Devices/{id}/Accounts/{id}/ pings this Account.

Table 427: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 428: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Accounts/{id}/

OPTIONS /Compute/Devices/{id}/Accounts/{id}/ returns the allowable HTTP Methods list for this resource.

Table 429: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 430: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.5 /Compute/Devices/{id}/OperatingSystem/

The methods and attributes of /Compute/Devices/{id}/OperatingSystem/ return information about the Operating System (Application) and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 431: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware name
Architecture	String	Optional	No	"x84_64"	The Operating System platform CPU Architecture information
OSDescription	String	Optional	No	Free-Form 128 Chars	The Operating System description information
OSType. Name	String	Mandatory	No	"Winnt" | "Linux"	The Operating System Type value
OSType. ID	Number	Mandatory	No	18 | 36	The Operating System Type value map
SerialNumber	String	Optional	No	Free-Form 128 Chars	The Operating System serial Number
Version	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Application Version
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries

GET /Compute/Devices/{id}/OperatingSystem/

GET /Compute/Devices/{id}/OperatingSystem/ returns the information of the Operating System resource.

Table 432: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/OperatingSystem/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/OperatingSystem/",
    "Name": "Linux",
    "Architecture": "x86_64",
    "OSDescription": "Ubuntu 22.04 (Jammy Jellyfish) 5.15.0-69-generic",
    "SerialNumber": "0123456789",
    "OSType": {
        "ID": 36,
        "Name": "Linux"
    },
    "Version": "Ubuntu 22.04 (Jammy Jellyfish) 5.15.0-69-generic",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    }
}
```

Table 433: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Operating System resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/OperatingSystem/

HEAD /Compute/Devices/{id}/OperatingSystem/ pings the Operating System resource.

Table 434: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 435: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/OperatingSystem/

OPTIONS /Compute/Devices/{id}/OperatingSystem/ returns the allowable HTTP Methods list for this resource.

Table 436: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 437: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.6 /Compute/Devices/{id}/Files/

The Collection-level methods and attributes of /Compute/Devices/{id}/Files/ return information about the Files collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 438: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Compute/Devices/{id}/Files/{id}/** ***(page 261)*** for Instance Attribute details

GET /Compute/Devices/{id}/Files/

GET /Compute/Devices/{id}/Files/ returns the Collection of Files.

Table 439: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Files/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/",
    "Members": [
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/auditlog/",
            "ID": "auditlog",
            "Description": "Audit Log"
        },
        {
```

```
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/syslog/",
            "ID": "syslog",
            "Description": "System Log"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/authlog/",
            "ID": "authlog",
            "Description": "Auth Log"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/kernlog/",
            "ID": "kernlog",
            "Description": "Kernel Log"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/notices/",
            "ID": "notices",
            "Description": "Notices"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/eula/",
            "ID": "eula",
            "Description": "EULA"
        },
        {
            "Self": "http://10.20.30.40/Compute/Devices/0123456789/Files/telemetry/",
            "ID": "telemetry",
            "Description": "Telemetry"
        }
    ],
    "TLSCertAndKey": {},
    "LDAPCert": {}
}
```


Note: If a custom TLS/SSL Certificate and/or an LDAP Client Certificate have been uploaded, these will also be listed in the GET /Files response with links to retrieve them.

Note: The Compute Device connection returns the GET /Telemetry JSON information (with local Media SmartLog info along with any in-band attached Storage Device GET /Telemetry JSON information with Media SmartLog info) and with the above Log Files within the GET /Files/telemetry download bundle.

Table 440: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Files + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Files/

HEAD /Compute/Devices/{id}/Files/ pings this Files collection.

Table 441: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 442: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Files/

OPTIONS /Compute/Devices/{id}/Files/ returns the allowable HTTP Methods list for this resource.

Table 443: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 444: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.7 /Compute/Devices/{id}/Files/{id}/

The Device-level methods and attributes of /Compute/Devices/{id}/Files/{id}/ return information about a specific File and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 445: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier
Name	String	Mandatory	No	Free-Form 256 Chars	The name of the file

GET /Compute/Devices/{id}/Files/{id}/

GET /Compute/Devices/{id}/Files/{id}/ returns information regarding a specific File based on the {id} in the URI address string.

Table 446: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command – Audit Log

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Files/auditlog
```

Example GET Response Body – Audit Log

```
datetime:20231003_140317+00
addr:[10.20.30.40:59634]
type:Response
status code:401
user:
path:/Compute/Devices/0123456789/
```

Table 447: Available Files

OS	File Type	Description
All	auditlog	Returns the current audit log entries as text to the requestor (content-type: text/plain)
	notices	Returns the 3rd Party open-source software entries and their licensing used in this application
	eula	Returns the End-User License Agreement documentation
	cert_pem (204 No Content or 200 OK)	If the User uploaded a custom Cert/Key pair, returns the current SSL certificate entry as text to the requestor (content-type: text/plain)
	key_pem (405 Method Not Allowed)	If the User uploaded a custom Cert/Key pair, SSL key GET not allowed for security of the key (403 Forbidden response)
Linux	syslog	Returns system log
	authlog	Returns authorization log
	kernlog	Returns the kernel log
Windows	syslog	Returns system log
	applicationlog	Returns the list of Open Source usage upon request
	securitylog	Returns the EULA license agreement

Note: Specifying "telemetry" in the GET operation will return a compressed file bundle of several system files and the GET /Telemetry JSON full data. The web client should wait for the HTTP response before trying to request telemetry again. All the other files will return their "text/plain" output to the web client.

Table 448: GET Response Codes

Response Code	Name	Definition
200	OK	OK & File information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Compute/Devices/{id}/Files/{id}/

POST /Compute/Devices/{id}/Files/{id}/ updates the requested file.

Table 449: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
Content-Type: multipart/form-data	Mandatory	The FW image file is of content type "multipart/form-data"

Example POST curl Command – Adding an SSL Certificate & Key

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Files/TLSCertAndKey/
    -H "Content-Type: multipart/form-data"
    -F "cert.pem=@cert.pem"
    -F "key.pem=@key.pem"
```

Example POST Request Body – Adding an SSL Certificate & Key

```
Content-Disposition: form-data; cert.pem="{file_name}"
{file body}
Content-Disposition: form-data; key.pem="{file_name}"
{file body}
```

Note: Both Certificate and Key files MUST be sent together in the same POST request.

Note: Once the Certificate and Key files are uploaded, the API will restart for the certificate/key to take effect. See PUT **/Query/InformationStructure/** *(page 16)* {"RestartAPI" : true}.

Note: The Certificate and Key files cannot be validated, therefore the user must make sure the files are valid before uploading. If they are invalid, the result is the HTTPS service will not operate.

Example POST curl Command – Adding an LDAP Client SSL Certificate

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Compute/Devices/0123456789/Files/LDAPCert/
    -H "Content-Type: multipart/form-data"
    -F "cert.pem=@ldapcert.pem"
```

Example POST Request Body – Adding an LDAP Client SSL Certificate

```
Content-Disposition: form-data; ldapcert.pem="{file_name}"
{file body}
```

Note: The Certificate file cannot be validated, therefore the user must make sure the files are valid before uploading. If they are invalid, the result is the LDAP service will not operate properly.

Table 450: POST Response Codes

Response Code	Name	Definition
201	Created	Created & File information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Files/{id}/

HEAD /Compute/Devices/{id}/Files/{id}/ pings a specific Compute File.

Table 451: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 452: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Files/{id}/

OPTIONS /Compute/Devices/{id}/Files/{id}/ returns the allowable HTTP Methods list for this resource.

Table 453: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 454: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
204	No Content	No Content & "Allow" list of methods in the response header. Some files will return 204 No Content indicating that there is no entity body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.8 /Compute/Devices/{id}/Location/

The methods and attributes of /Compute/Devices/{id}/Location/ return information about the resource Location and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 455: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Country	String	Optional	Yes	Free-Form 64 Chars	The Country name where this Device is located
Territory	String	Optional	Yes	Free-Form 64 Chars	The Territory name where this Device is located
State	String	Optional	Yes	Free-Form 64 Chars	The State name where this Device is located
City	String	Optional	Yes	Free-Form 64 Chars	The City name where this Device is located
Address1	String	Optional	Yes	Free-Form 64 Chars	The first Address information where this Device is located
Address2	String	Optional	Yes	Free-Form 64 Chars	The second Address information where this Device is located as needed
Address3	String	Optional	Yes	Free-Form 64 Chars	The third Address information where this Device is located as needed
PostalCode	String	Optional	Yes	Free-Form 32 Chars	The Postal or Zip Code name where this Device is located
SiteName	String	Optional	Yes	Free-Form 32 Chars	The friendly Site Name
Building	String	Optional	Yes	Free-Form 32 Chars	The Building name or number where this Device is located
Room	String	Optional	Yes	Free-Form 32 Chars	The Room name or number in the Building where this Device is located

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Pod	String	Optional	Yes	Free-Form 32 Chars	The Pod name or number in the Room this Device is located as needed
Row	String	Optional	Yes	Free-Form 32 Chars	The Row name or number in the Room or Pod this Device is located
Rack	String	Optional	Yes	Free-Form 32 Chars	The Rack name or number in the Row this Device is located
Shelf	String	Optional	Yes	Free-Form 32 Chars	The Shelf name or number of the Rack this Device is located
Device	String	Optional	Yes	Free-Form 32 Chars	The Device name or number of the Rack this Device is located
Item	String	Optional	Yes	Free-Form 32 Chars	The Item name indicating the Shelf units or dimension for this Device
GPSCoords	String	Optional	Yes	Decimal Latitude, Decimal Longitude Format	The GPS coordinates of the location of this Device
OtherLocationInfo	String	Optional	Yes	Free-Form 64 Chars	A place-holder for other location information

GET /Compute/Devices/{id}/Location/

GET /Compute/Devices/{id}/Location/ returns the Location information.

Table 456: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Location/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Location/",
    "Country": "USA",
    "Territory": "",
    "State": "Colorado",
    "City": "Colorado Springs",
    "Address1": "9950 Federal Drive",
    "Address2": "Suite 100",
    "Address3":"",
    "PostalCode": "80921",
    "SiteName": "WD COS DataCenter 1",
    "Building": "9950",
    "Room": "159",
    "Pod": "p2",
    "Rack": "r02",
    "Shelf" : "30",
    "Device":"5",
    "Item": "Rack Units",
    "GPSCoords": "38.9838684,-104.8040493",
    "OtherLocationInfo": "This is other info"
}
```

Table 457: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Location information + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Compute/Devices/{id}/Location/

PUT /Compute/Devices/{id}/Location/ modifies an attribute of the Location.

Table 458: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Compute/Devices/0123456789/Location/
    --data '{
        "Address1": "9950 Federal Drive",
        "Address2": "Suite 100",
        "Address3": "North",
        "Building": "9950",
        "City": "Colorado Springs",
        "Country": "USA",
        "Device": "255",
        "GPSCoords": "38.9838643,-104.806244",
        "Item": "Rack Units",
        "OtherLocationInfo": "More Info",
        "Pod": "2",
        "PostalCode": "80921",
        "Rack": "2",
        "Room": "159",
        "Row": "2",
        "Shelf": "24",
        "SiteName": "WDC COS DataCenter",
        "State": "Colorado",
        "Territory": "El Paso"
}'
```

Example PUT Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Location/"
    "Address1": "9950 Federal Drive",
    "Address2": "Suite 100",
    "Address3": "North",
    "Building": "9950",
    "City": "Colorado Springs",
    "Country": "USA",
    "Device": "255",
    "GPSCoords": "38.9838643,-104.806244",
    "Item": "Rack Units",
    "OtherLocationInfo": "More Info",
    "Pod": "2",
    "PostalCode": "80921",
    "Rack": "2",
    "Room": "159",
    "Row": "2",
    "Shelf": "24",
    "SiteName": "WDC COS DataCenter",
    "State": "Colorado",
    "Territory": "El Paso"
}
```

Table 459: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Location information with no response body
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	Resource could not execute the request

DELETE /Compute/Devices/{id}/Location/

DELETE /Compute/Devices/{id}/Location/ deletes Location information.

Table 460: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Compute/Devices/0123456789/Location/
```

Table 461: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn’t exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Compute/Devices/{id}/Location/

HEAD /Compute/Devices/{id}/Location/ pings the Location resource.

Table 462: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 463: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Location/

OPTIONS /Compute/Devices/{id}/Location/ returns the allowable HTTP Methods list for this resource.

Table 464: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 465: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.9 /Compute/Devices/{id}/SystemClock/

The methods and attributes of /Compute/Devices/{id}/SystemClock/ return information about the System Clock and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 466: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Uptime. Duration	String	Mandatory	No		The current uptime value in ISO 8601 format
Uptime. Days	Number	Mandatory	No	0 - n	The current uptime Days since boot
Uptime. Hours	Number	Mandatory	No	0 - 23	The current uptime Hours since boot
Uptime. Minutes	Number	Mandatory	No	0 - 59	The current uptime Minutes since boot
Uptime. Seconds	Number	Mandatory	No	0 - 59	The current uptime Seconds since boot

GET /Compute/Devices/{id}/SystemClock/

GET /Compute/Devices/{id}/SystemClock/ returns the System Clock information.

Table 467: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/SystemClock/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/SystemClock/",
    "Date": "17-Aug-2022",
    "Time": "12:34:13",
    "TimeZone": "(UTC-6.00) MDT",
    "TimeZoneSetting": "MDT",
    "DST": "Enabled",
    "Uptime": {
        "Duration": "P82DT1H26M51S",
        "Days": 82,
        "Hours": 1,
        "Minutes": 26,
        "Seconds": 51
    }
}
```

Table 468: GET Response Codes

Response Code	Name	Definition
200	OK	OK & System Clock information + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned. **Note**: Due to the time changing every second, catching a 304 response will be very rare
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/SystemClock/

HEAD /Compute/Devices/{id}/SystemClock/ pings the System Clock resource.

Table 469: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 470: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/SystemClock/

OPTIONS /Compute/Devices/{id}/SystemClock/ returns the allowable HTTP Methods list for this resource.

Table 471: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 472: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

4.3.10 /Compute/Devices/{id}/Performance/

The methods and attributes of /Compute/Devices/{id}/Performance/ return information about the Performance resource and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 473: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Diskstats[]. Name	String	Mandatory	No	Free-Form 64 Chars	Name of the device
Diskstats[]. UnixNano	Number	Mandatory	No	Integer epoch time	Epoch Nanoseconds
Diskstats[]. BusMaj	Number	Mandatory	No	0 - n	Major Bus Number
Diskstats[]. BusMin	Number	Mandatory	No	0 - n	Minor Bus Number
Diskstats[]. Device	URI String	Mandatory	No	Free-Form URI Format 256 Chars	URI to the Device as Volume or Media
Diskstats[]. DeviceHandle	String	Mandatory	No	Free-Form 64 Chars	The system connection to this Volume
Diskstats[]. ReadsComplete	Number	Mandatory	No	0 - n	Integer counter for number of bytes read
Diskstats[]. ReadsMerged	Number	Mandatory	No	0 - n	Number of requests collapsed into one single read request
Diskstats[]. SectorsRead	Number	Mandatory	No	0 - n	Number of sectors read
Diskstats[]. MillisecondsSpent Reading	Number	Mandatory	No	0 - n	Read time in milliseconds
Diskstats[]. WritesComplete	Number	Mandatory	No	0 - n	Integer counter for number of bytes written
Diskstats[]. WritesMerged	Number	Mandatory	No	0 - n	Number of requests collapsed into one single write request
Diskstats[]. SectorsWritten	Number	Mandatory	No	0 - n	Number of sectors written

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Diskstats[]. MillisecondsSpent Writing	Number	Mandatory	No	0 - n	Write time in milliseconds
Diskstats[]. IOsInProgress	Number	Mandatory	No	0 - n	Current number of IO operations in progress
Diskstats[]. MillisecondsSpent DoingIO	Number	Mandatory	No	0 - n	Time spent doing IO in milliseconds
Diskstats[]. WeightedNumber OfMillisecondsSpent DoingIO	Number	Mandatory	No	0 - n	Field is incremented by the time spent doing I/O multiplied by the number of I/O requests in progress, to provide time weighted by number of active requests

GET /Compute/Devices/{id}/Performance/

GET /Compute/Devices/{id}/Performance/ returns the information of the Performance resource.

Table 474: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Compute/Devices/0123456789/Performance
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Compute/Devices/0123456789/Performance/",
    "DiskStats": [
        {
            "UnixNano": 1697050529417774300,
            "BusMaj": 8,
            "BusMin": 1,
            "Device": "http://10.20.30.40/Compute/Devices/0123456789/Performance/
Volumes/4D4B-2427/",
            "DeviceHandle": "/dev/sda1",
            "ReadsComplete": 1359,
            "ReadsMerged": 8891,
            "SectorsRead": 135260,
            "MillisecondsSpentReading": 427,
            "WritesComplete": 2,
            "WritesMerged": 0,
            "SectorsWritten": 2,
            "MillisecondsSpentWriting": 0,
            "IOsInProgress": 0,
            "MillisecondsSpentDoingIO": 3896,
            "WeightedNumberOfMillisecondsSpentDoingIO": 2894
        }
    ]
}
```


Table 475: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Performance information in the response body
304	Not Modified	
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource

HEAD /Compute/Devices/{id}/Performance/

HEAD /Compute/Devices/{id}/Performance/ pings the Performance resource.

Table 476: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 477: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

OPTIONS /Compute/Devices/{id}/Performance/

OPTIONS /Compute/Devices/{id}/Performance/ returns the allowable HTTP Methods list for this resource.

Table 478: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

4.3.11 /Compute/Devices/{id}/Telemetry/

The methods and attributes of /Compute/Devices/{id}/Telemetry/ return information about the Telemetry resource and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 479: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
LastAccessed	String	Mandatory	No	datetime based on UTC	The UTC time of the request
Members[]	String Array	Mandatory	No	Member Entry	This is the instantaneous, fully expanded metadata from this Device

GET /Compute/Devices/{id}/Telemetry/

GET /Compute/Devices/{id}/Telemetry/ returns the information of the Telemetry resource.

Table 480: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.202.237.61:80/Compute/Devices/78dde6b28cc44503bbdc446d8c85b3e/
Telemetry/
```

Example GET Response Body

Note: In this example, the Compute Device is proxy managing 3 other Storage Devices, so this Telemetry response includes those as well.


```
{
    "Self": "http://10.202.237.61:80/Compute/Devices/78dde6b28cc44503bbdc446d8c85b3e/
Telemetry/",
    "LastAccessed": "26-Jan-2024 12:48:34 UTC",
    "Members": [
        {
            "Self": "http://10.202.237.61:80/Compute/
Devices/78dde6b28cc44503bbdc446d8c85b3e/",
            "SystemType": {...},
            "Name": "cos-r16-s2",
            "ID": "78dde6b28cc44503bbdc446d8c85b3e",
            "OperatingSystem": {...},
            "SerialNumber": "0123456789",
            "Model": "Super Server",
            "Manufacturer": "Supermicro",
            "Capabilities": {...},
            "Status": {...},
            "IPAddresses": {...},
            "Architecture": "x86_64",
            "InformationStructure": {...},
            "Domain": "",
            "Memory": 134951862272,
            "RemainingMemory": 130660556800,
            "Rackmountable": {...},
            "Location": {...},
            "Accounts": {...},
            "Adapters": {...},
            "Files": {...},
            "Media": {...},
            "Performance": {...},
            "Ports": {...},
            "PowerSupplies": {...},
            "Processors": {...},
            "SystemClock": {...},
            "Volumes": {...},
            "Telemetry": {...}
        },
        {...},
        {...},
        {...}
    ]
}
```


Table 481: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Telemetry resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned

Response Code	Name	Definition
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Compute/Devices/{id}/Telemetry/

HEAD /Compute/Devices/{id}/Telemetry/ pings the Telemetry resource.

Table 482: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 483: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Compute/Devices/{id}/Telemetry/

OPTIONS /Compute/Devices/{id}/Telemetry/ returns the allowable HTTP Methods list for this resource.

Table 484: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 485: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

Proxied Node Storage In-Band

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5.1 Physical Devices

The OCAPI's physical resource model includes the overall Compute Proxy Device. The topics in this section provide information for accessing and managing the features of these physical resources.

5.1.1 /Storage/Devices/

The Collection-level methods and attributes of /Storage/Devices/ return information about the Devices collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 486: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/** *(page 288)* for Instance Attribute details

GET /Storage/Devices/

GET /Storage/Devices/ returns the Collection of Devices.

Table 487: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/
```

Table 488: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Devices + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/

HEAD /Storage/Devices/ pings the Devices collection.

Table 489: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 490: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/

OPTIONS /Storage/Devices/ returns the allowable HTTP Methods list for this resource.

Table 491: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 492: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.2 /Storage/Devices/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/ return information about a specific Device and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 493: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 256 Chars	The user-defined Friendly Name for this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
SerialNumber	String	Mandatory	No	Free-Form 64 Chars	The serial number for this Resource
Model	String	Mandatory	No	Free-Form 64 Chars	The model name or number for this Resource
SystemType. Name	String	Mandatory	No	"Storage"	The domain type string indicating the type of Device Resource
SystemType. ID	Number	Mandatory	No	2	The domain type number indicating the type of Device Resource
Manufacturer	String	Mandatory	No	Free-Form 64 Chars	The vendor name for this Resource
OperatingSystem. Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
OperatingSystem. Name	String	Optional	No	Free-Form 128 Chars	The Operating System or Firmware Name
OperatingSystem. OSType. ID	Number	Mandatory	No	59	Indicates dedicated software
OperatingSystem. OSType. Name	String	Mandatory	No	Dedicated	Indicates dedicated software value

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
OperatingSystem. Version	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware Version
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries.
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536 - 65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
Capabilities. Members[]. Name	String	Mandatory	No	"Chassis Manager" | "Chassis Device Enclosure"	The capabilities information value(s) regarding the Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Capabilities. Members[]. ID	Number	Mandatory	No	29 | 31	The capabilities information value map(s) regarding the Resource
Capabilities. Members[]. CapabilityDescription	String	Mandatory	No	Free-form	Entry describing the capability
IPAddresses. Members[]. IPAddress	String	Mandatory	Yes	IPv4 or IPv6 pattern	The IP Address list for this Resource
TotalCapacity	Number	Mandatory	No	Bytes	Total Capacity of the Device in "Bytes"
DurableName	String	Optional	No	WWN or UUID Format	This is the unique ID for discovery in the In-Band Data Path
DeviceHandle	String	Mandatory	No	Handle from the Hardware point-of-view	The handle for communicating to this Device
SlotNumber	Number	Optional	No	1 or 2 for Devices with Controllers (IOMs)	This is the "side" that this connection is going to
PowerState. Name	String	Optional	No	"On" | "Sleep" | "Off"	The current Power State of the Device
PowerState. ID	Number	Optional	No	2 | 4 | 6	The current Power State of the Device value map
EnclosureCover	String	Optional	No	"OK" | "Critical"	The current status of the Enclosure Cover, if supported
Adapters. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Adapters Resource collection
Connectors. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Connectors Resource Collection
Controllers. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Controllers Resource Collection
Expanders. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Expanders Resource Collection
Ports. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Ports Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Fans.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Fan Resource Collection
PowerSupplies.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Power Supplies Resource
Sensors.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Sensor Resource
Location.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Location Resource
Jobs.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Jobs Resource
Files.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Files Resource
ZoneSets.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the ZoneSets Resource
Reboot	Boolean	WriteOnly	Yes	true	Directs this device to reboot
FirmwareActivate	Boolean	WriteOnly	Yes	true	Directs this device to Activate the previously uploaded Device OS image

GET /Storage/Devices/{id}/

GET /Storage/Devices/{id}/ returns information regarding a specific Device based on the {id} in the URI address string.

Table 494: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 495: GET Query Parameters

Parameter	Description
`?AllLog=true`	Downloads the "all_logs_bundle.tar" file, which contains log files from both Controllers (IOMs) as "all_logs_bundle_<device ID>_<timestamp>.tgz"

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/12345
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/12345/",
    "SystemType": {
        "ID": 2,
        "Name": "Storage"
    },
    "Name": "12345",
    "ID": "12345",
    "OperatingSystem": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/OperatingSystem/",
        "Name": "Vendor Firmware",
        "OSType": {
            "ID": 59,
            "Name": "Dedicated"
        },
        "Version": "4.0.86"
    },
    "SerialNumber": "THCLS03819EZ0090",
    "Model": "H4060-J",
    "Manufacturer": "HGST",
    "Capabilities": {
        "Members": [
            {
                "ID": 3,
                "Name": "Storage",
                "CapabilityDescription": "This device is a storage provider."
            },
            {
                "ID": 15,
                "Name": "Block Server",
                "CapabilityDescription": "This device provides block storage."
            },
            {
                "ID": 31,
                "Name": "Storage Device Enclosure",
                "CapabilityDescription": "This device is a storage-based enclosure
 type."
            },
            {
                "ID": 47202,
                "Name": "Disk Media Device",
                "CapabilityDescription": "This device provides disk-based storage
 volumes."
            }
        ]
    },
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
```

```
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "None"
        ]
    },
    "DeviceHandle": "/dev/sg2",
    "DurableName": "0123456789",
    "IPAddresses": {
        "Members": [
            {
                "IPAddress": "10.20.30.41"
            }
        ]
    },
    "PowerState": {
        "ID": 2,
        "Name": "On"
    },
    "SlotNumber": 1,
    "TotalCapacity": 600049880924160,
    "EnclosureCover": "OK",
    "Location": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Location/"
    },
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Adapters": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Adapters/"
    },
    "Connectors": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Connectors/"
    },
    "Controllers": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Controllers/"
    },
    "Expanders": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Expanders/"
    },
    "Fans": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Fans/"
    },
    "Files": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Files/"
    },
    "Jobs": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Jobs/"
    },
    "Media": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Media/"
    },
    "Ports": {
```

```
        "Self": "http://10.20.30.40/Storage/Devices/12345/Ports/"
    },
    "PowerSupplies": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/PowerSupplies/"
    },
    "Sensors": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Sensors/"
    },
    "Slots": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Slots/"
    },
    "ZoneSets": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/ZoneSets/"
    },
    "Telemetry": {
        "Self": "http://10.20.30.40/Storage/Devices/12345/Telemetry/"
    }
}
```


Table 496: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Device information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/

PUT /Storage/Devices/{id}/ modifies an attribute or executes an operation for this Device.

Table 497: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789
    --data '{
        "IndicatorLED": {
            "ID": 2
        }
    }'
```

Table 498: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Device information in the response body + ETag Header
202	Accepted	Accepted & Jobs URI returned in the header to get more information regarding the long-lived operation
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/

HEAD /Storage/Devices/{id}/ pings this Storage Device.

Table 499: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 500: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/

OPTIONS /Storage/Devices/{id}/ returns the allowable HTTP Methods list for this resource.

Note: There may be some attributes returned in the OPTIONS response schema that may not be directly supported by this resource.

Table 501: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	Optional	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 502: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.3 /Storage/Devices/{id}/Controllers/

The Collection-level methods and attributes of /Storage/Devices/{id}/Controllers/ return information about the Controllers collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 503: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Controllers/{id}/** *(page 300)* for Instance Attribute details

GET /Storage/Devices/{id}/Controllers/

GET /Storage/Devices/{id}/Controllers/ returns the Collection of Controllers.

Table 504: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 505: GET Query Parameters

Parameter	Description
`?E6Log=true`	Downloads a bundled, compressed file with one or both Controller Instances E6 Log to the requesting web client. **FILENAME FORMAT**: `e6log_<deviceID>_Controller_Collection_<timestamp>.tgz` **Note:** At the Collection level, both Controller E6 Logs are retrieved, bundled into a compressed TGZ file, and downloaded for convenience.

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Controllers/
```

Table 506: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Controllers + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Controllers/

HEAD /Storage/Devices/{id}/Controllers/ pings the Controllers collection.

Table 507: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 508: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Controllers/

OPTIONS /Storage/Devices/{id}/Controllers/ returns the allowable HTTP Methods list for this resource.

Table 509: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 510: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy, client should try again later

5.1.4 /Storage/Devices/{id}/Controllers/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Controllers/{id}/ return information about a specific Controller and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 511: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars	The device-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
PartNumber	String	Mandatory	No	Free-Form 32 Chars	PartNumber or Model Name
SerialNumber	String	Mandatory	No	Free-Form 32 Chars	Serial Number for the Controller
Version	String	Mandatory	No	Major.Minor.Patch	The current Device OS Version on this Controller
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
IndicatorLED. Name	String	Mandatory	No	"Off "| "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
ProtocolIdentifier. Name	String	Mandatory	No	"SAS Serial SCSI Protocol"	Indicates the communication protocol name
ProtocolIdendifier. ID	Number	Mandatory	No	6	Indicates the communication protocol value ID
RemovalConditions. Name	String	Mandatory	No	"Unknown" | "Not applicable" | "Removable when off" | "Removable when on or off" | "Removable, removal not currently safe"	Indicates removal capabilities
RemovalConditions. ID	Number	Mandatory	No	0 | 2 | 3 | 4 | 10	Indicates removal capabilities value ID
DeviceHandle	String	Mandatory	No	/dev/sgX (Linux) or SCSI0:1,2,3 (Windows)	Handle for communication to the Hardware

GET /Storage/Devices/{id}/Controllers/{id}/

GET /Storage/Devices/{id}/Controllers/{id}/ returns information regarding a specific Controller based on the {id} in the URI address string.

Table 512: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 513: GET Query Parameters

Parameter	Description
`?E6Log=true`	Downloads the binary E6 Log for this particular Controller Instance to the requesting web client **FILENAME FORMAT**: `e6log_<deviceID>_<controllerID>_<timestamp>.bin` **Note:** At the Instance level, the Controller E6 Log is retrieved into a BIN data file, and downloaded for convenience.

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storge/Devices/0123456789/Controllers/IOModuleBFRU/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storge/Devices/0123456789/Controllers/IOModuleBFRU/",
      "ID": "IOModuleBFRU",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "IOModuleBFRU",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "None"
        ]
    },
    "PartNumber": "1EB1049-A2",
```

```
    "SerialNumber": "THCLS00320EG0029",
    "Version": "4.0.86",
    "DurableName": "5000ccab0410a1fc",
    "ProtocolIdentifier": {
        "ID": 6,
        "Name": "SAS Serial SCSI Protocol"
    },
    "RemovalConditions": {
        "ID": 4,
        "Name": "Removable when on or off"
    },
    "DeviceHandle": "/dev/sg3"
}
```


Table 514: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Controller information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Controllers/{id}/

PUT /Storage/Devices/{id}/Controllers/{id}/ modifies an attribute or executes an operation for this Controller.

Table 515: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/Controller/1
    --data '{
        "Reboot": true
    }'
```

Table 516: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Controller information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	A new parameter value already exists (e.g., Name or NQN)
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Controllers/{id}/

HEAD /Storage/Devices/{id}/Controllers/{id}/ pings this Controller.

Table 517: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 518: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Controllers/{id}/

OPTIONS /Storage/Devices/{id}/Controllers/{id}/ returns the allowable HTTP Methods list for this resource.

Table 519: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 520: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.5 /Storage/Devices/{id}/Connectors/

The Collection-level methods and attributes of /Storage/Devices/{id}/Connectors/ return information about the Connectors collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 521: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Connectors/{id}/** *(page 309)* for Instance Attribute details

GET /Storage/Devices/{id}/Connectors/

GET /Storage/Devices/{id}/Connectors/ returns the Collection of Connectors.

Table 522: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Connectors/
```

Table 523: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Connectors + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Connectors/

HEAD /Storage/Devices/{id}/Connectors/ pings the Connectors collection.

Table 524: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 525: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Connectors/

OPTIONS /Storage/Devices/{id}/Connectors/ returns the allowable HTTP Methods list for this resource.

Table 526: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 527: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.6 /Storage/Devices/{id}/Connectors/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Connectors/{id}/ return information about a specific Connectors and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 528: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars	The device-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”.	The current Health Details of the Device
CablePresent	Boolean	Mandatory	No	true | false	Indicates whether or not a cable is plugged into the Connector
ConnectorDescription	String	Mandatory	No	"Mini SAS HD 4x receptacle (see SAS-4 and SFF-8644)[max number of phys: 4]"	Indicates the type of connector
Layout. Name	String	Mandatory	No	"PCI-E x4"	Indicates the type of connector protocol
Layout. ID	Number	Mandatory	No	4	Indicates the type of connector protocol value ID

GET /Storage/Devices/{id}/Connectors/{id}/

GET /Storage/Devices/{id}/Connectors/{id}/ returns the information regarding a specific Connector based on the {id} in the URI address string.

Table 529: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Connectors/CONN_HOST_02/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Connectors/CONN_HOST_02/",
    "ID": "CONN_HOST_02",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
```

```
    },
    "Name": "CONN HOST 02",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "ConnectorDescription": "Mini SAS HD 4x receptacle (see SAS-4 and SFF-8644)[max
 number of phys: 4]",
    "CablePresent": true,
    "Layout": {
        "ID": 4,
        "Name": "PCI-E x4"
    }
}
```


Table 530: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Connector information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Connectors/{id}/

HEAD /Storage/Devices/{id}/Connectors/{id}/ pings this Connector.

Table 531: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 532: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Connectors/{id}/

OPTIONS /Storage/Devices/{id}/Connectors/{id}/ returns the allowable HTTP Methods list for this resource.

Table 533: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 534: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.7 /Storage/Devices/{id}/Expanders/

The Collection-level methods and attributes of /Storage/Devices/{id}/Expanders/ return information about the Expanders collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 535: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Expanders/{id}/** *(page 316)* for Instance Attribute details

GET /Storage/Devices/{id}/Expanders/

GET /Storage/Devices/{id}/Expanders/ returns the Collection of Expanders.

Table 536: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Expanders
```

Table 537: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Expanders + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Expanders/

HEAD /Storage/Devices/{id}/Expanders/ pings this Expander collection.

Table 538: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 539: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Expanders/

OPTIONS /Storage/Devices/{id}/Expanders/ returns the allowable HTTP Methods list for this resource.

Table 540: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 541: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.8 /Storage/Devices/{id}/Expanders/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Expanders/{id}/ return information about a specific Expander and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 542: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars	The device-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	No	4 | 2	The current state value map of the Indicator LED for this Resource
Version	String	Mandatory	No	Major-Minor Format	The current Expander FW image version
DurableName	String	Mandatory	No	WWN or UUID Format	The in-band discovery identifier

GET /Storage/Devices/{id}/Expanders/{id}/

GET /Storage/Devices/{id}/Expanders/{id}/ returns the information regarding a specific Connector based on the {id} in the URI address string.

Table 543: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Expanders/EXP_IOMA_0/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Expanders/EXP_IOMA_0/",
    "ID": "EXP_IOMA_0",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "EXP IOMA 0",
    "Status": {
        "State": {
            "ID": 16,
```

```
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "Version": "3010-007",
    "DurableName": "5000ccab0410a1bd"
}
```


Table 544: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Expander information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Expanders/{id}/

HEAD /Storage/Devices/{id}/Expanders/{id}/ pings this Expander.

Table 545: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 546: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Expanders/{id}/

OPTIONS /Storage/Devices/{id}/Expanders/{id}/ returns the allowable HTTP Methods list for this resource.

Table 547: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 548: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.9 /Storage/Devices/{id}/Ports/

The Collection-level methods and attributes of /Storage/Devices/{id}/Ports/ return information about the Ports collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 549: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Ports/{id}/** ***(page 323)*** for Instance Attribute details

GET /Storage/Devices/{id}/Ports/

GET /Storage/Devices/{id}/Ports/ returns the Collection of Ports.

Table 550: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Ports
```

Table 551: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Ports + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Ports/

HEAD /Storage/Devices/{id}/Ports/ pings the Ports collection.

Table 552: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 553: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Ports/

OPTIONS /Storage/Devices/{id}/Ports/ returns the allowable HTTP Methods list for this resource.

Table 554: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 555: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.10 /Storage/Devices/{id}/Ports/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Ports/{id}/ return information about a specific Port and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 556: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
AddressOrigin.Name	String	Mandatory	Yes	"UNKNOWN" | "OTHER" | "NOT APPLICABLE" | "STATIC" | "DHCP" | "BOOTP" | "IPv4 Link Local" | "DHCPv6" | "IPv6 Link Local" | "DHCPv4"	Information value as to where the address is supplied
AddressOrigin.ID	Number	Mandatory	Yes	0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 10 | 65536	Information value map as to where the address is supplied
IPv4Address	String	Mandatory	Yes	IPv4 4-Octet pattern + CIDR mask	The Network IP Address for this Device; IPv4 type **Note:** 0.0.0.0/8 & 127.0.0.0/8 are reserved and rejected.
IPv4Gateway	String	Optional	Yes	IPv4 4-Octet pattern	The Network IP Gateway address setting for IPv4
IPv6Address	String	Mandatory	Yes	IPv6 Hex pattern + CIDR mask	The Network IP Address for this Device; IPv6 type
IPv6Gateway	String	Optional	Yes	IPv6 Hex pattern	The Network IP Gateway address setting for IPv6
MACAddress	String	Mandatory	No	6-hex digit HW address	Unique MAC Address reported by the local hardware adapter
NetworkType.Name	String	Mandatory	No	"LAN" | "WLAN" | "WAN" | "MAN" | "SAN" | "VLAN" | "Ethernet Network" | "IP Network" | "IPv4 Network" | "IPv6 Network"	The network type value
NetworkType.ID	Number	Mandatory	No	0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8	The network type value map
MTUBytes	Number	Mandatory	Yes	Range: 46 - 9216 (default: 1500)	The current Maximum Transfer Bytes value

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Adapters.Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Link to get the Adapter information hosting this Port

GET /Storage/Devices/{id}/Ports/{id}/

GET /Storage/Devices/{id}/Ports/{id}/ returns the information regarding a specific Port based on the {id} in the URI address string.

Table 557: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command (IPv4)

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Ports/01_0a_f7_95_79_3c/
```

Example GET Response Body (IPv4)


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Ports/01_0a_f7_95_79_3c/",
    "ID": "01_0a_f7_95_79_3c",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "AddressOrigin": {
        "ID": 4,
        "Name": "DHCP"
    },
    "IPv4Address": "10.20.30.40/22",
    "IPGateway": "",
    "MACAddress": "01:0a:f7:95:79:3c",
    "NetworkType": {
        "ID": 8,
        "Name": "IPv4 Network"
    },
```

```
    "MTUBytes": 1500,
    "Adapters": "http://10.20.30.40/Storage/Devices/0123456789/Adapters/?
portid=01_0a_f7_95_79_3c"
    }
}
```

Example GET curl Command (IPv6)

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.40:80/Storage/Devices/0123456789/
Ports/00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64/
```

Example GET Response Body (IPv6)

```
{
    "Self": "http://10.20.40:80/Storage/Devices/0123456789/
Ports/00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64/",
    "ID": "00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64",
    "Status": {
        "State": {
            "ID": 0,
            "Name": "Unknown"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "AddressOrigin": {
        "ID": 7,
        "Name": "DHCPv6"
    },
    "IPv6Address": "fdf4:cfad:91f3:6061::18/64",
    "MACAddress": "00:0c:ca:09:00:3c",
    "NetworkType": {
        "ID": 9,
        "Name": "IPv6 Network"
    },
    "MTUBytes": 1500,
    "Adapters": "http://10.20.30.40:80/Storage/Devices/0123456789/Adapters/?
portid=00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64"
}
```

Table 558: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Port information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Ports/{id}/

PUT /Storage/Devices/{id}/Ports/{id}/ modifies an attribute or executes an operation for this Port.

Table 559: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.41/Storage/Devices/9876543210/Ports/00_0c_ca_09_00_2d_inet
    --data '{
        "AddressOrigin": {
            "ID": 65536
        }
    }'
```

Table 560: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Port information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Storage/Devices/{id}/Ports/{id}/

HEAD /Storage/Devices/{id}/Ports/{id}/ pings this Port.

Table 561: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 562: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Ports/{id}/

OPTIONS /Storage/Devices/{id}/Ports/{id}/ returns the allowable HTTP Methods list for this resource.

Note: There may be some attributes returned in the OPTIONS response schema that may not be directly supported by this resource.

Table 563: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 564: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.11 /Storage/Devices/{id}/PowerSupplies/

The Collection-level methods and attributes of /Storage/Devices/{id}/PowerSupplies/ return information about the Power Supplies collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 565: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/PowerSupplies/{id}/** ***(page 333)*** for Instance Attribute details

GET /Storage/Devices/{id}/PowerSupplies/

GET /Storage/Devices/{id}/PowerSupplies/ returns the Collection of Power Supplies.

Table 566: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies
```

Table 567: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Power Supplies + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/PowerSupplies/

HEAD /Storage/Devices/{id}/PowerSupplies/ pings the Power Supplies collection.

Table 568: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 569: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/PowerSupplies/

OPTIONS /Storage/Devices/{id}/PowerSupplies/ returns the allowable HTTP Methods list for this resource.

Table 570: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 571: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.12 /Storage/Devices/{id}/PowerSupplies/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/PowerSupplies/{id}/ return information about a specific Power Supply and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 572: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Name	String	Mandatory	No	Free-Form 64 Chars	
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Power Supply
PartNumber	String	Mandatory	No	Free-Form 32 Characters	The Part Number for this Power Supply
SerialNumber	String	Mandatory	No	Free-Form 32 Characters	The Serial Number of this Power Supply
Model	String	Optional	No	Free-Form 32 Characters	The model of this Power Supply
Manufacturer	String	Mandatory	No	Free-Form 32 Characters	The manufacturer of this Power Supply
TotalOutputPower	Number	Optional	No	Number in Watts	The total output power rating
BaseUnits. ID	Number	Optional	No	7	BaseUnits value ID (7 = Watts)
BaseUnits. Name	String	Optional	No	Watts	BaseUnits name (Watts)
RateUnits. ID	Number	Optional	No	0	RateUnits ID
RateUnits. Name	String	Optional	No	None	RateUnits Name
UnitModifier	Number	Optional	No	0	Unit modifier (multiplier) based on 10 ^ x where x is the UnitModifier value to convert the TotalOutput Power to Watts

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
RemovalConditions.Name	String	Mandatory	No	"Unknown" | "Not applicable" | "Removable when off" | "Removable when on or off" | "Removable, removal not currently safe"	Indicates removal capabilities
RemovalConditions.ID	Number	Mandatory	No	0 | 2 | 3 | 4 | 10	Indicates removal capabilities value ID

GET /Storage/Devices/{id}/PowerSupplies/{id}/

GET /Storage/Devices/{id}/PowerSupplies/{id}/ returns information regarding a specific Power Supply based on the {id} in the URI address string.

Table 573: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/POWER_SUPPLY_A/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/
POWER_SUPPLY_A/",
    "ID": "PSU_A",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "PSU A",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
```

```
        ],
        "Details": [
            "None"
        ]
    },
    "PartNumber": "DPS-1600AB-12",
    "SerialNumber": "IQHD1939009335",
    "RemovalConditions": {
        "ID": 4,
        "Name": "Removable when on or off"
    },
    "Manufacturer": "DELTA",
    "Model": "DPS-1600AB-12",
    "TotalOutputPower": 1600,
    "BaseUnits": {
        "ID": 7,
        "Name": "Watts"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
    },
    "UnitModifier": 0
}
```


Table 574: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Power Supply information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/PowerSupplies/{id}/

PUT /Storage/Devices/{id}/PowerSupplies/{id}/ modifies an attribute or executes an operation for this Power Supply.

Table 575: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/PowerSupplyAFRU/
    --data '{
        "IndicatorLED": {
            "ID": 2
        }
    }'
```

Table 576: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Power Supply information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/PowerSupplies/{id}/

HEAD /Storage/Devices/{id}/PowerSupplies/{id}/ pings this Power Supply.

Table 577: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 578: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/PowerSupplies/{id}/

OPTIONS /Storage/Devices/{id}/PowerSupplies/{id}/ returns the allowable HTTP Methods list for this resource.

Table 579: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 580: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.13 /Storage/Devices/{id}/Fans/

The Collection-level methods and attributes of /Storage/Devices/{id}/Fans/ return information about the Fans collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 581: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Fans/{id}/** *(page 342)* for Instance Attribute details

GET /Storage/Devices/{id}/Fans/

GET /Storage/Devices/{id}/Fans/ returns the Collection of Fans.

Table 582: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Fans
```

Table 583: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Fans + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Fans/

HEAD /Storage/Devices/{id}/Fans/ pings the Fans collection.

Table 584: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 585: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Fans/

OPTIONS /Storage/Devices/{id}/Fans/ returns the allowable HTTP Methods list for this resource.

Table 586: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 587: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.14 /Storage/Devices/{id}/Fans/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Fans/{id}/ return information about a specific Fan and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 588: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Fan
CurrentSpeed	Number	Mandatory	No	0 - n	The current speed of the fan in RPM
BaseUnits. Name	String	Mandatory	No	"RPM"	The base units value for the speed value
BaseUnits. ID	Number	Mandatory	No	19	The base units value map for the speed value
RateUnits. Name	String	Mandatory	No	"None"	The rate units value for the fan
RateUnits. ID	Number	Mandatory	No	0	The rate units value map for the fan
UnitModifier	Number	Mandatory	No	0	The unit modifier for the fan speed
Removal Conditions. ID	Number	Mandatory	No	0 | 2 | 3 | 4 | 10	The value map of the removal conditions for this FRU
Removal Conditions. Name	String	Mandatory	No	"Unknown" | "Not Applicable" | "Removable when off" | "Removable when on or off" | "Removable, removal not currently safe"	The value of the removal conditions for this FRU
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource

GET /Storage/Devices/{id}/Fans/{id}/

GET /Storage/Devices/{id}/Fans/{id}/ returns the information regarding a specific Fan based on the {id} in the URI address string.

Table 589: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Fans/FAN_ENCL_1/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Fans/FAN_ENCL_1/",
    "ID": "FAN_ENCL_1",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "FAN ENCL 1",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "CurrentSpeed": 2980,
    "BaseUnits": {
        "ID": 19,
        "Name": "RPM"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
    },
    "UnitModifier": 0,
    "RemovalConditions": {
        "ID": 4,
        "Name": "Removable when on or off"
    }
}
```


Table 590: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Fan information in the response body + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Fans/{id}/

PUT /Storage/Devices/{id}/Fans/{id}/ modifies an attribute or executes an operation for this Fan.

Table 591: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/Fans/FAN_ENCL_1/
    --data '{
        "IndicatorLED": {
            "ID": 2
        }
    }'
```

Table 592: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Fan information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed

Response Code	Name	Definition
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Fans/{id}/

HEAD /Storage/Devices/{id}/Fans/{id}/ pings the Fans collection.

Table 593: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 594: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Fans/{id}/

OPTIONS /Storage/Devices/{id}/Fans/{id}/ returns the allowable HTTP Methods list for this resource.

Table 595: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text |
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 596: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.15 /Storage/Devices/{id}/Sensors/

The Collection-level methods and attributes of /Storage/Devices/{id}/Sensors/ return information about the Sensors collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 597: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Sensors/{id}/** *(page 351)* for Instance Attribute details

GET /Storage/Devices/{id}/Sensors/

GET /Storage/Devices/{id}/Sensors/ returns the Collection of Sensors.

Table 598: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Sensors
```

Table 599: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Sensors + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Sensors/

HEAD /Storage/Devices/{id}/Sensors/ pings the the Sensors collection.

Table 600: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 601: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Sensors/

OPTIONS /Storage/Devices/{id}/Sensors/ returns the allowable HTTP Methods list for this resource.

Table 602: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 603: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.16 /Storage/Devices/{id}/Sensors/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Sensors/{id}/ return information about a specific Sensor and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 604: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Name	String	Mandatory	No	Free-Form 256 Chars	The name of this resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the sensor
CurrentReading	Number	Mandatory	No	Signed Integer	The current reading value for the sensor
SensorType. Name	String	Mandatory	No	"Temperature"	The sensor type value
SensorType. ID	Number	Mandatory	No	2	The sensor type value map
BaseUnits. Name	String	Mandatory	No	"Degrees C" | "Degrees F"	The base units value for the sensor current reading
BaseUnits. ID	Number	Mandatory	No	2 | 3	The base units value map for the sensor current reading
RateUnits. Name	String	Mandatory	No	"None"	The rate units value for the sensor current reading
RateUnits. ID	Number	Mandatory	No	0	The rate units value map for the sensor current reading
UnitModifier	Number	Mandatory	No	Signed Integer	The unit modifier for the sensor current reading
HighThreshold Fatal	Number	Optional	No	Signed Integer	The high threshold fatal value usually for temperature
HighThreshold Critical	Number	Optional	No	Signed Integer	The high threshold critical value usually for temperature
LowThreshold Critical	Number	Optional	No	Signed Integer	The low threshold critical value usually for temperature
LowThreshold Fatal	Number	Optional	No	Signed Integer	The low threshold fatal value usually for temperature
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
IndicatorLED.ID	Number	Mandatory	No	4 | 2	The current state value map of the Indicator LED for this Resource

GET /Storage/Devices/{id}/Sensors/{id}/

GET /Storage/Devices/{id}/Sensors/{id}/ returns the information regarding a specific Sensor based on the {id} in the URI address string.

Table 605: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Sensors/TEMP_PSU_A_PRI/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Sensors/TEMP_PSU_A_PRI/",
    "ID": "TEMP_PSU_A_PRI",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "TEMP PSU A PRI",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "SensorType": {
        "ID": 2,
        "Name": "Temperature"
    },
    "CurrentReading": 62,
```

```
    "BaseUnits": {
        "ID": 2,
        "Name": "Degrees C"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
    },
    "UnitModifier": 0,
    "HighThresholdFatal": 110,
    "HighThresholdCritical": 107,
    "LowThresholdCritical": 5,
    "LowThresholdFatal": 1
}
```


Table 606: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Sensor information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Sensors/{id}/

HEAD /Storage/Devices/{id}/Sensors/{id}/ pings a specific Sensor.

Table 607: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 608: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Sensors/{id}/

OPTIONS /Storage/Devices/{id}/Sensors/{id}/ returns the allowable HTTP Methods list for this resource.

Table 609: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 610: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.17 /Storage/Devices/{id}/Slots/

The Collection-level methods and attributes of /Storage/Devices/{id}/Slots/ return information about the Slots collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 611: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Slots/** *(page 356)* for Instance Attribute details

GET /Storage/Devices/{id}/Slots/

GET /Storage/Devices/{id}/Slots/ returns the Collection of Slots.

Table 612: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Slots
```

Table 613: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Slots + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Slots/

HEAD /Storage/Devices/{id}/Slots/ pings the Slots collection.

Table 614: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 615: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Slots/

OPTIONS /Storage/Devices/{id}/Slots/ returns the allowable HTTP Methods list for this resource.

Table 616: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 617: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.18 /Storage/Devices/{id}/Slots/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Slots/{id}/ return information about a specific Slot and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 618: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
PowerState. Name	String	Optional	No	"On" | "Off"	The current Power State of the Slot
PowerState. ID	Number	Optional	No	2 | 6	The current Power State of the Slot value map
Associations	String Array	Mandatory	No	"Media" | "Slots"	The associations among resources based on ID
Medium	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Medium or Drive

GET /Storage/Devices/{id}/Slots/{id}/

GET /Storage/Devices/{id}/Slots/{id}/ returns information about a specific Slot based on the {id} in the URI address string .

Table 619: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Slots/SLOT_000_7PGL4LXG/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Slots/SLOT_000_7PGL4LXG/",
    "ID": "SLOT_000_7PGL4LXG",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "PowerState": {
        "ID": 2,
        "Name": "On"
    },
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "Associations": {
        "Media": [
            "5000cca25120fff2"
        ],
        "Slots": [
            "SLOT_000_7PGL4LXG"
        ]
    },
    "Medium": "http://10.202.239.39:80/Storage/Devices/5000ccab0410a180/
Media/5000cca25120fff2/"
}
```


Table 620: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Slot information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Slots/{id}/

HEAD /Storage/Devices/{id}/Slots/{id}/ pings a specific Slot.

Table 621: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 622: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Slots/{id}/

OPTIONS /Storage/Devices/{id}/Slots/{id}/ returns the allowable HTTP Methods list for this resource.

Table 623: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 624: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.19 /Storage/Devices/{id}/Media/

The Collection-level methods and attributes of /Storage/Devices/{id}/Media/ return information about the Media collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 625: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Media/{id}/** ***(page 367)*** for Instance Attribute details

GET /Storage/Devices/{id}/Media/

GET /Storage/Devices/{id}/Media/ returns the Collection of Media.

Table 626: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 627: GET Query Parameters

Parameter	Description
`?SmartLog=true`	Adds the "SmartLog" JSON object attributes based on the particular protocol type (e.g, ATA, SCSI, and NVMe) for all the Media Instances
`?E6Log=true`	Retrieves a collection bundle file of E6 Logs for all the currently discovered Media that support the E6 Log capability

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Media/
```

Table 628: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Media + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Storage/Devices/{id}/Media/?update=true

POST /Storage/Devices/{id}/Media/?update=true requests a drive firmware update.

Table 629: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Table 630: POST Query Parameters (Required)

Parameter	Description
`?update=true`	Requests a drive firmware update for the list of device handles

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Storage/Devices/0123456789/Media/?update=true
    --form 'FirmwareFileMedia=@"/root/LEGSA9G0.bin"'
    --form 'Media="[{\"DeviceHandle\": \"/dev/sg32\"}]'
```

Table 631: POST Response Codes

Response Code	Name	Definition
200	OK	OK & Medium information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed

Response Code	Name	Definition
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Media/

HEAD /Storage/Devices/{id}/Media/ pings the Media collection.

Table 632: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 633: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Media/

OPTIONS /Storage/Devices/{id}/Media/ returns the allowable HTTP Methods list for this resource.

Table 634: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 635: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.1.20 /Storage/Devices/{id}/Media/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Media/{id}/ return information about a specific Medium and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 636: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars; no spaces allowed	The user-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Capacity	Number	Mandatory	No	Integer Bytes	The total number of raw bytes for this Resource
Model	String	Mandatory	No	Free-Form 32 Chars	The model name or number for this Resource
Manufacturer	String	Mandatory	No	Free-Form 32 Chars	The manufacturer name for this Resource
SerialNumber	String	Mandatory	No	Alphanumeric values	The serial number for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
Associations	String Array	Mandatory	No	"Media" | "Slots"	Lists the associated Resource IDs
Version	String	Mandatory	No	Free-Form URI Format 16 Chars	FW Version of the Medium Device
Protocol. ID	Number	Mandatory	No	4 | 5 | 65536 | 65537	Protocol Interface Type ID
Protocol. Name	String	Mandatory	No	"Serial ATA" | "SAS" | "NVMe"	Protocol Interface Type Name
DiskType. ID	Number	Mandatory	No	2 | 3	Disk Type ID
DiskType. Name	String	Mandatory	No	"HDD" | "SSD"	Disk Type Name
DurableName	String	Mandatory	No	Typically a WWN	Durable Name/ID for the Medium
Slots	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Slots
Media[]. DeviceHandle	String	Mandatory	Yes	Free-Form string	Device Handle to be Firmware updated

GET /Storage/Devices/{id}/Media/{id}/

GET /Storage/Devices/{id}/Media/{id}/ returns information regarding a specific Medium based on the {id} in the URI address string.

Table 637: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 638: GET Query Parameters

Query Parameter	Description
`?SmartLog=true`	Adds the "SmartLog" JSON object attributes based on the particular protocol type (e.g, ATA, SCSI, and NVMe) for all the Media Instances
`?E6Log=true`	Retrieves a collection bundle file of E6 Log Mode Pages for the instances of Media that support the E6 Log capability

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Media/5000cca25103b366/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Media/5000cca25103b366/",
    "ID": "5000cca25103b366",
    "Name": "SLOT 006",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "Associations": {
        "Media": [
            "5000cca25103b366"
        ],
        "Slots": [
            "SLOT_006_7PG212XR"
        ]
    },
    "Capacity": 10000831348736,
    "AlternateDeviceHandle": "",
    "DeviceHandle": "/dev/sg10",
    "Manufacturer": "HGST",
    "Model": "HUH721010AL4204",
    "MultiplePathsDetected": false,
    "Protocol": {
        "ID": 65536,
        "Name": "SAS"
    },
    "DiskType": {
```

```
        "ID": 2,
        "Name": "HDD"
    },
    "Version": "C9G0",
    "SerialNumber": "7PG212XR",
    "DurableName": "5000cca25103b366",
    "Slots": "http://10.202.239.39:80/Storage/Devices/5000ccab0410a180/Slots/
SLOT_006_7PG212XR/"
}
```


Table 639: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Medium information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Media/{id}/

PUT /Storage/Devices/{id}/Media/{id}/ modifies an attribute or executes an operation for this Medium.

Table 640: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Table 641: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Medium information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Media/{id}/

HEAD /Storage/Devices/{id}/Media/{id}/ pings this Medium.

Table 642: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 643: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Media/{id}/

OPTIONS /Storage/Devices/{id}/Media/{id}/ returns the allowable HTTP Methods list for this resource.

Table 644: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 645: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2 Logical Devices

The OCAPI's logical and virtual resources model include those resources that are logical or virtual in nature. The topics in this section provide information for accessing and managing the features of these logical and virtual resources.

5.2.1 /Storage/Devices/{id}/Location/

The methods and attributes of /Storage/Devices/{id}/Location/ return information about the Location resource and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 646: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Country	String	Optional	Yes	Free-Form 64 Chars	The Country name where this Device is located
Territory	String	Optional	Yes	Free-Form 64 Chars	The Territory name where this Device is located
State	String	Optional	Yes	Free-Form 64 Chars	The State name where this Device is located
City	String	Optional	Yes	Free-Form 64 Chars	The City name where this Device is located
Address1	String	Optional	Yes	Free-Form 64 Chars	The first Address information where this Device is located
Address2	String	Optional	Yes	Free-Form 64 Chars	The second Address information where this Device is located as needed

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Address3	String	Optional	Yes	Free-Form 64 Chars	The third Address information where this Device is located as needed
PostalCode	String	Optional	Yes	Free-Form 32 Chars	The Postal or Zip Code name where this Device is located
SiteName	String	Optional	Yes	Free-Form 32 Chars	The friendly Site Name
Building	String	Optional	Yes	Free-Form 32 Chars	The Building name or number where this Device is located
Room	String	Optional	Yes	Free-Form 32 Chars	The Room name or number in the Building where this Device is located
Pod	String	Optional	Yes	Free-Form 32 Chars	The Pod name or number in the Room this Device is located as needed
Row	String	Optional	Yes	Free-Form 32 Chars	The Row name or number in the Room or Pod this Device is located
Rack	String	Optional	Yes	Free-Form 32 Chars	The Rack name or number in the Row this Device is located
Shelf	String	Optional	Yes	Free-Form 32 Chars	The Shelf name or number of the Rack this Device is located
Device	String	Optional	Yes	Free-Form 32 Chars	The Device name or number of the Rack this Device is located
Item	String	Optional	Yes	Free-Form 32 Chars	The Item name indicating the Shelf units or dimension for this Device
GPSCoords	String	Optional	Yes	Decimal Latitude, Decimal Longitude Format	The GPS coordinates of the location of this Device
OtherLocationInfo	String	Optional	Yes	Free-Form 64 Chars	A place-holder for other location information

GET /Storage/Devices/{id}/Location/

GET /Storage/Devices/{id}/Location/ returns the Location information.

Table 647: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Location/
```

Example GET Response Body


```
{
 "Self": "http://10.20.30.40/Storage/Devices/0123456789/Location/",
 "Country": "USA",
 "Territory": "",
 "State": "Colorado",
 "City": "Colorado Springs",
 "Address1": "9950 Federal Drive",
 "Address2": "Suite 100",
 "Address3":"",
 "PostalCode": "80921",
 "SiteName": "WD COS DataCenter 1",
 "Building": "100",
 "Room": "1",
 "Pod": "p2",
 "Rack": "r02",
 "Shelf" : "30",
 "Device":"5",
 "Item": "Rack Units",
 "GPSCoords": "38.9838684,-104.8040493",
 "OtherLocationInfo": "This is other info"
}
```


Table 648: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Location information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Location/

PUT /Storage/Devices/{id}/Location/ modifies an attribute of the Location.

Table 649: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/Location/
    --data '{
        "City": "COS"
    }'
```

Example PUT Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Location/"
    "City": "COS"
}
```

Table 650: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Location information with no response body
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	Resource could not execute the request

DELETE /Storage/Devices/{id}/Location/

DELETE /Storage/Devices/{id}/Location/ deletes all Location information for this resource.

Table 651: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Storage/Devices/0123456789/Location/
```

Table 652: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Storage/Devices/{id}/Location/

HEAD /Storage/Devices/{id}/Location/ pings the Location resource.

Table 653: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 654: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Location/

OPTIONS /Storage/Devices/{id}/Location/ returns the allowable HTTP Methods list for this resource.

Table 655: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 656: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.2 /Storage/Devices/{id}/OperatingSystem/

The methods and attributes of /Storage/Devices/{id}/OperatingSystem/ return information about the Operating System (Application) and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 657: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware name
OSType. Name	String	Mandatory	No	Depends on Operating System Type	The Operating System Type value
OSType. ID	Number	Mandatory	No	Depends on Operating System Type	The Operating System Type value map
Version	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware Version

GET /Storage/Devices/{id}/OperatingSystem/

GET /Storage/Devices/{id}/OperatingSystem/ returns the information of the Operating System resource.

Table 658: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/OperatingSystem/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/OperatingSystem/",
    "Name": "Vendor Firmware",
    "OSType": {
        "ID": 59,
        "Name": "Dedicated"
    },
    "Version": "4.0.13"
}
```

Table 659: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Operating System resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/OperatingSystem/

HEAD /Storage/Devices/{id}/OperatingSystem/ pings the Operating System resource.

Table 660: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 661: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/OperatingSystem/

OPTIONS /Storage/Devices/{id}/OperatingSystem/ returns the allowable HTTP Methods list for this resource.

Table 662: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 663: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.3 /Storage/Devices/{id}/Jobs/

The Collection-level methods and attributes of /Storage/Devices/{id}/Jobs/ return information about the Jobs collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 664: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Jobs/{id}/** ***(page 384)*** for Instance Attribute details

GET /Storage/Devices/{id}/Jobs/

GET /Storage/Devices/{id}/Jobs/ returns the Collection of Jobs.

Table 665: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Jobs/
```

Table 666: GET Response Codes

Response Code	Name	Definition
200	OK	OK & list of Jobs + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Jobs/

HEAD /Storage/Devices/{id}/Jobs/ pings the Jobs collection.

Table 667: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 668: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Jobs/

OPTIONS /Storage/Devices/{id}/Jobs/ returns the allowable HTTP Methods list for this resource.

Table 669: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 670: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.4 /Storage/Devices/{id}/Jobs/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Jobs/{id}/ return information about a specific Job and enable management operations.

HTTP Methods: GET DELETE HEAD OPTIONS

Table 671: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	"FirmwareUpdate_{id}" | "Reboot_{id}" | "FirmwareActivate_{id}"	The Name of this particular Job
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service" | "Inoperative" | "Write Protected" | "Reboot Needed" | "Activate Needed" | "Activate with reboot needed" | "Activate invalidated by reboot"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0-16 | 65537-65542	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then "None"	The current Health Details of the Device
PercentComplete	Number	Mandatory	No	0-100	The percent complete status of the Job

GET /Storage/Devices/{id}/Jobs/{id}/

GET /Storage/Devices/{id}/Jobs/{id}/ returns information regarding a specific Job based on the {id} in the URI address string.

Table 672: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Jobs/FirmwareUpdate_0123456789
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Jobs/",
    "Members": [
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Jobs/
FirmwareUpdate_0123456789/",
            "ID": "FirmwareUpdate_0123456789",
            "StartTime": "09-Oct-2023 23:05:59 UTC",
            "ElapsedTime": "09-Oct-2023 23:08:37 UTC",
            "Owner": "http://10.20.30.40/Storage/Devices/0123456789/",
            "PercentComplete": 100,
            "Status": {
                "State": {
                    "ID": 65540,
                    "Name": "Activate needed"
                },
                "Health": [
                    {
                        "ID": 5,
```

```
                        "Name": "OK"
                    }
                ],
                "Details": [
                    "FW update completed. Waiting for activation."
                ]
            }
        }
    ]
}
```

Table 673: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Job information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Storage/Devices/{id}/Jobs/{id}/

DELETE /Storage/Devices/{id}/Jobs/{id}/ deletes a specific Job.

Note: NOTE: When deleting firmware-related jobs, the recommended order is to delete FirmwareUpdate before FirmwareActivate.

Note: Jobs cannot be deleted when in-progress (i.e. when PercentComplete is less than 100%).

Table 674: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Storage/Devices/0123456789/Jobs/Reboot_0123456789
```

Table 675: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	If job is still in progress
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Jobs/{id}/

HEAD /Storage/Devices/{id}/Jobs/{id}/ pings a specific Job.

Table 676: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 677: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Jobs/{id}/

OPTIONS /Storage/Devices/{id}/Jobs/{id}/ returns the allowable HTTP Methods list for this resource.

Table 678: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 679: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.5 /Storage/Devices/{id}/Files/

The Collection-level methods and attributes of /Storage/Devices/{id}/Files/ return information about the Files collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 680: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Files/{id}/** *(page 392)* for Instance Attribute details

GET /Storage/Devices/{id}/Files/

GET /Storage/Devices/{id}/Files/ returns the Collection of Files.

Table 681: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Files/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/",
    "Members": [
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/telemetry",
            "ID": "telemetry",
            "Description": "Telemetry"
        }
    ]
```

```
}
```

Note: The In-Band Storage device connection returns the GET /Telemetry JSON information (with Media SmartLog info) and any Controller E6 Logs and Media E6 Logs.

Table 682: GET Response Codes

Response Code	Name	Definition
200	OK	OK & list of Files + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Files/

HEAD /Storage/Devices/{id}/Files/ pings the Files collection.

Table 683: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 684: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Files/

OPTIONS /Storage/Devices/{id}/Files/ returns the allowable HTTP Methods list for this resource.

Table 685: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json

Header	Mandatory or Optional	Description
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 686: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.6 /Storage/Devices/{id}/Files/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Files/{id}/ return information about a specific File and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 687: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier
Name	String	Mandatory	No	Free-Form 256 Chars	The name of the file

GET /Storage/Devices/{id}/Files/{id}/

GET /Storage/Devices/{id}/Files/{id}/ returns information regarding a specific File based on the {id} in the URI address string.

Table 688: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Files/telemetry
```

Table 689: GET Response Codes

Response Code	Name	Definition
200	OK	OK & File information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Files/{id}/

HEAD /Storage/Devices/{id}/Files/{id}/ pings a specific File.

Table 690: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 691: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Files/{id}/

OPTIONS /Storage/Devices/{id}/Files/{id}/ returns the allowable HTTP Methods list for this resource.

Table 692: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 693: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
204	No Content	No Content & "Allow" list of methods in the response header. Some files will return 204 No Content indicating that there is no entity body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.7 /Storage/Devices/{id}/ZoneSets/

The Collection-level methods and attributes of /Storage/Devices/{id}/ZoneSets/ return information about the Zone Sets collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 694: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Type. Name	String	Mandatory	No	"Fixed Configuration" | "Custom File Configuration"	Indicates the supported type of ZoneSet configuration of the Device. "Fixed" indicates using native static configurations (e.g., Config 1, 2 or 3 or Disabled). "Custom" indicates supporting "custom file" upload to change the ZoneSet configuration.
Type. ID	Number	Mandatory	No	1 | 2	The ZoneSet configuration type value ID
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/ZoneSets/{id}/** *(page 399)* for Instance Attribute details

GET /Storage/Devices/{id}/ZoneSets/

GET /Storage/Devices/{id}/ZoneSets/ returns the Collection of ZoneSets.

Table 695: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/
```

Table 696: GET Response Codes

Response Code	Name	Definition
200	OK	OK & list of Files + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Storage/Devices/{id}/ZoneSets/

POST /Storage/Devices/{id}/ZoneSets/ uploads a custom configuration file for ZoneSets.

Table 697: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -X POST http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/
    -H "Content-Type: multipart/form-data"
    -F "ZoneSetsFile=@customZoneConfigFile.bin"
```

Table 698: POST Response Codes

Response Code	Name	Definition
201	OK	Created & ZoneSet information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/ZoneSets/

HEAD /Storage/Devices/{id}/ZoneSets/ pings the ZoneSets collection.

Table 699: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 700: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/ZoneSets/

OPTIONS /Storage/Devices/{id}/ZoneSets/ returns the allowable HTTP Methods list for this resource.

Table 701: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 702: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.8 /Storage/Devices/{id}/ZoneSets/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/ZoneSets/{id}/ return information about a specific Zone Set and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 703: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Name	String	Mandatory	No	Free-Form 64 Chars	The Name of this ZoneSet
ID	String	Mandatory	No	Free-Form 64 Chars. Valid characters include: a-z | A-Z | 0-9 | _ | -	The Identifier of this particular ZoneSet
Description	String	Mandatory	No	Free-Form 128 Characters	Description of this ZoneSet
Active	Boolean	Mandatory	Yes	true | false	Determines if the ZoneSet is currently active. When a ZoneSet is set to "true", all other ZoneSets are set to "false" automatically.
MaxNumZone	Number	Mandatory	No	Data102 = 102 Data60 = 60	The maximum number of media members per ZoneSet for this Configuration
MaxNumZone	Number	Mandatory	No	1 | 2 | 3 | 6	The maximum number of zones per ZoneSet for this Configuration
Associations	String Array	Mandatory	No	Device | ZoneSets	Association between the Enclosure and the ZoneSets
Zones[]. ID	String	Mandatory	No	ZoneSet 0 : 1 (1 Zone) ZoneSet 1 : 1 - 6 (6 Zones) ZoneSet 2 : 1 - 3 (3 Zone) ZoneSet 3 : 1 - 2 (2 Zones)	The Zone ID for each of the Zones within a ZoneSet

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Zones[]. NumZoneMe	Number	Mandatory	No	**Data102 Media Members Per Zone** ZoneSet 0 : 102 ZoneSet 1 : 17 ZoneSet 2 : 34 ZoneSet 3 : 51 **Data60 Media Per Zone** ZoneSet 0 : 60 ZoneSet 1 : 10 ZoneSet 2 : 20 ZoneSet 3 : 30	Media Member arrangement. There are 102 Media (Drives) in the Data102 and 60 Media (Drives) in the Data60
Zones[]. Connectors	String Array	Mandatory	No	**Host Connectors Per Zone** ZoneSet 0 : 12 ZoneSet 1 : 2 ZoneSet 2 : 4 ZoneSet 3 : 6	Host Connector zoning arrangement. There are 12 Host Connectors, 6 per Controller (IOM).
Zones[]. Media	String Array	Mandatory	No	**Data102 Media Per Zone** ZoneSet 0 : 1 - 102 ZoneSet 1 : 1 - 17, 18 - 34, 35 - 51, 52- 68 , 69 - 85, 86 - 102 ZoneSet 2 : 1 - 34, 35 - 68, 69 - 102 ZoneSet 3 : 1 - 51, 52 - 102 **Data60 Media Per Zone** ZoneSet 0 : 1 - 60 ZoneSet 1 : 1 - 10, 11 - 20, 21 - 30, 31- 40 , 41 - 50, 51 - 60 ZoneSet 2 : 1 - 20, 21 - 40, 41 - 60 ZoneSet 3 : 1 - 30, 31 - 60	

GET /Storage/Devices/{id}/ZoneSets/{id}/

GET /Storage/Devices/{id}/ZoneSets/{id}/ returns information about a specific ZoneSet based on the {id} in the URI address string.

Table 704: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/1/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/1/",
    "Name": "Zone Set 1",
    "ID": "1",
    "Description": "Configuration 1",
    "Active": false,
    "MaxNumZoneMembers": 17,
    "MaxNumZonesPerZoneSet": 6,
    "Associations": {
        "Device": [
            "5000CCAB04110A00"
        ],
        "ZoneSets": [
            "1"
        ]
    },
    "Zones": [
        {
            "ID": "1",
            "NumZoneMembers": 17,
            "Connectors": [
                "1",
                "7"
            ],
            "Media": [
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
                "14",
                "15",
                "16",
                "17"
            ]
        },
```

```
        {
            "ID": "2",
            "NumZoneMembers": 17,
            "Connectors": [
                "2",
                "8"
            ],
            "Media": [
                "18",
                "19",
                "20",
                "21",
                "22",
                "23",
                "24",
                "25",
                "26",
                "27",
                "28",
                "29",
                "30",
                "31",
                "32",
                "33",
                "34"
            ]
        },
        {
            "ID": "3",
            "NumZoneMembers": 17,
            "Connectors": [
                "3",
                "9"
            ],
            "Media": [
                "35",
                "36",
                "37",
                "38",
                "39",
                "40",
                "41",
                "42",
                "43",
                "44",
                "45",
                "46",
                "47",
                "48",
                "49",
                "50",
                "51"
            ]
        },
        {
            "ID": "4",
            "NumZoneMembers": 17,
            "Connectors": [
                "4",
```

```
                "10"
            ],
            "Media": [
                "52",
                "53",
                "54",
                "55",
                "56",
                "57",
                "58",
                "59",
                "60",
                "61",
                "62",
                "63",
                "64",
                "65",
                "66",
                "67",
                "68"
            ]
        },
        {
            "ID": "5",
            "NumZoneMembers": 17,
            "Connectors": [
                "5",
                "11"
            ],
            "Media": [
                "69",
                "70",
                "71",
                "72",
                "73",
                "74",
                "75",
                "76",
                "77",
                "78",
                "79",
                "80",
                "81",
                "82",
                "83",
                "84",
                "85"
            ]
        },
        {
            "ID": "6",
            "NumZoneMembers": 17,
            "Connectors": [
                "6",
                "12"
            ],
            "Media": [
                "86",
                "87",
```

```
                "88",
                "89",
                "90",
                "91",
                "92",
                "93",
                "94",
                "95",
                "96",
                "97",
                "98",
                "99",
                "100",
                "101",
                "102"
            ]
        }
    ]
}
```

Table 705: GET Response Codes

Response Code	Name	Definition
200	OK	OK & ZoneSet information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/ZoneSets/{id}/

HEAD /Storage/Devices/{id}/ZoneSets/{id}/ pings a specific Zone File.

Table 706: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 707: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/ZoneSets/{id}/

OPTIONS /Storage/Devices/{id}/ZoneSets/{id}/ returns the allowable HTTP Methods list for this resource.

Table 708: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 709: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

5.2.9 /Storage/Devices/{id}/Telemetry/

The methods and attributes of /Storage/Devices/{id}/Telemetry/ return information about the Telemetry resource and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 710: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
LastAccessed	String	Mandatory	No	Datetime based on UTC	The UTC time of the request
Members[]	String Array	Mandatory	No	Member Entry	This is the instantaneous, fully expanded metadata from this Device

GET /Storage/Devices/{id}/Telemetry/

GET /Storage/Devices/{id}/Telemetry/ returns information about the Telemetry resource.

Table 711: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Telemetry/
```

Example GET Response Body


```
{
    "Self": "http://10.202.237.61:80/Compute/Devices/0123456789/Telemetry/",
    "LastAccessed": "26-Jan-2024 12:48:34 UTC",
    "Members": [
        {
            "Self": "http://10.202.237.61:80/Compute/Devices/0123456789/",
            "SystemType": { ... },
            "Name": "cos-r16-s2",
            "ID": "0123456789",
            "OperatingSystem": { ... },
            "SerialNumber": "0123456789",
            "Model": "H4102-J",
            "Manufacturer": "WDC",
            "Capabilities": { ... },
            "Status": { ... },
            "DurableName": "0123456789",
            "DeviceHandle": "/dev/sg2",
            "IPAddresses": { ... },
            "PowerState": { ... },
            "TotalCapacity": 424024316084224,
            "EnclosureCover": "OK",
            "Location": { ... },
            "IndicatorLED": { ... },
            "Adapters": { ... },
            "Connectors": { ... },
            "Controllers": { ... },
            "Expanders": { ... },
            "Fans": { ... },
            "Files": { ... },
            "Jobs": { ... },
            "Media": { ... },
            "Ports": { ... },
            "PowerSupplies": { ... },
            "Sensors": { ... },
            "Slots": { ... },
            "ZoneSets": { ... },
            "Telemetry": { ... }
        }
    ]
}
```


Table 712: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Telemetry resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Telemetry/

HEAD /Storage/Devices/{id}/Telemetry/ pings the Telemetry resource.

Table 713: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 714: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Telemetry/

OPTIONS /Storage/Devices/{id}/Telemetry/ returns the allowable HTTP Methods list for this resource.

Table 715: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 716: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

Proxied Node Storage Out-of-Band

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- Logical Devices..... 489

6.1 Physical Devices

The OCAPI's physical resource model includes the overall Compute Proxy Device. The topics in this section provide information for accessing and managing the features of these physical resources.

6.1.1 /Storage/Devices/

The Collection-level methods and attributes of /Storage/Devices/ return information about the Devices collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 717: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/** *(page 414)* for Instance Attribute details

GET /Storage/Devices/

GET /Storage/Devices/ returns the Collection of Devices.

Table 718: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/
```

Table 719: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Devices + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no response body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/

HEAD /Storage/Devices/ pings the Devices collection.

Table 720: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 721: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/

OPTIONS /Storage/Devices/ returns the allowable HTTP Methods list for this resource.

Table 722: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	Etag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 723: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no response body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.2 /Storage/Devices/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/ return information about a specific Device and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 724: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	Yes	Free-Form 256 Chars	The user-defined Friendly Name for this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Serial Number	String	Mandatory	No	Free-Form 64 Chars	The serial number for this resource
Model	String	Mandatory	No	Free-Form 64 Chars	The model name or number for this Resource
SystemType. Name	String	Mandatory	No	"Storage"	The domain type string indicating the type of Device Resource
SystemType. ID	Number	Mandatory	No	2	The domain type number indicating the type of Device Resource
ChassisPackageType. Name	String	Optional	No	"RackMount"	The type of chassis for this Resource
ChassisPackageType. ID	Number	Optional	No	2	The type of chassis for this Resource value map
Manufacturer	String	Mandatory	No	Free-Form 64 Chars	The vendor name for this Resource
OperatingSystem. Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
OperatingSystem. Name	String	Optional	No	Free-Form 128 Chars	The Operating System or Firmware Name

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
OperatingSystem. OSType. ID	Number	Mandatory	No	59	Indicates dedicated software
OperatingSystem. OSType. Name	String	Mandatory	No	Dedicated	Indicates dedicated software value
OperatingSystem. Version	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware Version
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries.
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536 - 65538	The current health value map of this Resource. There can be 1 or more ID entries.

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”.	The current Health Details of the Device
Capabilities. Members[]. Name	String	Mandatory	No	"Chassis Manager" | "Chassis Device Enclosure"	The capabilities information value(s) regarding the Resource
Capabilities. Members[]. ID	Number	Mandatory	No	29 | 31	The capabilities information value map(s) regarding the Resource
Capabilities. Members[]. CapabilityDescription	String	Mandatory	No	Free-Form	Entry describing the capability
IpAddresses. Members[]. IpAddress	String	Mandatory	Yes	IPv4 or IPv6 pattern	The IP Address list for this Resource
TotalCapacity	Number	Mandatory	No	Bytes	This is the Total Capacity of the Device in "Bytes"
DurableName	String	Optional	No	WWN or UUID Format	This is the unique ID for discovery in the In-Band Data Path
SlotNumber	Number	Optional	No	1 or 2 for Devices with Controllers (IOMs)	This is the "side" that this connection is going to
PowerState. Name	String	Optional	No	"On" | "Sleep" | "Off"	This is the current Power State of the Device
PowerState. ID	Number	Optional	No	2 | 4 | 6	This is the current Power State of the Device value map
Hostname	String	Optional	No	Server Hostname Formatting	This is the Hostname as provided by the DNS based on the IP Address
InformationStructure	String	Optional	No	Free-Form URI	Fully qualified link to the InformationStructure Resource Collection
Accounts. Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Account Resource Collection

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Adapters.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Adapter Resource Collection
Connectors.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Connectors Resource Collection
Controllers.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Controllers Resource Collection
Expanders.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Expanders Resource Collection
Ports.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Ports Resource Collection
Fans.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Fan Resource Collection
PowerSupplies.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to Power Supply Resource Collection
Sensors.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Sensor Resource
Location.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Location Resource
Jobs.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Jobs Resource
Files.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the Files Resource
ZoneSets.Self	String	Mandatory	No	Free-Form URI	Fully qualified link to the ZoneSets Resource
Reboot	Boolean	WriteOnly	Yes	true	Directs this device to reboot
FirmwareActivate	Boolean	WriteOnly	Yes	true	Directs this device to Activate the previously uploaded Device OS image

GET /Storage/Devices/{id}/

GET /Storage/Devices/{id}/ returns information regarding a specific Device based on the {id} in the URI address string.

Table 725: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/?IPAddress=10.20.30.41",
    "SystemType": {
        "ID": 2,
        "Name": "Storage"
    },
    "Name": "0123456789",
    "ID": "0123456789",
    "OperatingSystem": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/OperatingSystem/?
IPAddress=10.20.30.41",
        "Name": "Vendor Firmware",
        "OSType": {
            "ID": 59,
            "Name": "Dedicated"
        },
        "Version": "3.1.11"
    },
    "SerialNumber": "THCLS03819EZ0090",
    "Model": "H4060-J",
    "Manufacturer": "HGST",
    "Capabilities": {
        "Members": [
            {
                "ID": 3,
                "Name": "Storage",
                "CapabilityDescription": "This device is a storage provider."
            },
            {
                "ID": 15,
                "Name": "Block Server",
                "CapabilityDescription": "This device provides block storage."
            },
            {
                "ID": 31,
                "Name": "Storage Device Enclosure",
                "CapabilityDescription": "This device is a storage-based enclosure
 type."
            },
            {
```

```
                "ID": 47202,
                "Name": "Disk Media Device",
                "CapabilityDescription": "This device provides disk-based storage
 volumes."
            }
        ]
    },
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "Ports: IOM-A: Cannot read property 'value' of null",
            "Operating System: IOM-A: Version: Cannot read property 'value' of
 undefined"
        ]
    },
    "Hostname": "oobm-00-0c-ca-08-30-e0.hgst.com.",
    "DurableName": "0123456789",
    "IPAddresses": {
        "Members": [
            {
                "IPAddress": "10.20.30.41"
            }
        ]
    },
    "ChassisPackageType": {
        "ID": 2,
        "Name": "RackMount"
    },
    "InformationStructure": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/InformationStructure/?
IPAddress=10.20.30.41"
    },
    "PowerState": {
        "ID": 2,
        "Name": "On"
    },
    "SlotNumber": 2,
    "TotalCapacity": 600049880924160,
    "Location": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Location/?
IPAddress=10.20.30.41"
    },
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Accounts": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Accounts/?
IPAddress=10.20.30.41"
    },
```

```
    "Adapters": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Adapters/?
IPAddress=10.20.30.41"
    },
    "Connectors": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Connectors/?
IPAddress=10.20.30.41"
    },
    "Controllers": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Controllers/?
IPAddress=10.20.30.41"
    },
    "Expanders": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Expanders/?
IPAddress=10.20.30.41"
    },
    "Fans": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Fans/?
IPAddress=10.20.30.41"
    },
    "Files": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/?
IPAddress=10.20.30.41"
    },
    "Jobs": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Jobs/?
IPAddress=10.20.30.41"
    },
    "Media": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Media/?
IPAddress=10.20.30.41"
    },
    "Ports": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Ports/?
IPAddress=10.20.30.41"
    },
    "PowerSupplies": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/?
IPAddress=10.20.30.41"
    },
    "Sensors": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/Sensors/?
IPAddress=10.20.30.41"
    },
    "ZoneSets": {
        "Self": "http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/?
IPAddress=10.20.30.41"
    }
}
```


Table 726: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Device information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned

Response Code	Name	Definition
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/

PUT /Storage/Devices/{id}/ modifies an attribute or executes an operation for this Device.

Table 727: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789
    --data '{
        "IndicatorLED": {
            "ID": 2
        }
    }'
```

Table 728: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Device information in the response body + ETag Header
	Accepted	Accepted & Jobs URI returned in the header to get more information regarding the long-lived operation
	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/

HEAD /Storage/Devices/{id}/ pings this Storage Device.

Table 729: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 730: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/

OPTIONS /Storage/Devices/{id}/ returns the allowable HTTP Methods list for this resource.

Note: There may be some attributes returned in the OPTIONS response schema that may not be directly supported by this resource.

Table 731: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	Optional	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 732: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.3 /Storage/Devices/{id}/Controllers/

The Collection-level methods and attributes of /Storage/Devices/{id}/Controllers/ return information about the Controllers collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 733: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Controllers/{id}/** *(page 427)* for Instance Attribute details

GET /Storage/Devices/{id}/Controllers/

GET /Storage/Devices/{id}/Controllers/ returns the collection of Controllers.

Table 734: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Controllers/
```

Table 735: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Controllers + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no response body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Controllers/

HEAD /Storage/Devices/{id}/Controllers/ pings the Controllers collection.

Table 736: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 737: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Controllers/

OPTIONS /Storage/Devices/{id}/Controllers/ returns the allowable HTTP Methods list for this resource.

Table 738: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 739: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides information based on the Request Header setting
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is Busy, client should try again later

6.1.4 /Storage/Devices/{id}/Controllers/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Controllers/{id}/ return information about a specific Controller and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 740: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars	The device-defined name of this Resource
Id	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
HostName	String	Mandatory	No	Free-Form 256 Chars	The resource HostName based on the Device Name with suffix of "-ioma" or "-iomb"
PartNumber	String	Mandatory	No	Free-Form 32 Chars	
SerialNumber	String	Mandatory	No	Free-Form 32 Chars	
Model	String	Mandatory	No	Free-Form 32 Chars	
Manufacturer	String	Mandatory	No	Free-Form 32 Chars	
Version	String	Mandatory	No	Major.Minor.Patch	The current Device OS Version on this Controller
Associations	String Array Entries	Mandatory	No	Ports[List of IDs] | Controllers[List of IDs]	Lists the associated Resource IDs along with this Resource ID

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries.
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536 - 65538	The current health value map of this Resource. There can be 1 or more Id entries.
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”.	The current Health Details of the Device
Ports	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Ports
Sensors	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Ports

GET /Storage/Devices/{id}/Controllers/{id}/

GET /Storage/Devices/{id}/Controllers/{id}/ returns information regarding a specific Controller based on the {id} in the URI address string.

Table 741: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storge/Devices/0123456789/Controllers/IOModuleBFRU/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Controllers/IOModuleBFRU/?
IPAddress=10.20.30.41",
    "ID": "IOModuleBFRU",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "IOModuleBFRU",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "Associations": {
        "Controllers": [
            "IOModuleBFRU"
        ],
        "Ports": [
            "00_0c_ca_08_30_e0_inet"
        ]
    },
    "HostName": "oobm-00-0c-ca-08-30-e0",
    "PartNumber": "1EB1049-A2",
    "SerialNumber": "THCLS03719EG006B",
    "Model": "1EB1049-A2",
    "Manufacturer": "HGST",
    "Version": "3.1.11",
    "Ports": "http://10.20.30.40/Storage/Devices/0123456789/Ports/?
Controllers=IOModuleBFRU/\\u0026IPAddress=10.20.30.41/",
    "Sensors": "http://10.20.30.40/Storage/Devices/0123456789/Sensors/?
Controllers=IOModuleBFRU/\\u0026IPAddress=10.20.30.41/"
```

```
}
```

Table 742: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Controller information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Controllers/{id}/

PUT /Storage/Devices/{id}/Controllers/{id}/modifies an attribute or executes an operation for this Controller.

Table 743: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Table 744: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Controller information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	A new parameter value already exists (e.g., Name or NQN)
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Controllers/{id}/

HEAD /Storage/Devices/{id}/Controllers/{id}/ pings this Controller.

Table 745: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 746: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Controllers/{id}/

OPTIONS /Storage/Devices/{id}/Controllers/{id}/ returns the allowable HTTP Methods list for this resource.

Table 747: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 748: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.5 /Storage/Devices/{id}/Connectors/

The Collection-level methods and attributes of /Storage/Devices/{id}/Connectors/ return information about the Connectors collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 749: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Controllers/{id}/** *(page 427)* for Instance Attribute details

GET /Storage/Devices/{id}/Connectors/

GET /Storage/Devices/{id}/Connectors/ returns the collection of Controllers.

Table 750: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Connectors/
```

Table 751: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Controllers + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Connectors/

HEAD /Storage/Devices/{id}/Connectors/ pings the Controllers collection.

Table 752: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 753: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Connectors/

OPTIONS /Storage/Devices/{id}/Connectors/ returns the allowable HTTP Methods list for this resource.

Table 754: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 755: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides information based on the Request Header setting.
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.6 /Storage/Devices/{id}/Connectors/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Connectors/{id}/ return information about a specific Connector and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 756: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars	The device-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”.	The current Health Details of the Device
CablePresent	Boolean	Mandatory	No	true | false	Indicates whether or not a cable is plugged into the Connector
Controllers	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Controllers

GET /Storage/Devices/{id}/Connectors/{id}/

GET /Storage/Devices/{id}/Connectors/{id}/ returns the information regarding a specific Connector based on the {id} in the URI address string.

Table 757: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Connectors/3
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Connectors/3/?
IPAddress=10.20.30.41",
    "ID": "3",
    "Name": "CONN HOST 02",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
```

```
                "Name": "OK"
            }
        ]
    },
    "CablePresent": true,
    "Controllers": "http://10.20.30.40/Storage/Devices/0123456789/Controllers/?
Connectors=3/\\u0026IPAddress=10.20.30.41/"
}
```


Table 758: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Connector information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Connectors/{id}/

HEAD /Storage/Devices/{id}/Connectors/{id}/ pings a specific controller.

Table 759: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 760: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Connectors/{id}/

OPTIONS /Storage/Devices/{id}/Connectors/{id}/ returns the allowable HTTP Methods list for this resource.

Table 761: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 762: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.7 /Storage/Devices/{id}/Expanders/

The Collection-level methods and attributes of /Storage/Devices/{id}/Expanders/ return information about the Expanders collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 763: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Expanders/{id}/** *(page 443)* for Instance Attribute details

GET /Storage/Devices/{id}/Expanders/

GET /Storage/Devices/{id}/Expanders/ returns the collection of Expanders.

Table 764: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Expanders
```

Table 765: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Expanders + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Expanders/

HEAD /Storage/Devices/{id}/Expanders/ pings the Expanders collection.

Table 766: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 767: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Expanders/

OPTIONS /Storage/Devices/{id}/Expanders/ returns the allowable HTTP Methods list for this resource.

Table 768: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 769: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides information based on the Request Header setting
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.8 /Storage/Devices/{id}/Expanders/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Expanders/{id}/ return information about a specific Expander and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 770: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars	The device-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then "None".	The current Health Details of the Device
Version	String	Mandatory	No	Major-Minor Format	The current Expander FW image version
DurableName	String	Mandatory	No	WWN or UUID Format	The in-band discovery identifier
Controllers	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Sensors

GET /Storage/Devices/{id}/Expanders/{id}/

GET /Storage/Devices/{id}/Expanders/{id}/ returns information about a specific expander based on the {id} in the URI address string.

Table 771: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Expanders/PrimaryA
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Expanders/PrimaryA/?
IPAddress=10.20.30.41",
    "ID": "PrimaryA",
    "Name": "EXP IOMA 0",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
```

```
                "Name": "OK"
            }
        ]
    },
    "Version": "4008-020",
    "Controllers": "http://10.20.30.40/Storage/Devices/0123456789/Controllers/?
Expanders=PrimaryA/\\u0026IPAddress=10.20.30.41/",
    "DurableName": "5000CCAB040E7FBD"
}
```


Table 772: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Expander information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Expanders/{id}/

HEAD /Storage/Devices/{id}/Expanders/{id}/ pings a specific expander.

Table 773: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 774: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Expanders/{id}/

OPTIONS /Storage/Devices/{id}/Expanders/{id}/ returns the allowable HTTP Methods list for this resource.

Table 775: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 776: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.9 /Storage/Devices/{id}/Ports/

The Device-level methods and attributes of /Storage/Devices/{id}/Ports/ return information about the Ports collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 777: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Ports/{id}/** ***(page 450)*** for Instance Attribute details

GET /Storage/Devices/{id}/Ports/

GET /Storage/Devices/{id}/Ports/ returns the Collection of Ports.

Table 778: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Ports
```

Table 779: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Ports + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Ports/

HEAD /Storage/Devices/{id}/Ports/ pings the Ports collection.

Table 780: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 781: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Ports/

OPTIONS /Storage/Devices/{id}/Ports/ returns the allowable HTTP Methods list for this resource.

Table 782: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 783: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.10 /Storage/Devices/{id}/Ports/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Ports/{id}/ return information about a specific Port and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 784: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”.	The current Health Details of the Device
AddressOrigin. Name	String	Mandatory	Yes	"UNKNOWN" | "OTHER" | "NOT APPLICABLE" | "STATIC" | "DHCP" | "BOOTP" | "IPv4 Link Local" | "DHCPv6" | "IPv6 Link Local" | "DHCPv4"	Information value as to where the address is supplied
AddressOrigin. ID	Number	Mandatory	Yes	0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 10 | 65536	Information value map as to where the address is supplied
IPv4Address	String	Mandatory	Yes	IPv4 4-Octet pattern + CIDR mask	The Network IP Address for this Device; IPv4 type **Note:** 0.0.0.0/8 & 127.0.0.0/8 are reserved and rejected.
IPv4Gateway	String	Optional	Yes	IPv4 4-Octet pattern	The Network IP Gateway address setting for IPv4
IPv6Address	String	Mandatory	Yes	IPv6 Hex pattern + CIDR mask	The Network IP Address for this Device; IPv6 type
IPv6Gateway	String	Optional	Yes	IPv6 Hex pattern	The Network IP Gateway address setting for IPv6
MACAddress	String	Mandatory	No	6-hex digit HW address	Unique MAC Address reported by the local hardware adapter
NetworkType. Name	String	Mandatory	No	"LAN" | "WLAN" | "WAN" | "MAN" | "SAN" | "VLAN" | "Ethernet Network" | "IP Network" | "IPv4 Network" | "IPv6 Network"	The network type value

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
NetworkType.ID	Number	Mandatory	No	0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8	The network type value map
MTUBytes	Number	Mandatory	Yes	Range: 46 - 9216 (default: 1500)	The current Maximum Transfer Bytes value
Adapters.Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Link to get the Adapter information hosting this Port

GET /Storage/Devices/{id}/Ports/{id}/

GET /Storage/Devices/{id}/Ports/{id}/ returns information regarding a specific Port based on the {id} in the URI address string.

Table 785: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command (IPv4)

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Ports/01_0a_f7_95_79_3c
```

Example GET Response Body (IPv4)

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Ports/01_0a_f7_95_79_3c/",
    "ID": "01_0a_f7_95_79_3c",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "AddressOrigin": {
        "ID": 4,
        "Name": "DHCP"
    },
    "IPv4Address": "10.20.30.40/22",
```

```
    "IPGateway": "",
    "MACAddress": "01:0a:f7:95:79:3c",
    "NetworkType": {
        "ID": 8,
        "Name": "IPv4 Network"
    },
    "MTUBytes": 1500,
    "Adapters": "http://10.20.30.40/Storage/Devices/0123456789/Adapters/?
portid=01_0a_f7_95_79_3c"
}
```

Example GET curl Command (IPv6)

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/
Ports/00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64
```

Example GET Response Body (IPv6)

```
{
    "Self": "http://10.20.40:80/Storage/Devices/0123456789/
Ports/00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64/",
    "ID": "00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64",
    "Status": {
        "State": {
            "ID": 0,
            "Name": "Unknown"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "AddressOrigin": {
        "ID": 7,
        "Name": "DHCPv6"
    },
    "IPv6Address": "fdf4:cfad:91f3:6061::18/64",
    "MACAddress": "00:0c:ca:09:00:3c",
    "NetworkType": {
        "ID": 9,
        "Name": "IPv6 Network"
    },
    "MTUBytes": 1500,
    "Adapters": "http://10.20.30.40:80/Storage/Devices/0123456789/Adapters/?
portid=00_0c_ca_09_00_3c_fdf4_cfad_91f3_6061_18_64"
}
```

Table 786: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Port information in the response body + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Ports/{id}/

PUT /Storage/Devices/{id}/Ports/{id}/ modifies an attribute or executes an operation for this Port.

Table 787: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.41/Storage/Devices/9876543210/Ports/00_0c_ca_09_00_2d_inet
    --data '{
        "AddressOrigin": {
            "ID": 65536
        }
    }'
```

Table 788: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Port information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Storage/Devices/{id}/Ports/{id}/

HEAD /Storage/Devices/{id}/Ports/{id}/ pings a specific Port.

Table 789: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 790: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Ports/{id}/

OPTIONS /Storage/Devices/{id}/Ports/{id}/ returns the allowable HTTP Methods list for this resource.

Table 791: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 792: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later **Note:** There may be some attributes returned in the OPTIONS response schema that may not be directly supported by this resource.

6.1.11 /Storage/Devices/{id}/PowerSupplies/

The Collection-level methods and attributes of /Storage/Devices/{id}/PowerSupplies/ return information about the Power Supplies collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 793: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/PowerSupplies/{id}/** ***(page 460)*** for Instance Attribute details

GET /Storage/Devices/{id}/PowerSupplies/

GET /Storage/Devices/{id}/PowerSupplies/ returns the Collection of Power Supplies.

Table 794: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies
```

Table 795: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Power Supplies + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/PowerSupplies/

HEAD /Storage/Devices/{id}/PowerSupplies/ pings the Power Supplies collection.

Table 796: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 797: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/PowerSupplies/

OPTIONS /Storage/Devices/{id}/PowerSupplies/ returns the allowable HTTP Methods list for this resource.

Table 798: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 799: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.12 /Storage/Devices/{id}/PowerSupplies/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/PowerSupplies/{id}/ return information about a specific Power Supply and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 800: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Name	String	Mandatory	No	Free-Form 64 Chars	
IndicatorLED. Name	String	Mandatory	No	"Off" | "Lit"	The current state value of the Indicator LED for this Resource
IndicatorLED. ID	Number	Mandatory	Yes	4 | 2	The current state value map of the Indicator LED for this Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Power Supply
PartNumber	String	Mandatory	No	Free-Form 32 Characters	The Part Number for this Power Supply
SerialNumber	String	Mandatory	No	Free-Form 32 Characters	The Serial Number of this Power Supply
Model	String	Optional	No	Free-Form 32 Characters	The model of this Power Supply
TotalOutputPower	Number	Optional	No	Number in Watts	The total output power in watts
BaseUnits. ID	Number	Optional	No	7	The Base Units value map
BaseUnits. Name	String	Optional	No	Watts	The Base Units value as Watts
RateUnits. ID	Number	Optional	No	0	The Rate Units value map
RateUnits. Name	String	Optional	No	None	The Rate Units value
UnitModifier	Number	Optional	No	0	The Unit Modifier to determine order of magnitude to multiply the Total Power Output as Base 10 * Unit Modifier
Sensors	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Sensors Link

GET /Storage/Devices/{id}/PowerSupplies/{id}/

GET /Storage/Devices/{id}/PowerSupplies/{id}/ returns information regarding a specific Power Supply based on the {id} in the URI address string.

Table 801: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/
PowerSupplyAFRU/?IPAddress=10.20.30.41
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/
PowerSupplyAFRU/?IPAddress=10.20.30.41",
    "ID": "PowerSupplyAFRU",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "PSU A",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "PartNumber": "DPS-1600AB-12",
    "SerialNumber": "IQHD1937008122",
    "Manufacturer": "DELTA",
    "Model": "DPS-1600AB-12",
    "TotalOutputPower": 1600,
    "BaseUnits": {
        "ID": 7,
        "Name": "Watts"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
```

```
    },
    "UnitModifier": 0,
 "Sensors": "http://10.20.30.40/Storage/Devices/0123456789/Sensors/?
PowerSupplies=PowerSupplyAFRU/?IPAddress=10.20.30.41/",
}
```

Table 802: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Power Supply information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/PowerSupplies/{id}/

PUT /Storage/Devices/{id}/PowerSupplies/{id}/ modifies an attribute or executes an operation for this Power Supply.

Table 803: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/
PowerSupplyAFRU/?IPAddress=10.20.30.41
    --data '{
        "IndicatorLED": {
            "ID": 2
        }
    }'
```

Table 804: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Power Supply information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/PowerSupplies/{id}/

HEAD /Storage/Devices/{id}/PowerSupplies/{id}/ pings this Power Supply.

Table 805: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 806: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/PowerSupplies/{id}/

OPTIONS /Storage/Devices/{id}/PowerSupplies/{id}/ returns the allowable HTTP Methods list for this resource.

Table 807: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text

Header	Mandatory or Optional	Description
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 808: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.13 /Storage/Devices/{id}/Fans/

The Collection-level methods and attributes of /Storage/Devices/{id}/Fans/ return information about the Fans collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 809: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Fans/{id}/** ***(page 469)*** for Instance Attribute details

GET /Storage/Devices/{id}/Fans/

GET /Storage/Devices/{id}/Fans/ returns the Collection of Fans.

Table 810: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Fans
```

Table 811: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Fans + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Fans/

HEAD /Storage/Devices/{id}/Fans/ pings the Fans collection.

Table 812: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 813: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Fans/

OPTIONS /Storage/Devices/{id}/Fans/ returns the allowable HTTP Methods list for this resource.

Table 814: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 815: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.14 /Storage/Devices/{id}/Fans/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Fans/{id}/ return information about a specific Fan and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 816: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Device
CurrentSpeed	Number	Mandatory	No	0 - n	The current speed of the fan in Percent
BaseUnits. Name	String	Mandatory	No	"Percent"	The base units value for the speed value
BaseUnits. ID	Number	Mandatory	No	65	The base units value map for the speed value
RateUnits. Name	String	Mandatory	No	"None"	The rate units value for the fan
RateUnits. ID	Number	Mandatory	No	0	The rate units value map for the fan
UnitModifier	Number	Mandatory	No	0	The unit modifier for the fan speed
Removal Conditions. ID	Number	Mandatory	No	0 | 2 | 3 | 4	The value map of the removal conditions for this FRU
Removal Conditions. Name	String	Mandatory	No	"Unknown" | "Not Applicable" | "Removable when off" | "Removable when on or off"	The value of the removal conditions for this FRU

GET /Storage/Devices/{id}/Fans/{id}/

GET /Storage/Devices/{id}/Fans/{id}/ returns the information regarding a specific Fan based on the {id} in the URI address string.

Table 817: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Fans/FAN_ENCL_0
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Fans/FAN_ENCL_0/",
    "ID": "FAN_ENCL_0",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Status": {
        "State": {
            "ID": 0,
            "Name": "Unknown"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "CurrentSpeed": 45,
    "BaseUnits": {
        "ID": 65,
        "Name": "Percent"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
    },
    "UnitModifier": 0,
    "RemovalConditions": {
        "ID": 4,
        "Name": "Removable when on or off"
    }
}
```


Table 818: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Fan information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Fans/{id}/

HEAD /Storage/Devices/{id}/Fans/{id}/ pings a specific Fan.

Table 819: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 820: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Fans/{id}/

OPTIONS /Storage/Devices/{id}/Fans/{id}/ returns the allowable HTTP Methods list for this resource.

Table 821: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 822: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.15 /Storage/Devices/{id}/Sensors/

The Collection-level methods and attributes of /Storage/Devices/{id}/Sensors/ return information about the Sensors collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 823: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Sensors/{id}/** *(page 476)* for Instance Attribute details

GET /Storage/Devices/{id}/Sensors/

GET /Storage/Devices/{id}/Sensors/ returns the Collection of Sensors.

Table 824: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Sensors
```

Table 825: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Sensors + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Sensors/

HEAD /Storage/Devices/{id}/Sensors/ pings the Sensor collection.

Table 826: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 827: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Sensors/

OPTIONS /Storage/Devices/{id}/Sensors/ returns the allowable HTTP Methods list for this resource.

Table 828: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)		Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 829: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.16 /Storage/Devices/{id}/Sensors/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Sensors/{id}/ return information about a specific Sensor and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 830: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Name	String	Mandatory	No	Free-Form 256 Chars	The name of this resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more ID entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Details[]	String Array	Optional	No	Free-Form 32 Characters per Array entry. If no entry, then “None”	The current Health Details of the Device
CurrentReading	Number	Mandatory	No	Signed Integer	The current reading value for the sensor
SensorType. Name	String	Mandatory	No	"Temperature"	The sensor type value
SensorType. ID	Number	Mandatory	No	2	The sensor type value map
BaseUnits. Name	String	Mandatory	No	"Degrees C" | "Degrees F"	The base units value map for the sensor current reading
BaseUnits. ID	Number	Mandatory	No	2 | 3	The base units value for the sensor current reading
RateUnits. Name	String	Mandatory	No	"None"	The rate units value for the sensor current reading
RateUnits. ID	Number	Mandatory	No	0	The rate units value map for the sensor current reading
UnitModifier	Number	Mandatory	No	Signed Integer	The unit modifier for the sensor current reading
HighThreshold Fatal	Number	Optional	No	Signed Integer	The high threshold fatal value usually for temperature
HighThreshold Critical	Number	Optional	No	Signed Integer	The high threshold critical value usually for temperature
LowThreshold Critical	Number	Optional	No	Signed Integer	The low threshold critical value usually for temperature
LowThreshold Fatal	Number	Optional	No	Signed Integer	The low threshold fatal value usually for temperature
Associations	String Array Entries	Optional	No	Sensors["Self.ID"] | PowerSupplies[List of IDs]	Lists the associated Resource IDs along with this Resource ID
PowerSupplies	String	Optional	No	Free-Form URI	Fully qualified link to the Power Supplies Resource

GET /Storage/Devices/{id}/Sensors/{id}/

GET /Storage/Devices/{id}/Sensors/{id}/ returns the information regarding a specific Sensor based on the {ID} in the URI address string.

Table 831: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Sensors/TEMP_PSU_A_PRI_3_1
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40:80/Storage/Devices/0123456789/Sensors/
TEMP_PSU_A_PRI_3_1/",
    "ID": "TEMP_PSU_A_PRI_3_1",
    "Name": "TEMP PSU A PRI",
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ]
    },
    "Associations": {
        "PowerSupplies": [
            "1"
        ]
    },
    "SensorType": {
        "ID": 2,
        "Name": "Temperature"
    },
    "CurrentReading": 44,
    "BaseUnits": {
        "ID": 2,
        "Name": "Degrees C"
    },
    "RateUnits": {
        "ID": 0,
        "Name": "None"
```

```
    },
    "UnitModifier": 0,
    "HighThresholdFatal": 125,
    "HighThresholdCritical": 120,
    "LowThresholdCritical": 0,
    "LowThresholdFatal": 0,
    "PowerSupplies": "http://10.20.30.40/Storage/Devices/0123456789/PowerSupplies/?
Sensors=TEMP_PSU_A_PRI_3_1"
}
```

Table 832: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Sensor information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Sensors/{id}/

HEAD /Storage/Devices/{id}/Sensors/{id}/ pings a specific Sensor.

Table 833: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 834: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Sensors/{id}/

OPTIONS /Storage/Devices/{id}/Sensors/{id}/ returns the allowable HTTP Methods list for this resource.

Table 835: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 836: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.17 /Storage/Devices/{id}/Media/

The Collection-level methods and attributes of /Storage/Devices/{id}/Media/ return information about the Media collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 837: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Media/{id}/** ***(page 484)*** for Instance Attribute details

GET /Storage/Devices/{id}/Media/

GET /Storage/Devices/{id}/Media/ returns the Collection of Media.

Table 838: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Media
```

Table 839: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Media + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Media/

HEAD /Storage/Devices/{id}/Media/ pings the Media collection.

Table 840: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 841: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Media/

OPTIONS /Storage/Devices/{id}/Media/ returns the allowable HTTP Methods list for this resource.

Table 842: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)		Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 843: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.1.18 /Storage/Devices/{id}/Media/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Media/{id}/ return information about a specific Medium and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 844: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 32 Chars; no spaces allowed	The user-defined name of this Resource
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier that is unique in space and time
Capacity	Number	Mandatory	No	Integer Bytes; on 1024^3 boundary	The total number of raw bytes for this Resource
Model	String	Mandatory	No	Free-Form 32 Chars	The model name for this Resource
Manufacturer	String	Mandatory	No	Free-Form 32 Chars	The manufacturer name for this Resource
SerialNumber	String	Mandatory	No	Alphanumeric values	The serial number for the Resource
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0 - 16	The current state value map of this Resource

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then “None”	The current Health Details of the Device
Associations	String Array Entries	Mandatory	No	Associations[List of IDs]	Lists the associated Resource IDs
Version	String	Mandatory	No	Free-Form URI Format 16 Chars	FW Version of the Medium Device
Protocol. ID	Number	Mandatory	No	4 | 5 | 65536 | 65537	Protocol Interface Type ID
Protocol. Name	String	Mandatory	No	"Serial ATA" | "SAS" | "NVMe"	Protocol Interface Type Name
DiskType. ID	Number	Mandatory	No	2 | 3	Disk Type ID
DiskType. Name	String	Mandatory	No	"HDD" | "SSD"	Disk Type Name
PowerState. ID	Number	Mandatory	Yes	2 | 4 | 6	The Power State identifier indicating the current power state
PowerState. Name	String	Mandatory	No	"On" | "Sleep" | "Off"	The Power State string indicating the current power state
IndicatorLED. ID	Number	Mandatory	Yes	2 | 4	The Indicator LED identifier indicating the current LED state
IndicatorLED. Name	String	Mandatory	No	"Lit" | "Off"	The Indicator LED string indicating the current LED state
DurableName	String	Mandatory	No	Typically a WWN	Durable Name/ID for the Medium

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Sensors	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Link to the participating Sensors

GET /Storage/Devices/{id}/Media/{id}/

GET /Storage/Devices/{id}/Media/{id}/ returns information regarding a specific Medium based on the {ID} in the URI address string.

Table 845: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Media/1/?IPAddress=10.20.30.41
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Media/1/?
IPAddress=10.20.30.41",
    "ID": "1",
    "IndicatorLED": {
        "ID": 4,
        "Name": "Off"
    },
    "Name": "SLOT 000",
    "PowerState": {
        "ID": 2,
        "Name": "On"
    },
    "Status": {
        "State": {
            "ID": 16,
            "Name": "In service"
        },
        "Health": [
            {
                "ID": 5,
                "Name": "OK"
            }
        ],
        "Details": [
            "None"
        ]
```

```
    },
    "Associations": {
        "Device": [
            "0123456789"
        ],
        "Media": [
            "1"
        ]
    },
    "Capacity": 10000831348736,
    "Manufacturer": "HGST",
    "Model": "HUH721010AL4200",
    "Protocol": {
        "ID": 65536,
        "Name": "SAS"
    },
    "DiskType": {
        "ID": 2,
        "Name": "HDD"
    },
    "Version": "A9G0",
    "SerialNumber": "7PGL4LXG",
    "DurableName": "5000CCA25120FFF2",
    "Sensors": "http://10.20.30.40/Storage/Devices/0123456789/Sensors/?Media=1/?
IPAddress=10.20.30.41/"
}
```


Table 846: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Medium information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Media/{id}/

PUT /Storage/Devices/{id}/Media/{id}/ modifies an attribute or executes an operation for this Medium.

Table 847: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Header	Mandatory or Optional	Description
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/Media/1/?IPAddress=10.20.30.41
    --data '{
        "PowerState": {
            "ID": 2
        }
    }'
```

Table 848: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Medium information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource does not exist
412	Precondition Failed	Indicates the “If-Match” ETag check failed
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Media/{id}/

HEAD /Storage/Devices/{id}/Media/{id}/ pings this Medium.

Table 849: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 850: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Media/{id}/

OPTIONS /Storage/Devices/{id}/Media/{id}/ returns the allowable HTTP Methods list for this resource.

Table 851: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)		Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 852: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2 Logical Devices

The OCAPI's logical and virtual resources model include those resources that are logical or virtual in nature. The topics in this section provide information for accessing and managing the features of these logical and virtual resources.

6.2.1 /Storage/Devices/{id}/Location/

The methods and attributes of /Storage/Devices/{id}/Location/ return information about the Location resource and enable management operations.

HTTP Methods: GET PUT DELETE HEAD OPTIONS

Table 853: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Country	String	Optional	Yes	Free-Form 64 Chars	The Country name where this Device is located
Territory	String	Optional	Yes	Free-Form 64 Chars	The Territory name where this Device is located
State	String	Optional	Yes	Free-Form 64 Chars	The State name where this Device is located
City	String	Optional	Yes	Free-Form 64 Chars	The City name where this Device is located
Address1	String	Optional	Yes	Free-Form 64 Chars	The first Address information where this Device is located
Address2	String	Optional	Yes	Free-Form 64 Chars	The second Address information where this Device is located as needed
Address3	String	Optional	Yes	Free-Form 64 Chars	The third Address information where this Device is located as needed
PostalCode	String	Optional	Yes	Free-Form 32 Chars	The Postal or Zip Code name where this Device is located
SiteName	String	Optional	Yes	Free-Form 32 Chars	The friendly Site Name
Building	String	Optional	Yes	Free-Form 32 Chars	The Building name or number where this Device is located
Room	String	Optional	Yes	Free-Form 32 Chars	The Room name or number in the Building where this Device is located

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Pod	String	Optional	Yes	Free-Form 32 Chars	The Pod name or number in the Room this Device is located as needed
Row	String	Optional	Yes	Free-Form 32 Chars	The Row name or number in the Room or Pod this Device is located
Rack	String	Optional	Yes	Free-Form 32 Chars	The Rack name or number in the Row this Device is located
Shelf	String	Optional	Yes	Free-Form 32 Chars	The Shelf name or number of the Rack this Device is located
Device	String	Optional	Yes	Free-Form 32 Chars	The Device name or number of the Rack this Device is located
Item	String	Optional	Yes	Free-Form 32 Chars	The Item name indicating the Shelf units or dimension for this Device
GPSCoords	String	Optional	Yes	Decimal Latitude, Decimal Longitude Format	The GPS coordinates of the location of this Device
OtherLocationInfo	String	Optional	Yes	Free-Form 64 Chars	A place-holder for other location information

GET /Storage/Devices/{id}/Location/

GET /Storage/Devices/{id}/Location/ returns the Location information.

Table 854: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Location/
```

Example GET Response Body

```
{
 "Self": "http://10.20.30.40/Storage/Devices/12345/Location/",
 "Country": "USA",
 "Territory": "",
 "State": "Colorado",
 "City": "Colorado Springs",
 "Address1": "9950 Federal Drive",
 "Address2": "Suite 100",
 "Address3":"",
 "PostalCode": "80921",
 "SiteName": "WD COS DataCenter 1",
 "Building": "100",
 "Room": "1",
 "Pod": "p2",
 "Rack": "r02",
 "Shelf" : "30",
 "Device":"5",
 "Item": "Rack Units",
 "GPSCoords": "38.9838684,-104.8040493",
 "OtherLocationInfo": "This is other info"

}
```

Table 855: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Location information + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Location/

PUT /Storage/Devices/{id}/Location/ modifies an attribute of the Location.

Table 856: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/Location/
    --data '{
        "Address1": "9950 Federal Drive",
        "Address2": "Suite 100",
        "Address3": "North",
        "Building": "9950",
        "City": "Colorado Springs",
        "Country": "USA",
        "Device": "255",
        "GPSCoords": "38.9838643,-104.806244",
        "Item": "Rack Units",
        "OtherLocationInfo": "More Info",
        "Pod": "2",
        "PostalCode": "80921",
        "Rack": "2",
        "Room": "159",
        "Row": "2",
        "Shelf": "24",
        "SiteName": "WDC COS DataCenter",
        "State": "Colorado",
        "Territory": "El Paso"
}'
```

Example PUT Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Location/"
    "Address1": "9950 Federal Drive",
    "Address2": "Suite 100",
    "Address3": "North",
    "Building": "9950",
    "City": "Colorado Springs",
    "Country": "USA",
    "Device": "255",
    "GPSCoords": "38.9838643,-104.806244",
    "Item": "Rack Units",
    "OtherLocationInfo": "More Info",
    "Pod": "2",
    "PostalCode": "80921",
    "Rack": "2",
    "Room": "159",
    "Row": "2",
    "Shelf": "24",
    "SiteName": "WDC COS DataCenter",
    "State": "Colorado",
    "Territory": "El Paso"
}
```

Table 857: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Location information with no response body
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	Resource could not execute the request

DELETE /Storage/Devices/{id}/Location/

DELETE /Storage/Devices/{id}/Location/ deletes Location information.

Table 858: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Storage/Devices/0123456789/Location/
```

Table 859: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful, no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later
500	Internal Server Error	System could not process the request

HEAD /Storage/Devices/{id}/Location/

HEAD /Storage/Devices/{id}/Location/ pings the Location resource.

Table 860: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 861: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Location/

OPTIONS /Storage/Devices/{id}/Location/ returns the allowable HTTP Methods list for this resource.

Table 862: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
Documentation	Optional	Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 863: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.2 /Storage/Devices/{id}/InformationStructure/

The methods and attributes of /Storage/Devices/{id}/InformationStructure/ return information about the Information Structure resource and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 864: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
SystemQuery	String	Mandatory	No	Free-Form URI Format with IP plus "/System/Query"	Link to ask this Device to discover other devices
InformationStructure.Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Link to use to display or modify the InformationStructure
InformationStructure.Name	String	Mandatory	No	Free-Form 64 Chars	The API name
InformationStructure.HTTPPort	Number	Mandatory	Yes	80, or 1024 - 65535 range	Port number for HTTP listener; default = 80
InformationStructure.HTTPSPort	Number	Mandatory	Yes	443, or 1024 - 65535 range	Port number for HTTPS listener; default = 443
InformationStructure.ID	String	Mandatory	No	Free-Form 256 Chars	The query identifier represents the main device ID
InformationStrucuture.AuthenticationType.ID	Number	Mandatory	Yes	0 = Basic	The current authentication type ID
InformationStrucuture.AuthenticationType.Name	String	Mandatory	No	Basic	The current authentication type Name
InformationStructure.StructureDescription	String	Mandatory	No	Free-Form 256 Chars	The API Schema Description
InformationStructure.OwningOrganization	String	Mandatory	No	Free-Form 64 Chars	The API Schema Owner
InformationStructure.Status	String	Mandatory	No	Free-Form 64 Chars	The API Schema Version Release Status

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
InformationStructure.URI	String	Mandatory	No	Free-Form URI Format	The API Schema URI starting point
InformationStructure.Version	String	Mandatory	No	Major.Minor.Release Format	The API Schema Version Number

GET /Storage/Devices/{id}/InformationStructure/

GET /Storage/Devices/{id}/InformationStructure/ returns the Information Structure.

Table 865: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/InformationStructure/?
IPAddress=10.20.30.41
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/InformationStructure/?
IPAddress=10.20.30.41",
    "AuthenticationType": {
        "ID": 0,
        "Name": "Basic"
    },
    "HTTPPort": 0,
    "HTTPSPort": 443,
    "Name": "Redfish® API converted to Open Composable API (OCAPI)",
    "OwningOrganization": "DMTF for Redfish® API; OCP for OCAPI",
    "Status": "Released",
    "StructureDescription": "Converts from DMTF's Redfish® to the OCP/WDC Open
 Composable API.",
    "URI": "/Query/ for OCAPI (/redfish/v1 for Redfish® API)",
    "Version": "v1.2.0, OCAPI: 1.5.30-277"
}
```

Table 866: GET Response Codes

Response Code	Name	Definition
200	OK	OK & InformationStructure resource information in the response body

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/InformationStructure/

HEAD /Storage/Devices/{id}/InformationStructure/ pings the Information Structure resource.

Table 867: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body + Header
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/InformationStructure/

OPTIONS /Storage/Devices/{id}/InformationStructure/ returns the allowable HTTP Methods list for this resource.

Table 868: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 869: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.3 /Storage/Devices/{id}/Accounts/

The Collection-level methods and attributes of /Storage/Devices/{id}/Accounts/ return information about the Accounts collection and enable management operations.

HTTP Methods: GET POST HEAD OPTIONS

Table 870: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Accounts/{id}** ***(page 502)*** for Instance Attribute details.

GET /Storage/Devices/{id}/Accounts/

GET /Storage/Devices/{id}/Accounts/ returns the Collection of Accounts.

Table 871: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Accounts/?
IPAddress=10.20.30.41
```

Table 872: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Accounts + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

POST /Storage/Devices/{id}/Accounts/

POST /Storage/Devices/{id}/Accounts/ creates a new Account.

Table 873: POST Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Example POST curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -X POST http://10.20.30.40/Storage/Devices/01234567890/Accounts/
    --data '{
        "UserID":"test",
        "BasicAuthentication": {
            "UserPassword":"password"
        },
        "Role": {
            "ID": 0
        }
    }'
```

Table 874: POST Response Codes

Response Code	Name	Definition
201	Created	Created & Account information in the response body + ETag Header + new URI in Location Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
409	Conflict	A new parameter value already exists (e.g., Name)
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Accounts/

HEAD /Storage/Devices/{id}/Accounts/ pings this Account Collection.

Table 875: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 876: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Accounts/

OPTIONS /Storage/Devices/{id}/Accounts/ returns the allowable HTTP Methods list for this resource.

Table 877: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text

Header	Mandatory or Optional	Description
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 878: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.4 /Storage/Devices/{id}/Accounts/{id}

The Device-level methods and attributes of /Storage/Devices/{id}/Accounts/{id} return information about a specific Account and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 879: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 64 Chars. Valid characters include: a-z | A-Z | 0-9 | _ | -	The Identifier this particular Account
BasicAuthentication. UserPasswordSHA1Chec	String	Mandatory	No	Free-Form 128 Characters	The basic authentication checksum
BasicAuthentication. UserPassword	String	Optional	Yes	Free-Form 128 Characters	The user password used to create the checksum (not shown in the GET response)
Role. Name	String	Mandatory	No	"Admin"	The Role type string for this Account

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Role. ID	Number	Mandatory	No	0	The Role type value for this Account
UserID	String	Mandatory	Yes	Free-Form 64 Chars	The user identifier for this Account

GET /Storage/Devices/{id}/Accounts/{id}

GET /Storage/Devices/{id}/Accounts/{id} returns the information regarding a specific Account based on the {id} in the URI address string.

Table 880: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Accounts/admin/?
IPAddress=10.20.30.41
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Accounts/admin/?
IPAddress=10.20.30.41",
    "ID": "admin",
    "BasicAuthentication": {
        "UserPasswordSHA1Checksum": "{SHA}UnknownForThisDevice"
    },
    "Role": {
        "ID": 0,
        "Name": "Admin"
    },
    "UserID": "admin",
    "RemoteID": "1"
}
```

Table 881: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/Accounts/{id}

PUT /Storage/Devices/{id}/Accounts/{id} modifies an attribute or executes an operation for this Account.

Table 882: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/Accounts/test/?
IPAddress=10.20.30.41
    --data '{
        "UserID":"test",
        "BasicAuthentication": {
            "UserPassword":"new_password"
        },
        "Role": {
            "ID": 1
        }
    }'
```

Table 883: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & Account information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	No Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed

Response Code	Name	Definition
428	Precondition Required	Indicates the need for an “If-Match” conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Accounts/{id}

HEAD /Storage/Devices/{id}/Accounts/{id} pings this Account.

Table 884: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 885: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Accounts/{id}

OPTIONS /Storage/Devices/{id}/Accounts/{id} returns the allowable HTTP Methods list for this resource.

Table 886: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 887: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.5 /Storage/Devices/{id}/OperatingSystem/

The methods and attributes of /Storage/Devices/{id}/OperatingSystem/ return information about the Operating System (Application) and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 888: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
Name	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Firmware name
OSType. Name	String	Mandatory	No	"Winnt" | "Linux"	The Operating System Type value
OSType. ID	Number	Mandatory	No	18 | 36	The Operating System Type value map
Version	String	Mandatory	No	Free-Form 128 Chars	The Operating System or Application Version

GET /Storage/Devices/{id}/OperatingSystem/

GET /Storage/Devices/{id}/OperatingSystem/ returns the information of the Operating System resource.

Table 889: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/OperatingSystem/
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/OperatingSystem/",
    "Name": "Vendor Firmware",
    "OSType": {
        "ID": 59,
        "Name": "Dedicated"
    },
    "Version": "4.0.13"
}
```

Table 890: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Operating System resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/OperatingSystem/

HEAD /Storage/Devices/{id}/OperatingSystem/ pings the Operating System resource.

Table 891: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 892: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/OperatingSystem/

OPTIONS /Storage/Devices/{id}/OperatingSystem/ returns the allowable HTTP Methods list for this resource.

Table 893: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 894: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.6 /Storage/Devices/{id}/Jobs/

The Collection-level methods and attributes of /Storage/Devices/{id}/Jobs/ return information about the Jobs collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 895: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Jobs/{id}/** ***(page 513)*** for Instance Attribute details

GET /Storage/Devices/{id}/Jobs/

GET /Storage/Devices/{id}/Jobs/ returns the Collection of Jobs.

Table 896: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Jobs/
```

Table 897: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Jobs + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Jobs/

HEAD /Storage/Devices/{id}/Jobs/ pings the Jobs collection.

Table 898: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 899: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Jobs/

OPTIONS /Storage/Devices/{id}/Jobs/ returns the allowable HTTP Methods list for this resource.

Table 900: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 901: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the "Allow" list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource collection does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.7 /Storage/Devices/{id}/Jobs/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Jobs/{id}/ return information about a specific Job and enable management operations.

HTTP Methods: GET HEAD DELETE OPTIONS

Table 902: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	"FactoryReset"	The Name of this particular Job
Status. State. Name	String	Mandatory	No	"Unknown" | "Not available" | "Servicing" | "Starting" | "Stopping" | "Stopped" | "Aborted" | "Dormant" | "Completed" | "Migrating" | "Emigrating" | "Immigrating" | "Snapshotting" | "Shutting down" | "In test" | "Transitioning" | "In service" | "Inoperative" | "Write Protected" | "Reboot Needed" | "Activate Needed" | "Activate with reboot needed" | "Activate invalidated by reboot"	The current state value of this Resource
Status. State. ID	Number	Mandatory	No	0-16 | 65537-65542	The current state value map of this Resource
Status. Health[]. Name	String (1..*)	Mandatory	No	"Unknown" | "OK" | "Degraded/Warning" | "Minor failure" | "Major failure" | "Critical failure" | "Non-recoverable error" | "Not installed" | "Not available" | "No access allowed"	The current health value of this Resource. There can be 1 or more Name entries

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Status. Health[]. ID	Number (1..*)	Mandatory	No	0 | 5 | 10 | 15 | 20 | 25 | 30 | 65536-65538	The current health value map of this Resource. There can be 1 or more Id entries
Status. Details[]	String Array	Optional	No	Free-Form Array entry. If no entry, then "None"	The current Health Details of the Device
PercentComplete	Number	Mandatory	No	0-100	The percent complete status of the Job

GET /Storage/Devices/{id}/Jobs/{id}/

GET /Storage/Devices/{id}/Jobs/{id}/ returns information regarding a specific Job based on the {id} in the URI address string.

Table 903: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET /Storage/Devices/0123456789/Jobs/FirmwareUpdate_0123456789
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Jobs/?
IPAddress=10.20.30.41",
    "Members": [
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Jobs/
FirmwareUpdate_0123456789/?IPAddress=10.20.30.41",
            "ID": "FirmwareUpdate_0123456789",
            "StartTime": "09-Oct-2023 23:05:59 UTC",
            "ElapsedTime": "09-Oct-2023 23:08:37 UTC",
            "Owner": "http://10.20.30.40/Storage/Devices/0123456789/?
IPAddress=10.20.30.41",
            "PercentComplete": 100,
            "Status": {
                "State": {
                    "ID": 65540,
                    "Name": "Activate needed"
                },
                "Health": [
```

```
                    {
                        "ID": 5,
                        "Name": "OK"
                    }
                ],
                "Details": [
                    "FW update completed. Waiting for activation."
                ]
            }
        }
    ]
}
```


Table 904: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Job information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Jobs/{id}/

HEAD /Storage/Devices/{id}/Jobs/{id}/ pings this specific Job.

Table 905: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 906: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

DELETE /Storage/Devices/{id}/Jobs/{id}/

DELETE /Storage/Devices/{id}/Jobs/{id}/ deletes a specific Job.

Note: When deleting firmware-related jobs, the recommended order is to delete FirmwareUpdate before FirmwareActivate.

Note: Jobs cannot be deleted when in-progress (i.e. when PercentComplete is less than 100%).

Table 907: DELETE Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example DELETE curl Command

Note: The `If-Match` conditional in this command requires the **ETag** from the GET Response Header.

```
curl
    -i
    -u admin:admin
    -H "If-Match: 3677651d2011428f32120a029b015e81"
    -X DELETE http://10.20.30.40/Storage/Devices/0123456789/Jobs/Reboot_0123456789/?
IPAddress=10.20.30.41
```

Table 908: DELETE Response Codes

Response Code	Name	Definition
204	No Content	Delete successful; no content in the response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
409	Conflict	If job is still in progress
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Jobs/{id}/

OPTIONS /Storage/Devices/{id}/Jobs/{id}/ returns the allowable HTTP Methods list for this resource.

Table 909: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 910: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.8 /Storage/Devices/{id}/Files/

The Collection-level methods and attributes of /Storage/Devices/{id}/Files/ return information about the Files collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 911: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Optional	No	Resource Collection (0 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/Files/{id}/** *(page 521)* for Instance Attribute details

GET /Storage/Devices/{id}/Files/

GET /Storage/Devices/{id}/Files/ returns the Collection of Files.

Table 912: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET /Storage/Devices/0123456789/Files/?IPAddress=10.20.30.41
```

Example GET Response Body

```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/?
IPAddress=10.20.30.41",
    "Members": [
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/datamgr.log?
IPAddress=10.20.30.41",
            "ID": "datamgr.log",
            "Description": "Data Manager Log"
```

```
        },
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/
sep_interface.log?IPAddress=10.20.30.41",
            "ID": "sep_interface.log",
            "Description": "SEP Interface Log"
        },
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/messages?
IPAddress=10.20.30.41",
            "ID": "messages",
            "Description": "Messages"
        },
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Files/telemetry?
IPAddress=10.20.30.41",
            "ID": "telemetry",
            "Description": "Telemetry"
        }
    ]
}
```


Table 913: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of Files + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Files/

HEAD /Storage/Devices/{id}/Files/ pings this Files collection.

Table 914: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 915: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing

Response Code	Name	Definition
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Files/

OPTIONS /Storage/Devices/{id}/Files/ returns the allowable HTTP Methods list for this resource.

Table 916: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 917: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.9 /Storage/Devices/{id}/Files/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/Files/{id}/ return information about a specific File and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 918: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
ID	String	Mandatory	No	Free-Form 256 Chars	The resource identifier
Name	String	Mandatory	No	Free-Form 256 Chars	The name of the file

GET /Storage/Devices/{id}/Files/{id}/

GET /Storage/Devices/{id}/Files/{id}/ returns information regarding a specific File based on the {id} in the URI address string.

Table 919: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

File Options:

- `datamgr.log` – Returns API audit log of HTTP activities
- `sep_interface.log` – Returns current FW build information
- `messages` – Returns customer-facing device log
- `telemetry` – Returns the 3 files above plus the GET /Telemetry JSON full information in a compressed bundle file

Note: Specifying `telemetry` in the GET operation will return a compressed file bundle of several system files and the GET /Telemetry JSON full data. The web client should wait for the HTTP response before trying to request telemetry again. All the other files will return their "text/plain" output to the web client.

Example GET curl Command – Audit Log

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Files/datamgr.log?
IPAddress=10.20.30.41
```

Example GET Response Body

```
1970-01-01 00:00:19,363386 INFO :  int main(int, char**) Starting the Storage Server
 Data Manager.
1970-01-01 00:00:19,363673 TRACE:  bool DataManager::run()
1970-01-01 00:00:19,432442 TRACE:  virtual bool TableCommon::create()
1970-01-01 00:00:19,444341 TRACE:  virtual bool TableFlags::create()
1970-01-01 00:00:19,450680 TRACE:  virtual bool TableDescriptors::create()
1970-01-01 00:00:19,458991 TRACE:  virtual bool TableSensors::create()
1970-01-01 00:00:19,469541 TRACE:  virtual bool TableFans::create()
1970-01-01 00:00:19,476196 TRACE:  virtual bool TableSysInfo::create()
1970-01-01 00:00:19,483058 TRACE:  bool SysInfo::init()
1970-01-01 00:00:19,607778 TRACE:  bool SysInfo::init()
1970-01-01 00:00:19,774472 INFO :  bool DataManager::run() Completed initial database
 build.
1970-01-01 00:00:19,788992 INFO :  bool DataSubscriber::subscribe(std::__cxx11::string,
 std::__cxx11::string, DataSubscriber::CBFuncRx) Subscriber connecting to tcp://
localhost:8888
1970-01-01 00:00:19,790512 TRACE:  void DataSubscriber::subscriberThread()
1970-01-01 00:00:19,790648 INFO :  void DataSubscriber::subscriberThread() Subscriber
 thread running
1970-01-01 00:00:19,790705 TRACE:  void DataSubscriber::sendReady()
1970-01-01 00:00:40,985707 INFO :  void DataSubscriber::sendReady() Received ready
 confirmation.
1970-01-01 00:11:34,915634 INFO :  void signalHandler(int) Received signal 15 signal
 name is: Terminated
1970-01-01 00:00:21,143899 INFO :  int main(int, char**) Starting the Storage Server
 Data Manager.
1970-01-01 00:00:21,144819 TRACE:  bool DataManager::run()
1970-01-01 00:00:21,212564 TRACE:  virtual bool TableCommon::create()
1970-01-01 00:00:21,223864 TRACE:  virtual bool TableFlags::create()
1970-01-01 00:00:21,230221 TRACE:  virtual bool TableDescriptors::create()
1970-01-01 00:00:21,237367 TRACE:  virtual bool TableSensors::create()
1970-01-01 00:00:21,247608 TRACE:  virtual bool TableFans::create()
1970-01-01 00:00:21,254900 TRACE:  virtual bool TableSysInfo::create()
1970-01-01 00:00:21,262911 TRACE:  bool SysInfo::init()
1970-01-01 00:00:21,387893 TRACE:  bool SysInfo::init()
1970-01-01 00:00:21,554522 INFO :  bool DataManager::run() Completed initial database
 build.
.
.
.
```

Table 920: GET Response Codes

Response Code	Name	Definition
200	OK	OK & File information in the response body + ETag Header

Response Code	Name	Definition
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Files/{id}/

HEAD /Storage/Devices/{id}/Files/{id}/ pings a specific Compute File.

Table 921: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 922: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Files/{id}/

OPTIONS /Storage/Devices/{id}/Files/{id}/ returns the allowable HTTP Methods list for this resource.

Table 923: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 924: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
204	No Content	No Content & "Allow" list of methods in the response header. Some files will return 204 No Content indicating that there is no entity body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.10 /Storage/Devices/{id}/ZoneSets/

The Collection-level methods and attributes of /Storage/Devices/{id}/ZoneSets/ return information about the Zone Sets collection and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 925: Collection Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Members	String Array	Mandatory	No	Resource Collection (1 - n entries)	The array of Resource Instances. See **/Storage/Devices/{id}/ZoneSets/{id}/** *(page 528)* for Instance Attribute details

GET /Storage/Devices/{id}/ZoneSets/

GET /Storage/Devices/{id}/ZoneSets/ returns the Collection of ZoneSets.

Table 926: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 927: GET Response Codes

Response Code	Name	Definition
200	OK	OK & List of ZoneSets + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/ZoneSets/

HEAD /Storage/Devices/{id}/ZoneSets/ pings this ZoneSet collection.

Table 928: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 929: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource Collection does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/ZoneSets/

OPTIONS /Storage/Devices/{id}/ZoneSets/ returns the allowable HTTP Methods list for this resource.

Table 930: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 931: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource

Response Code	Name	Definition
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.11 /Storage/Devices/{id}/ZoneSets/{id}/

The Device-level methods and attributes of /Storage/Devices/{id}/ZoneSets/{id}/ return information about a specific Zone Set and enable management operations.

HTTP Methods: GET PUT HEAD OPTIONS

Table 932: Instance Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	URI String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link for the Collection
Name	String	Mandatory	No	Free-Form 64 Chars	The Name of this ZoneSet
ID	String	Mandatory	No	Free-Form 64 Chars. Valid characters include: a-z | A-Z | 0-9 | _ | -	The Identifier this particular ZoneSet
Description	String	Mandatory	No	Free-Form 128 Characters	Description of this ZoneSet
Active	Boolean	Mandatory	Yes	true | false	Determines if the ZoneSet is currently active. When a ZoneSet is set to "true", all other ZoneSets are set to "false" automatically.
MaxNumZoneMembers	Number	Mandatory	No	Data102 = 102 Data60 = 60	The maximum number of media members per ZoneSet for this Configuration
MaxNumZonesPerZoneS	Number	Mandatory	No	1 | 2 | 3 | 6	The maximum number of zones per ZoneSet for this Configuration
Associations	String Array	Mandatory	No	Device | ZoneSets	Association between the Enclosure and the ZoneSets
Zones[]. ID	String	Mandatory	No	ZoneSet 0 : 1 (1 Zone) ZoneSet 1 : 1 - 6 (6 Zones) ZoneSet 2 : 1 - 3 (3 Zone) ZoneSet 3 : 1 - 2 (2 Zones)	The Zone ID for each of the Zones within a ZoneSet

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Zones[]. NumZoneMembers	Number	Mandatory	No	Data102 Media Members Per Zone: ZoneSet 0 : 102 ZoneSet 1 : 17 ZoneSet 2 : 34 ZoneSet 3 : 51 Data60 Media Per Zone: ZoneSet 0 : 60 ZoneSet 1 : 10 ZoneSet 2 : 20 ZoneSet 3 : 30	Media Member arrangement. There are 102 Media (Drives) in the Data102 and 60 Media (Drives) in the Data60
Zones[]. Connectors	String Array	Mandatory	No	Host Connectors Per Zone ZoneSet 0 : 12 ZoneSet 1 : 2 ZoneSet 2 : 4 ZoneSet 3 : 6	Host Connector zoning arrangement. There are 12 Host Connectors, 6 per Controller (IOM)
Zones[]. Media	String Array	Mandatory	No	Data102 Media Per Zone: ZoneSet 0 : 1 - 102 ZoneSet 1 : 1 - 17, 18 - 34, 35 - 51, 52- 68 , 69 - 85, 86 - 102 ZoneSet 2 : 1 - 34, 35 - 68, 69 - 102 ZoneSet 3 : 1 - 51, 52 - 102 Data60 Media Per Zone: ZoneSet 0 : 1 - 60 ZoneSet 1 : 1 - 10, 11 - 20, 21 - 30, 31- 40 , 41 - 50, 51 - 60 ZoneSet 2 : 1 - 20, 21 - 40, 41 - 60 ZoneSet 3 : 1 - 30, 31 - 60	

GET /Storage/Devices/{id}/ZoneSets/{id}/

GET /Storage/Devices/{id}/ZoneSets/{id}/ returns the information regarding a specific ZoneSet based on the {ID} in the URI address string.

Table 933: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/1/?
IPAddress=10.20.30.41
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/1/?
IPAddress=10.20.30.41",
    "Name": "Zone Set 1",
    "ID": "1",
    "Description": "Configuration 1",
    "Active": false,
    "MaxNumZoneMembers": 17,
    "MaxNumZonesPerZoneSet": 6,
    "Associations": {
        "Device": [
            "5000CCAB04110A00"
        ],
        "ZoneSets": [
            "1"
        ]
    },
    "Zones": [
        {
            "ID": "1",
            "NumZoneMembers": 17,
            "Connectors": [
                "1",
                "7"
            ],
            "Media": [
                "1",
                "2",
                "3",
                "4",
                "5",
                "6",
                "7",
                "8",
                "9",
                "10",
                "11",
                "12",
                "13",
```

```
                "14",
                "15",
                "16",
                "17"
            ]
        },
        {
            "ID": "2",
            "NumZoneMembers": 17,
            "Connectors": [
                "2",
                "8"
            ],
            "Media": [
                "18",
                "19",
                "20",
                "21",
                "22",
                "23",
                "24",
                "25",
                "26",
                "27",
                "28",
                "29",
                "30",
                "31",
                "32",
                "33",
                "34"
            ]
        },
        {
            "ID": "3",
            "NumZoneMembers": 17,
            "Connectors": [
                "3",
                "9"
            ],
            "Media": [
                "35",
                "36",
                "37",
                "38",
                "39",
                "40",
                "41",
                "42",
                "43",
                "44",
                "45",
                "46",
                "47",
                "48",
                "49",
                "50",
                "51"
            ]
```

```
        },
        {
            "ID": "4",
            "NumZoneMembers": 17,
            "Connectors": [
                "4",
                "10"
            ],
            "Media": [
                "52",
                "53",
                "54",
                "55",
                "56",
                "57",
                "58",
                "59",
                "60",
                "61",
                "62",
                "63",
                "64",
                "65",
                "66",
                "67",
                "68"
            ]
        },
        {
            "ID": "5",
            "NumZoneMembers": 17,
            "Connectors": [
                "5",
                "11"
            ],
            "Media": [
                "69",
                "70",
                "71",
                "72",
                "73",
                "74",
                "75",
                "76",
                "77",
                "78",
                "79",
                "80",
                "81",
                "82",
                "83",
                "84",
                "85"
            ]
        },
        {
            "ID": "6",
            "NumZoneMembers": 17,
            "Connectors": [
```

```
                "6",
                "12"
            ],
            "Media": [
                "86",
                "87",
                "88",
                "89",
                "90",
                "91",
                "92",
                "93",
                "94",
                "95",
                "96",
                "97",
                "98",
                "99",
                "100",
                "101",
                "102"
            ]
        }
    ]
}
```


Table 934: GET Response Codes

Response Code	Name	Definition
200	OK	OK & ZoneSet information in the response body + ETag Header
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

PUT /Storage/Devices/{id}/ZoneSets/{id}/

PUT /Storage/Devices/{id}/ZoneSets/{id}/ modifies an attribute or executes an operation for this ZoneSet.

Table 935: PUT Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires Basic authorization
If-Match	Mandatory	The request is conditionally processed only if the current ETag for the resource matches the ETag passed in this header

Example PUT curl Command

```
curl
    -i
    -u admin:admin
    -H "Content-Type: application/json"
    -H "If-Match: 3677651d2011428f32120a029b015e80"
    -X PUT http://10.20.30.40/Storage/Devices/0123456789/ZoneSets/2?
IPAddress=10.20.30.41
    --data '{
        "Active": true
    }'
```

Table 936: PUT Response Codes

Response Code	Name	Definition
200	OK	OK & ZoneSet information in the response body + ETag Header
400	Bad Request	Bad Request (typically a faulty parameter)
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	The Resource doesn't exist
412	Precondition Failed	Indicates the "If-Match" ETag check failed
428	Precondition Required	Indicates the need for an "If-Match" conditional with an ETag value in the Request Header
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/ZoneSets/{id}/

HEAD /Storage/Devices/{id}/ZoneSets/{id}/ pings this ZoneSet.

Table 937: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 938: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/ZoneSets/{id}/

OPTIONS /Storage/Devices/{id}/ZoneSets/{id}/ returns the allowable HTTP Methods list for this resource.

Table 939: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 940: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

6.2.12 /Storage/Devices/{id}/Telemetry/

The methods and attributes of /Storage/Devices/{id}/Telemetry/ return information about the Telemetry resource and enable management operations.

HTTP Methods: GET HEAD OPTIONS

Table 941: Resource Attributes

Attribute	Type	Mandatory or Optional	User Config-urable	Possible Values	Description
Self	String	Mandatory	No	Free-Form URI Format 256 Chars	Describes the fully qualified URI link
LastAccessed	String	Mandatory	No	datetime based on UTC	The UTC time of the request
Members[]	String Array	Mandatory	No	Member Entry	This is the instantaneous, fully expanded metadata from this Device

GET /Storage/Devices/{id}/Telemetry/

GET /Storage/Devices/{id}/Telemetry/ returns the information of the Telemetry resource.

Table 942: GET Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization
If-None-Match	Optional	The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Example GET curl Command

```
curl
    -i
    -u admin:admin
    -X GET http://10.20.30.40/Storage/Devices/0123456789/Telemetry/
```

Example GET Response Body


```
{
    "Self": "http://10.20.30.40/Storage/Devices/0123456789/Telemetry/",
    "LastAccessed": "26-Jan-2024 12:48:34 UTC",
    "Members": [
        {
            "Self": "http://10.20.30.40/Storage/Devices/0123456789/Telemetry/",
            "SystemType": { ... },
            "Name": "WDAP-LGWPTOMIDK",
            "ID": "A0A9CB7F-7D63-4075-8826-48E0ADF3F2C0",
            "OperatingSystem": { ... },
            "Model": "Resource Manager Data Center",
            "Manufacturer": "WDC",
            "Capabilities": { ... },
            "Status": { ... },
            "IPAddresses": { ... },
            "InformationStructure": { ... },
            "ApplicationSettings": { ... },
            "Location": { ... },
            "Accounts": { ... },
            "Files": { ... },
            "Jobs": { ... },
            "SystemClock": { ... },
            "Assets": { ... },
            "Analytics": { ... },
            "Notifications": { ... },
            "Policies": { ... },
            "Reports": { ... },
            "Telemetry": { ... }
        }
    ]
}
```


Table 943: GET Response Codes

Response Code	Name	Definition
200	OK	OK & Telemetry resource information in the response body
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

HEAD /Storage/Devices/{id}/Telemetry/

HEAD /Storage/Devices/{id}/Telemetry/ pings the Telemetry resource.

Table 944: HEAD Request Headers

Header	Mandatory or Optional	Description
Authorization	Mandatory	The request requires either Basic or Digest authorization

Table 945: HEAD Response Codes

Response Code	Name	Definition
200	OK	OK with no response body
401	Unauthorized	Unauthorized; credential entry failed or missing
404	Not Found	Resource does not exist; this IP Address is running an HTTP service but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

OPTIONS /Storage/Devices/{id}/Telemetry/

OPTIONS /Storage/Devices/{id}/Telemetry/ returns the allowable HTTP Methods list for this resource.

Table 946: OPTIONS Request Headers

Header	Mandatory or Optional	Description
(None)	N/A	Returns the full resource schema as Content-Type application/json
Documentation	Optional	Info - Returns documentation information in the form of plain text
		Schema - Returns the schema information in "pretty" JSON text as plain text
If-None-Match	Optional	ETag - The request is conditionally processed only if the current ETag for the resource does not match the ETag passed in this header

Table 947: OPTIONS Response Codes

Response Code	Name	Definition
200	OK	OK & the “Allow” list of methods in the response header. The response body provides the JSON Schema for this resource
304	Not Modified	If-None-Match ETag value matched, therefore no Response Body will be returned.
404	Not Found	Resource does not exist; this IP Address is running an HTTP service, but does not recognize this resource
429	Too Many Requests	Indicates API service is busy; client should try again later

Appendices

In This Chapter:

- References.....540

7.1 References

- IETF RFC3986, Uniform Resource Identifier (URI) General Syntax
- IETF RFC4627, The application/json Media Type for JavaScript Object Notation (JSON)
- IETF RFC5988, Web Linking
- IETF RFC6901, JavaScript Object Notation (JSON) Pointer
- IETF RFC6906, Link Relation Type
- IETF RFC6749, OAuth 2.0 Authorization Framework; Token Format and Usage
- IETF RFC7230, Hypertext Transfer Protocol (HTTP/1.1): Message Syntax and Routing
- IETF RFC7231, Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content
- IETF RFC7232, Hypertext Transfer Protocol (HTTP/1.1): Conditional Requests
- IETF RFC7235, Hypertext Transfer Protocol (HTTP/1.1): Authentication
- ISO 8601, Data elements and interchange formats — Representation of dates and times
- DMTF CIM_2.50.0, Common Information Model Schema
- DMTF DSP0266_1.2.0, Redfish API Specification
- SNIA Swordfish_1.0.3, Swordfish API Specification